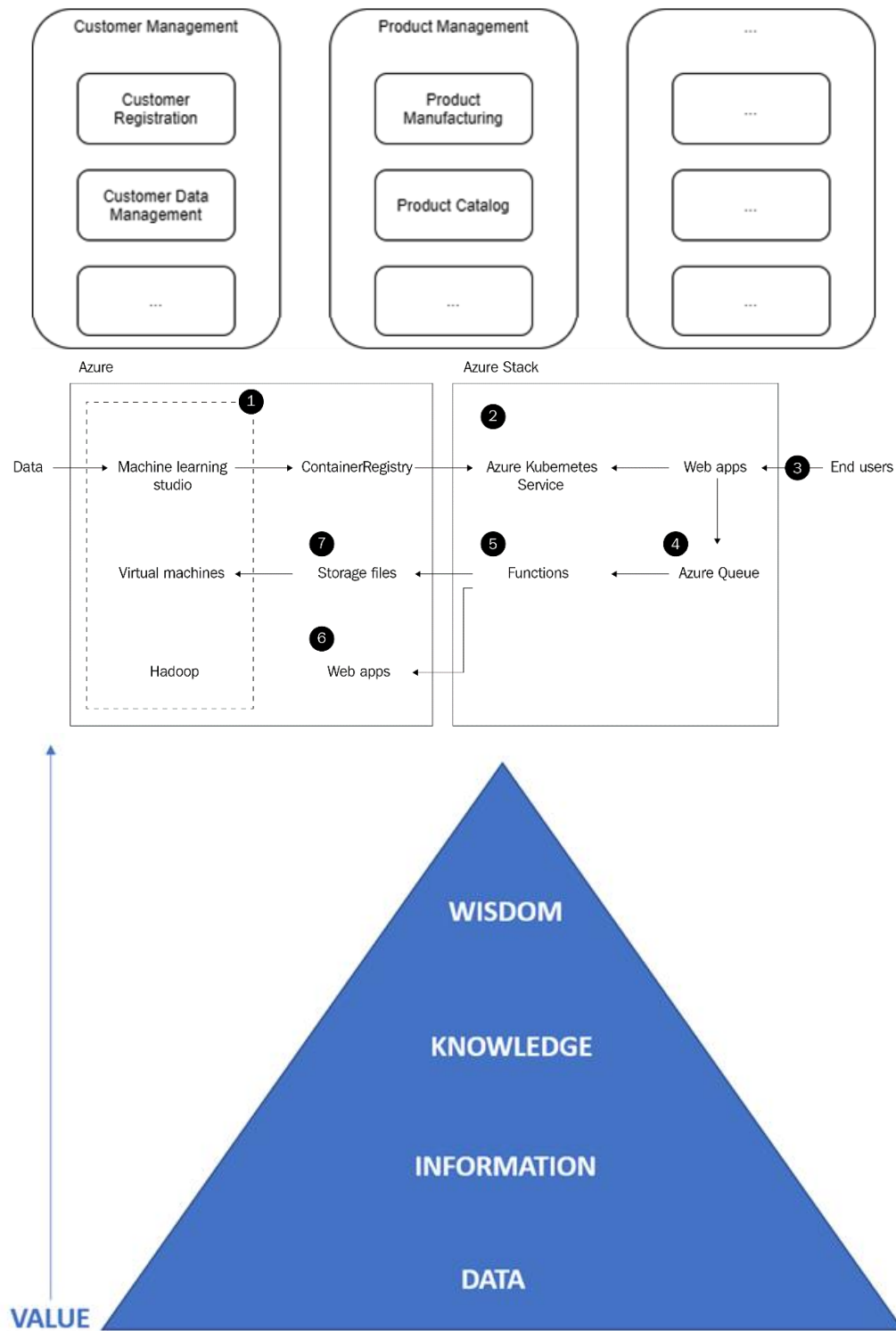
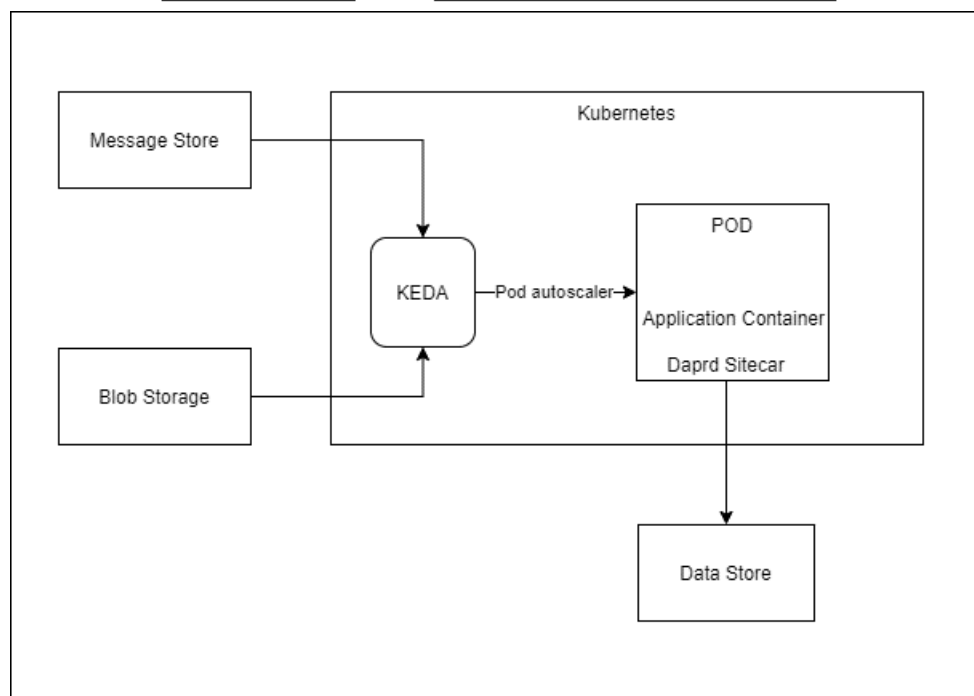
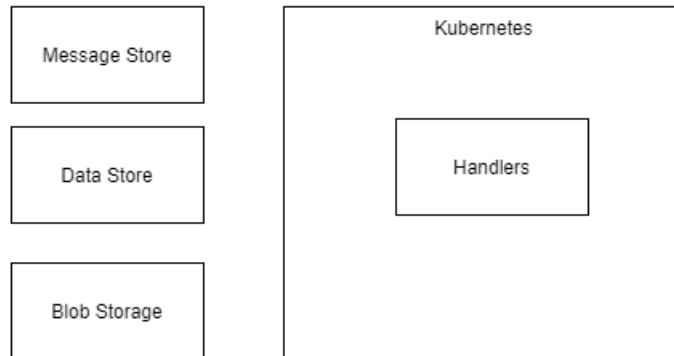
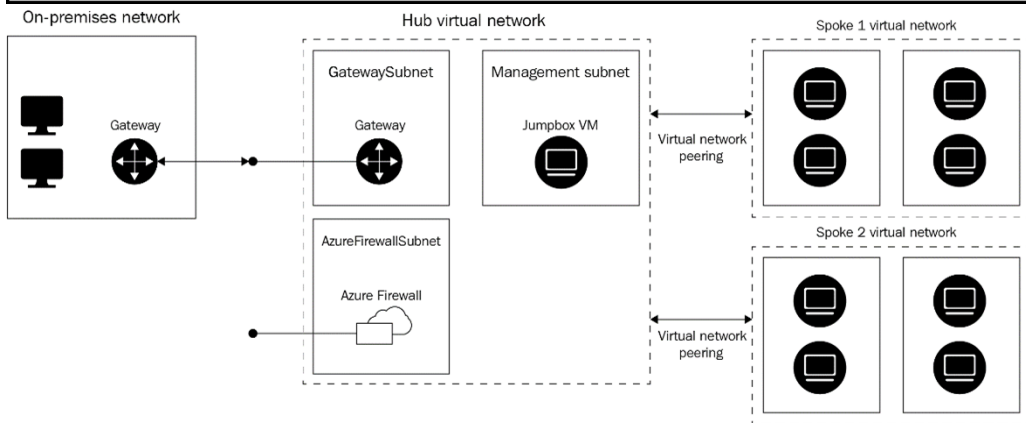
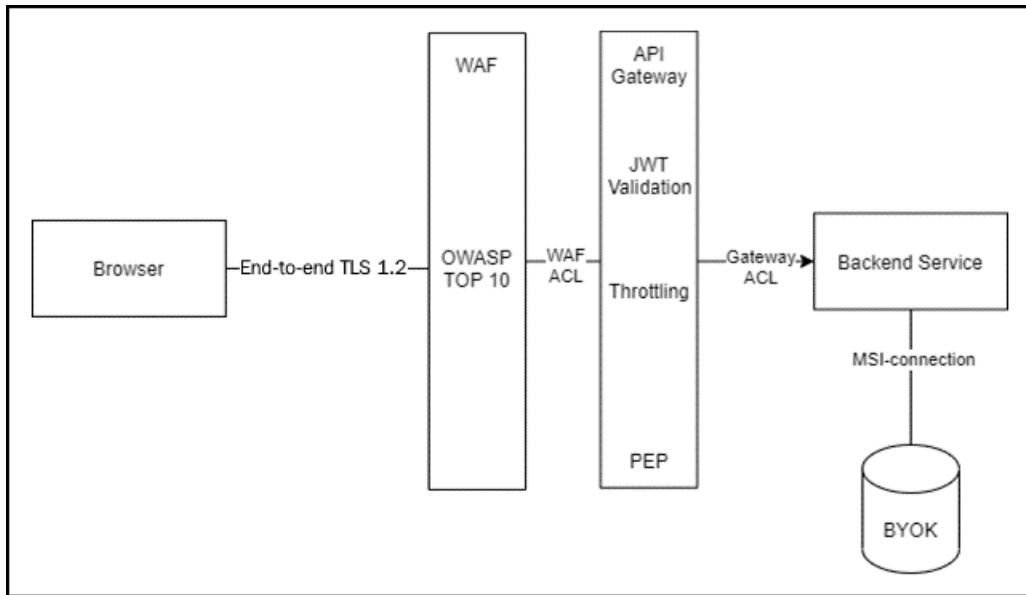


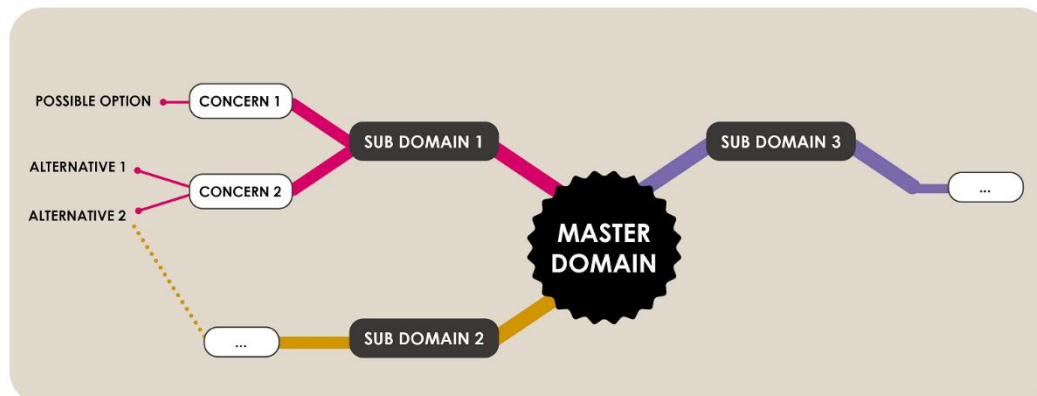
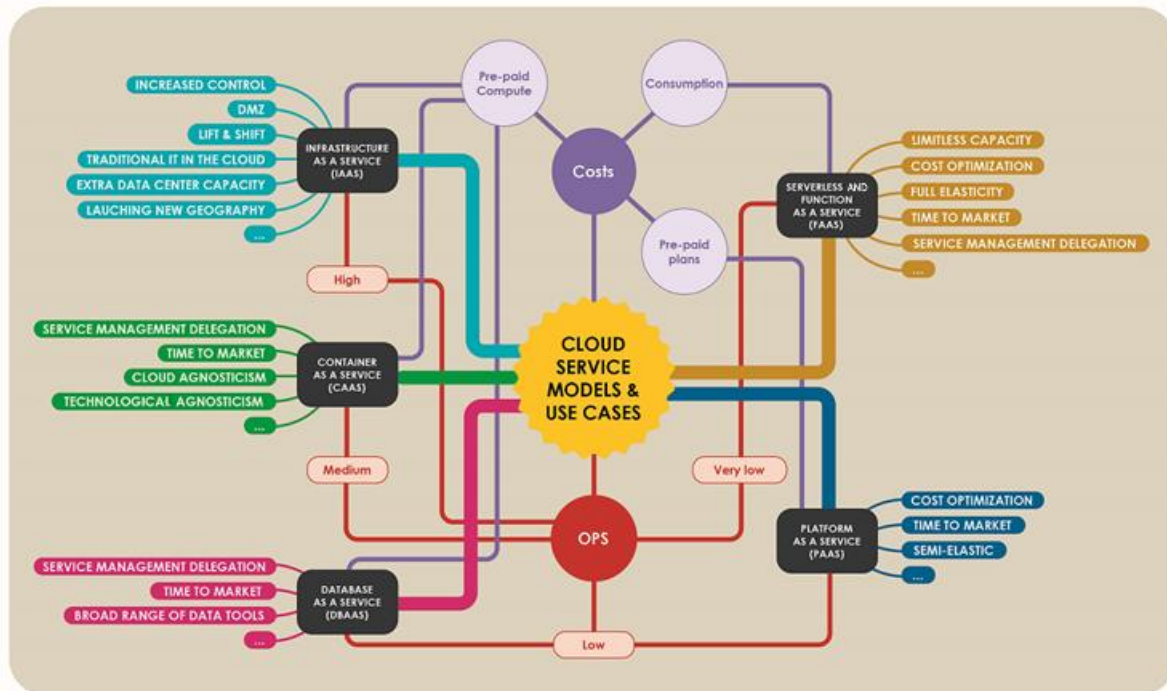
Chapter 1: Getting Started as an Azure Architect

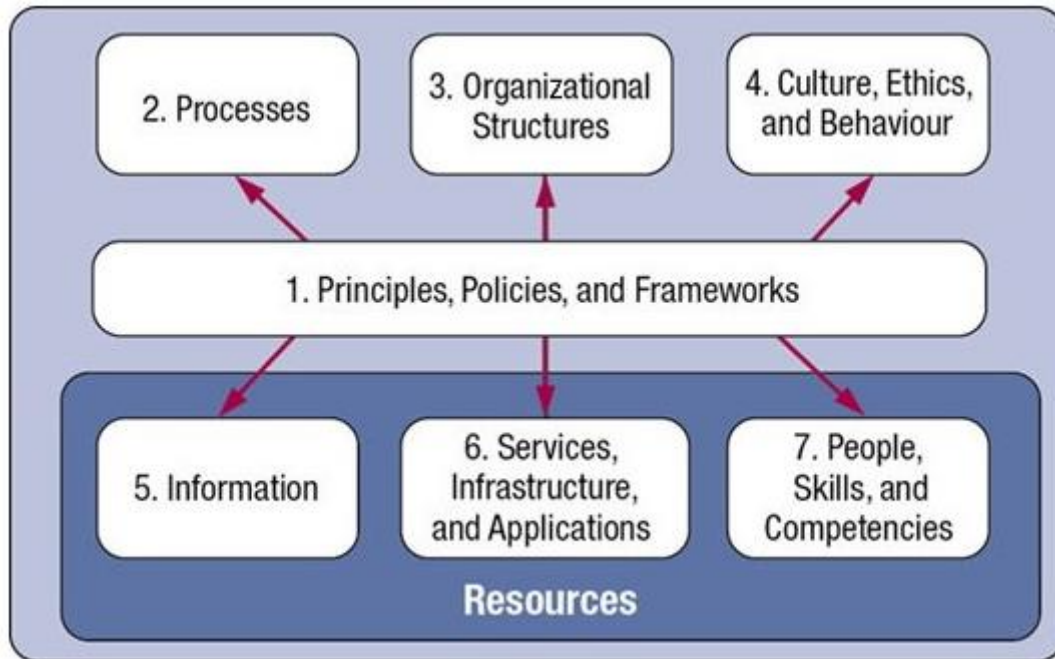


DATA	31, 3, 3,000
INFORMATION	DAY: 31, MONTH: MARCH, VISITS: 3000
KNOWLEDGE	Up to 3000 users visited our website on March 31, which is way above the usual daily average of 650. March 31 seems always busy, year after year.
WISDOM	We will make sure to restock warehouses before March 31.

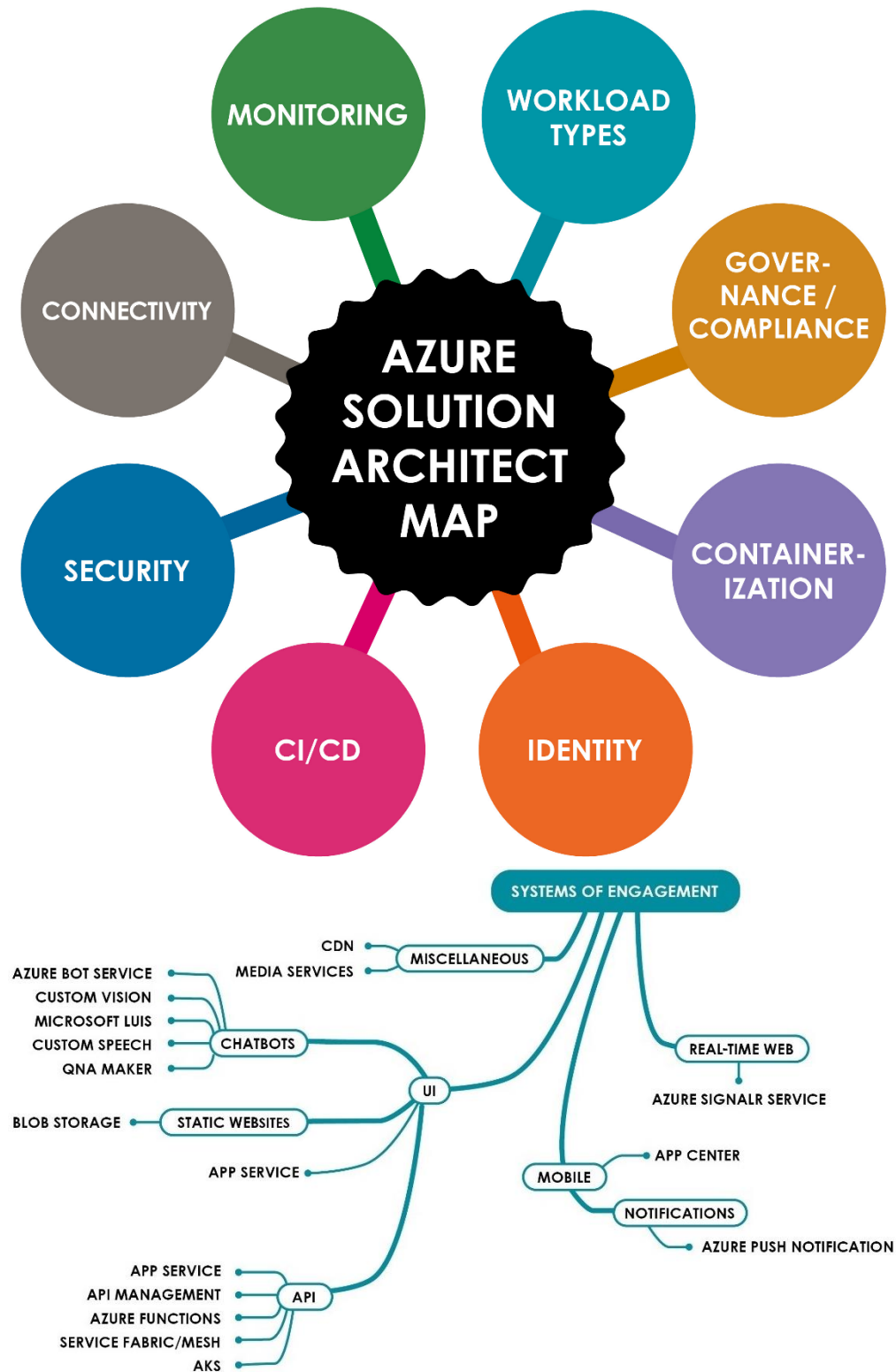


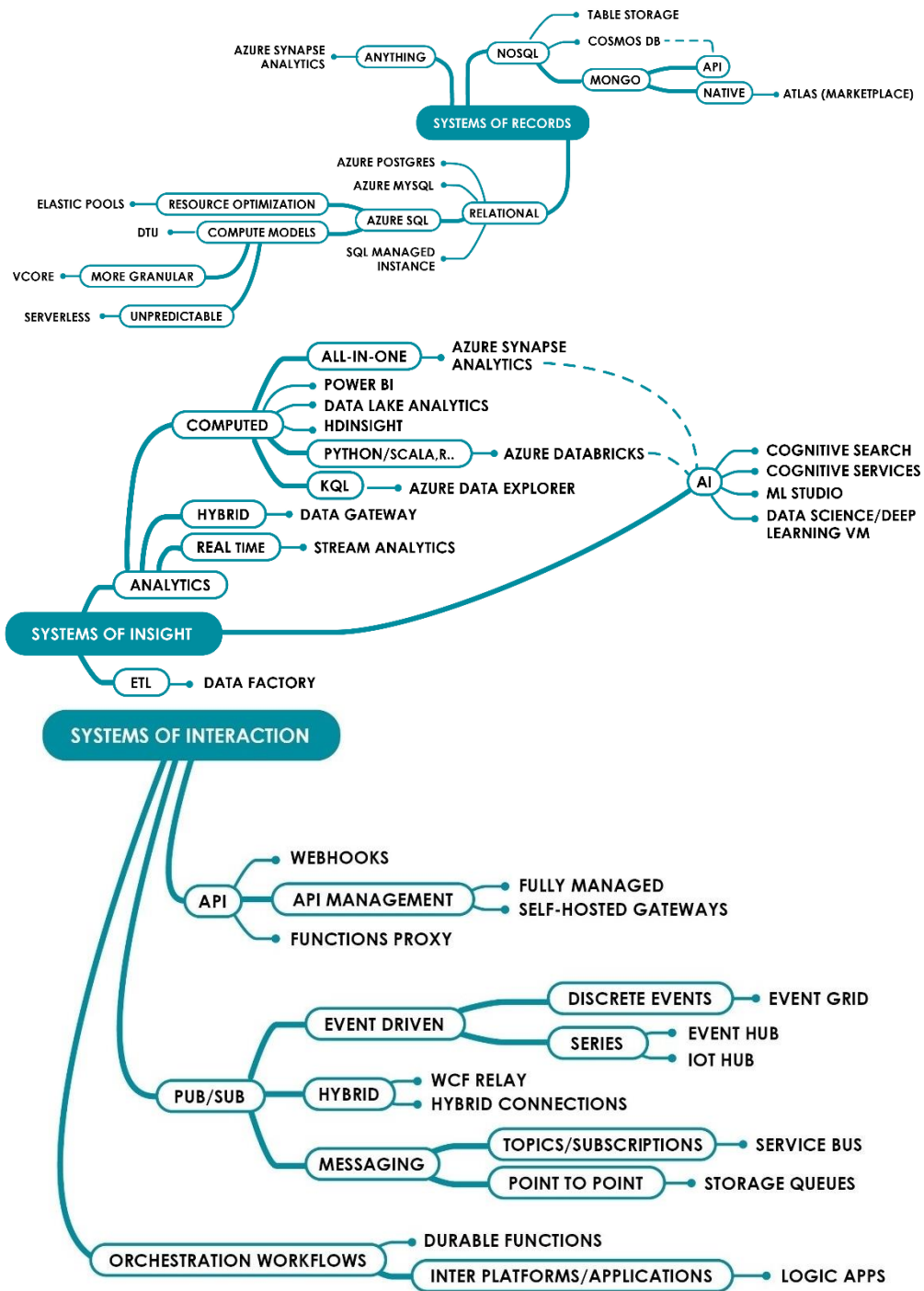


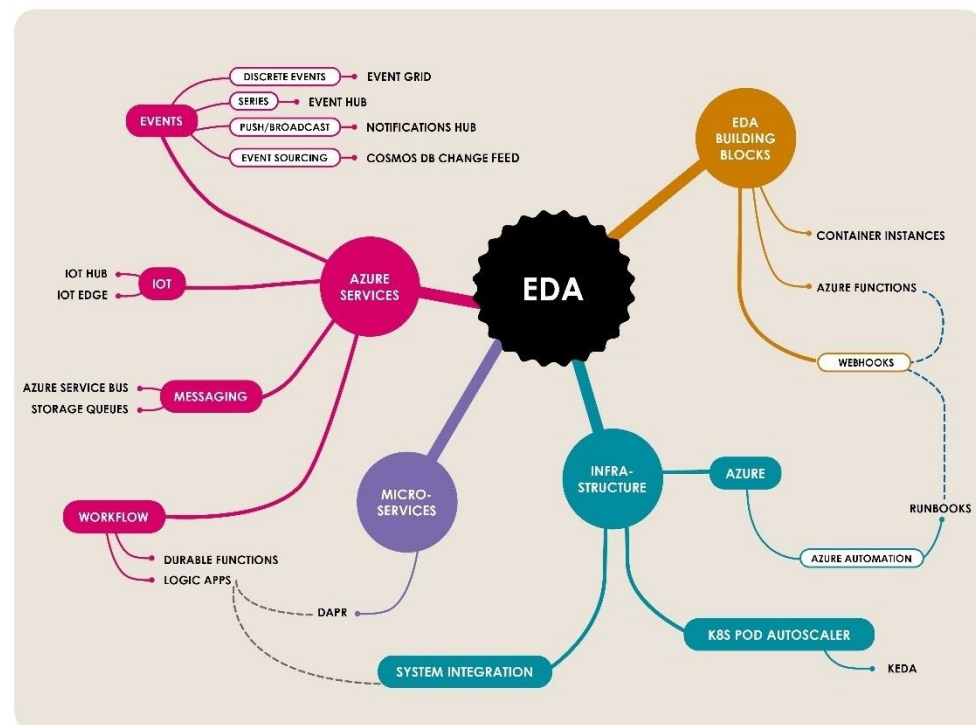
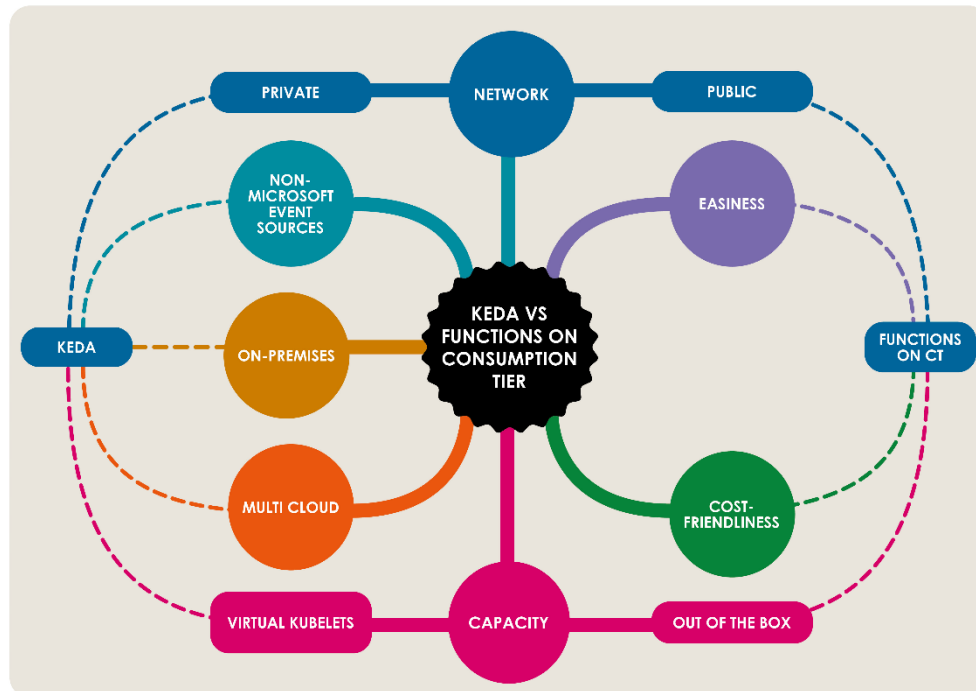


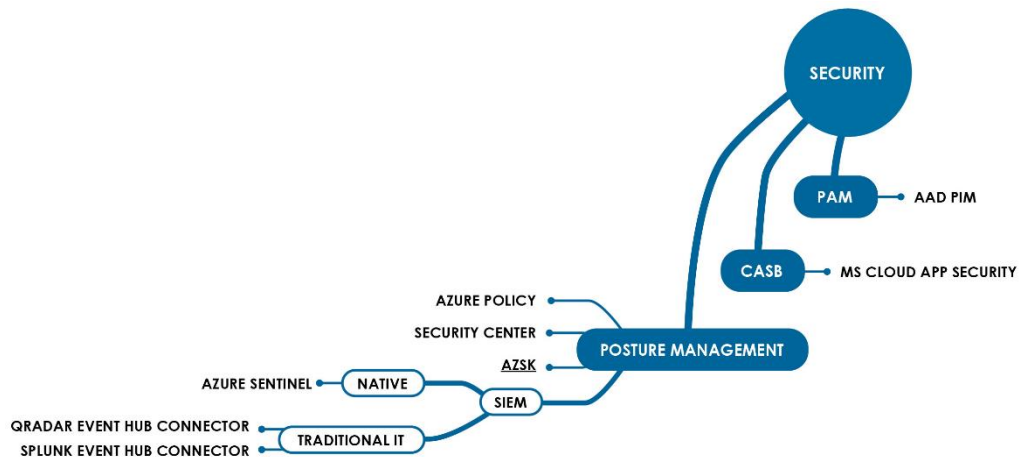
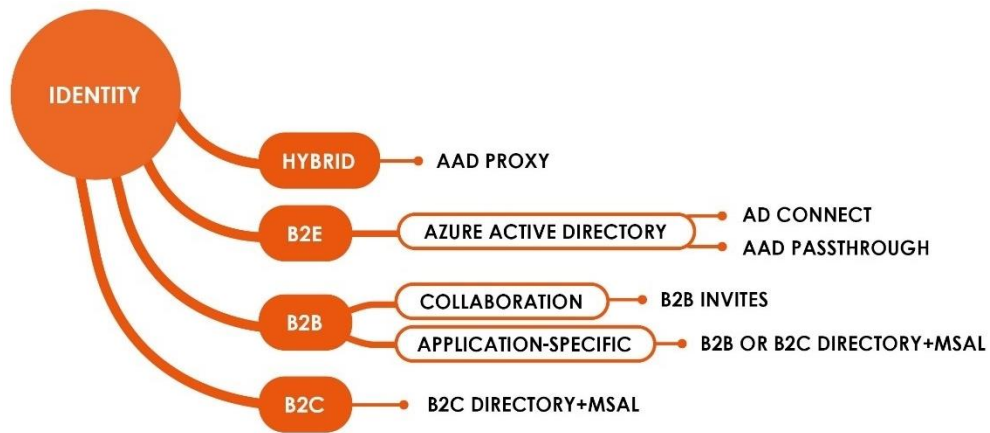
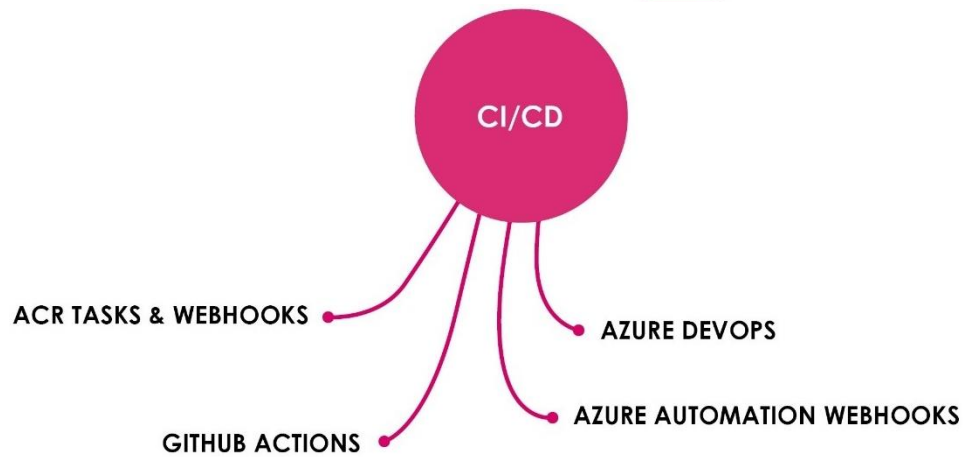
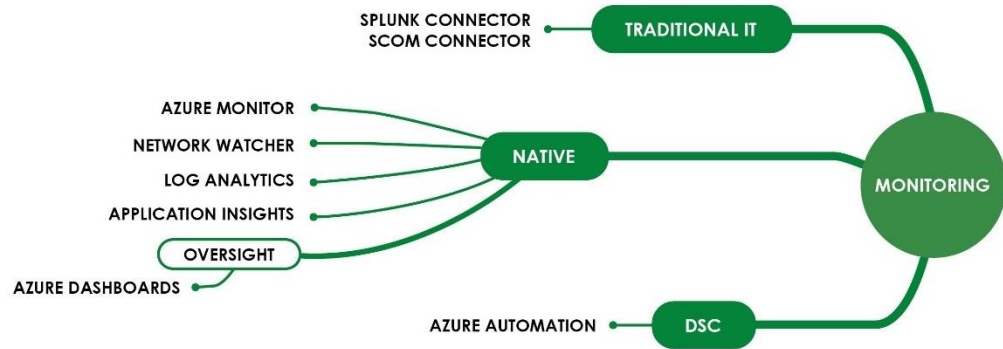


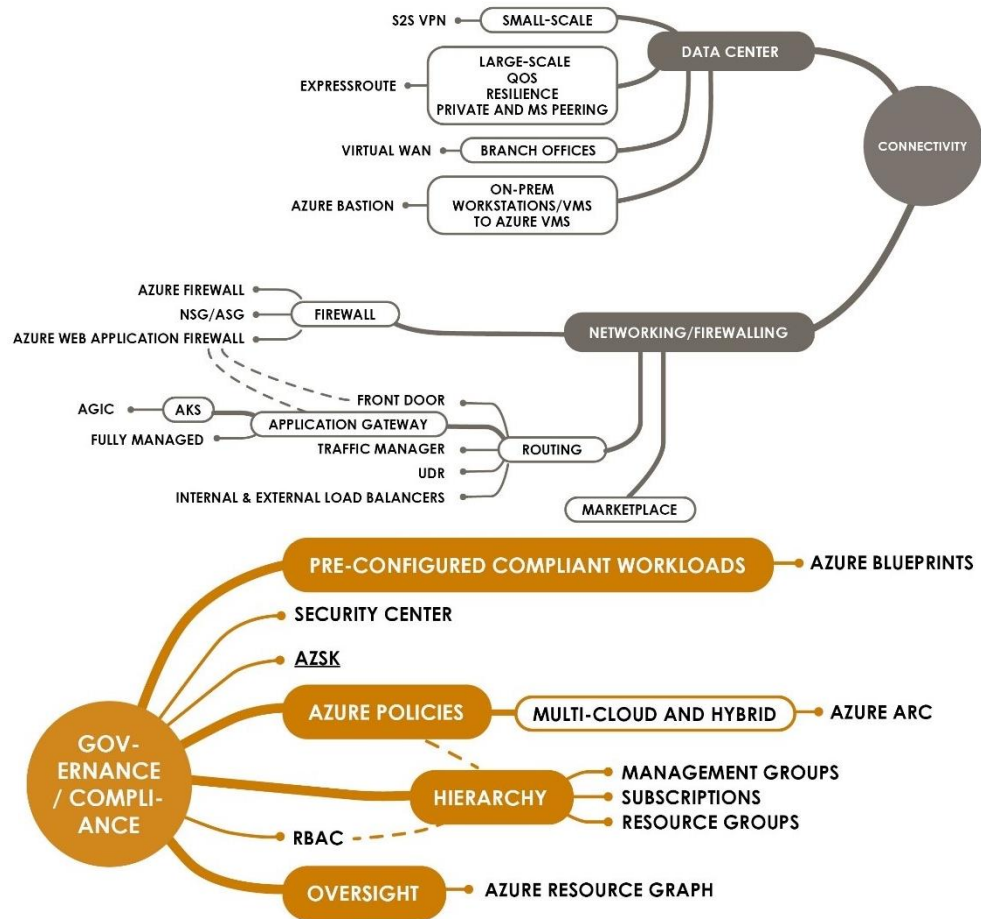
Chapter 2: Solution Architecture

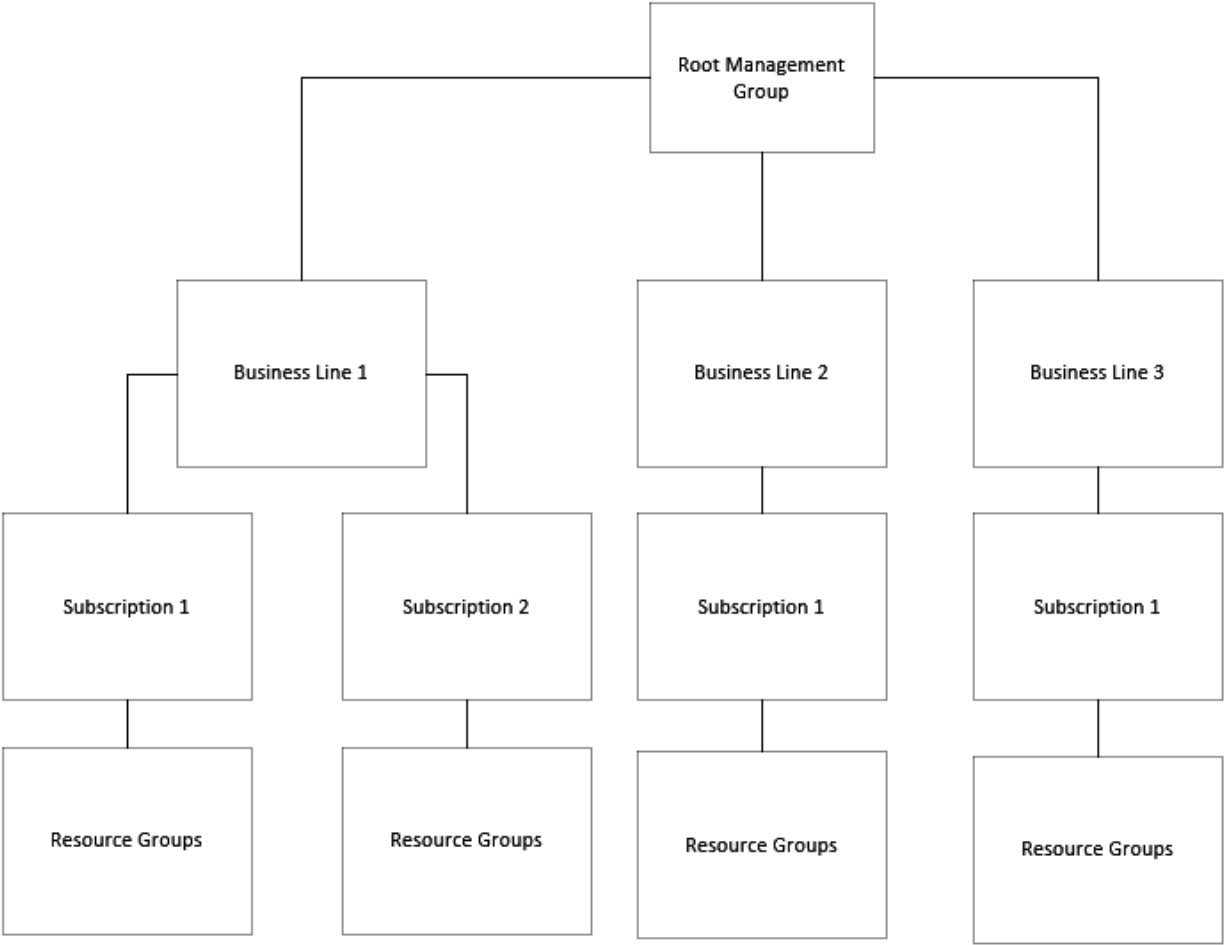


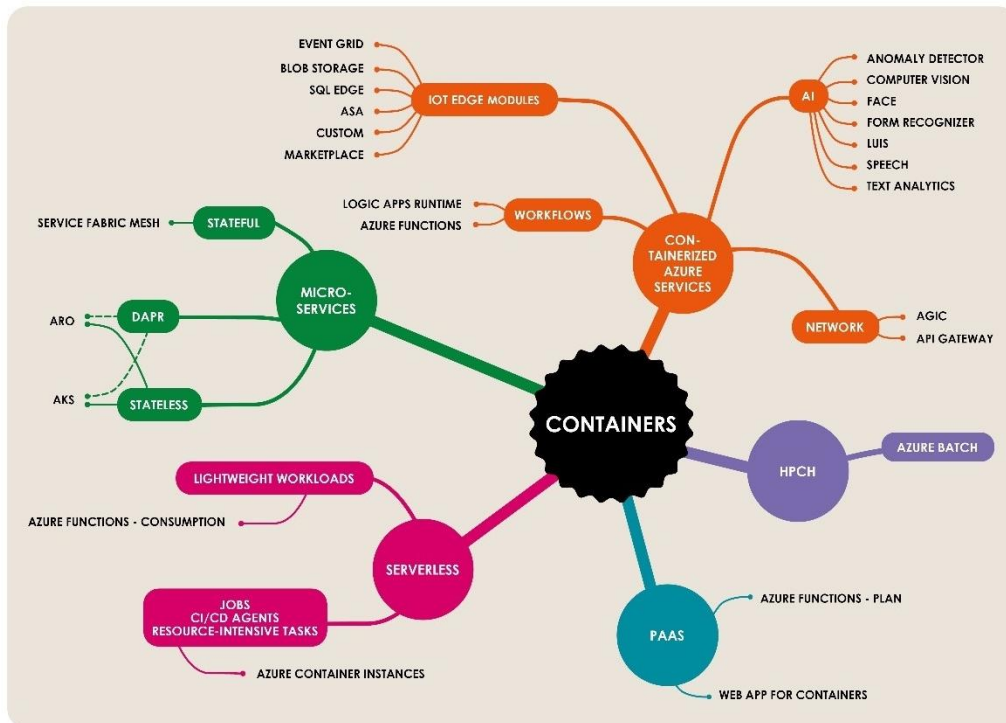
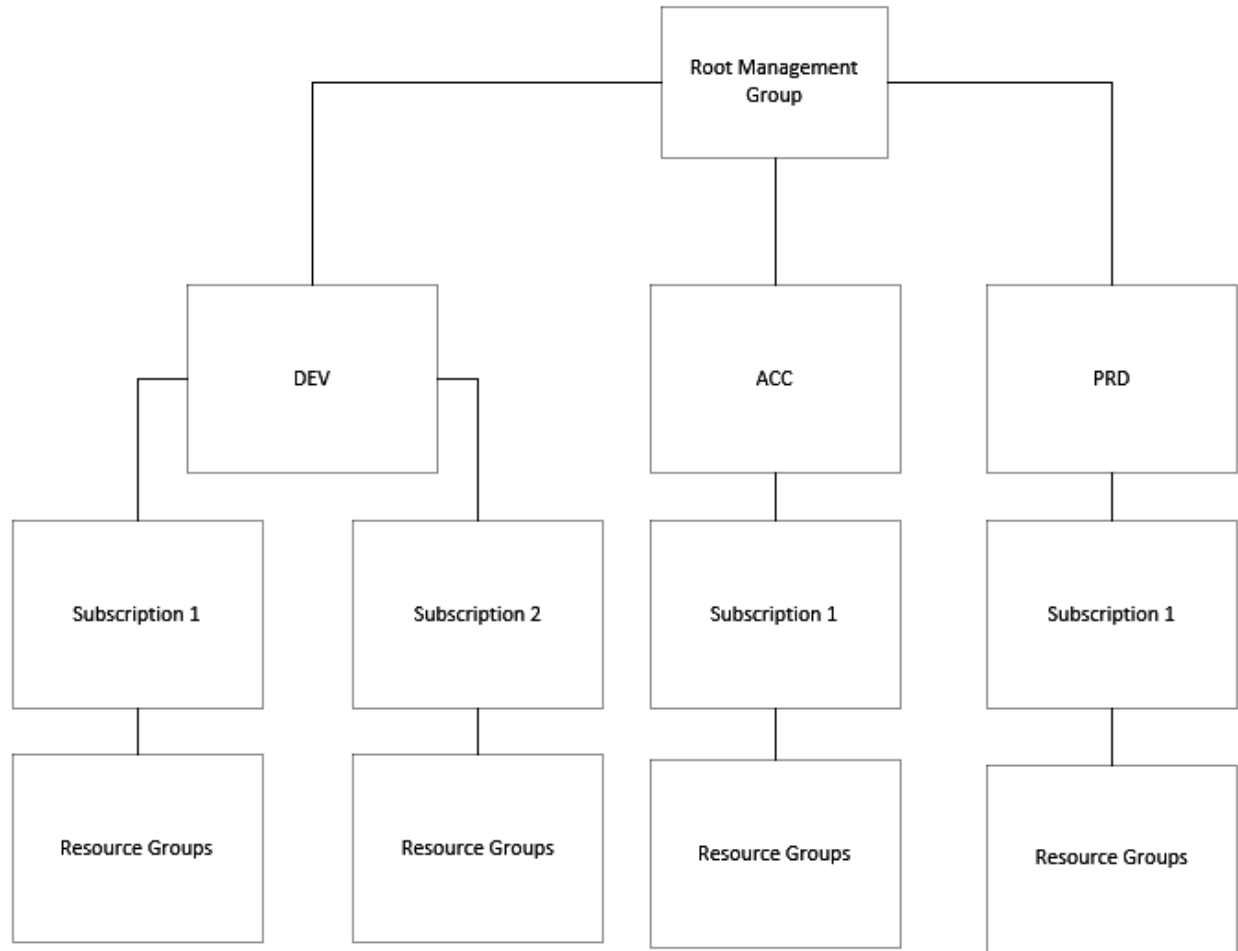


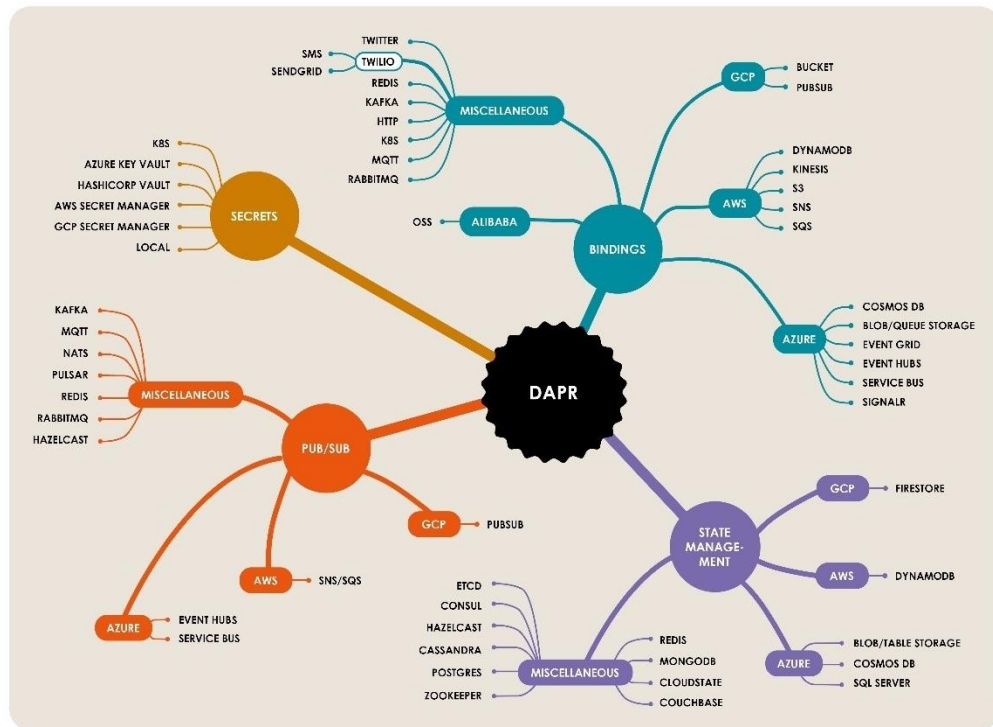




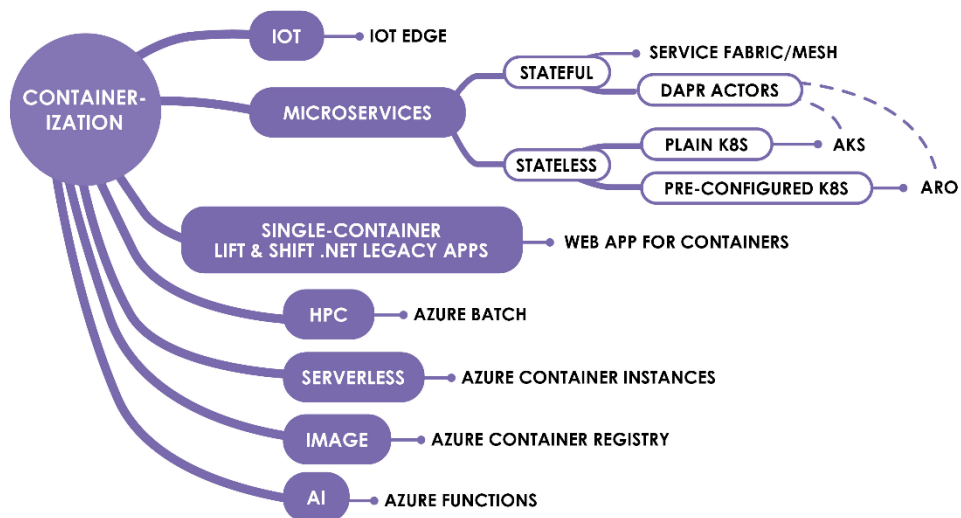
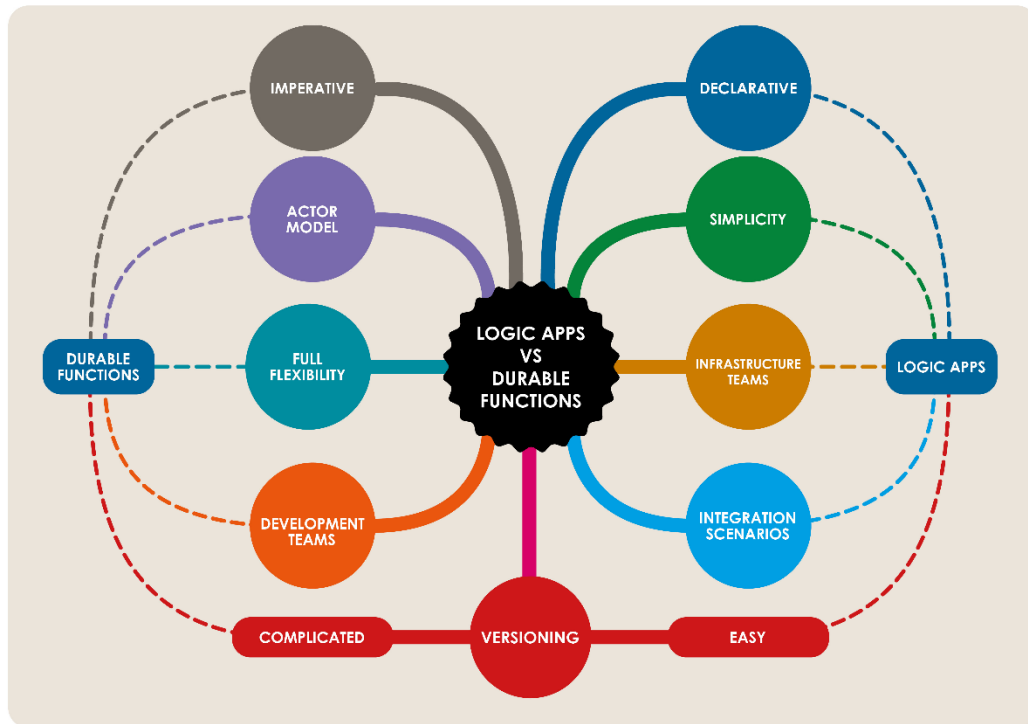
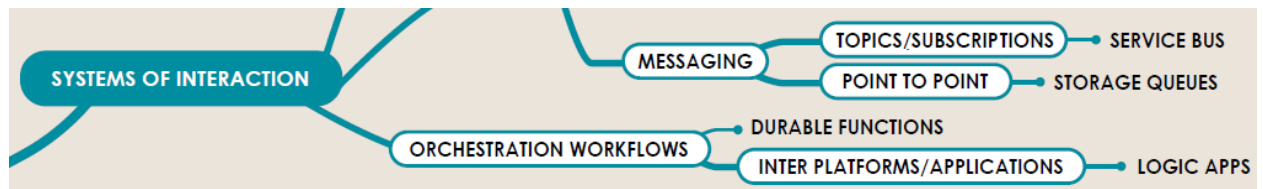


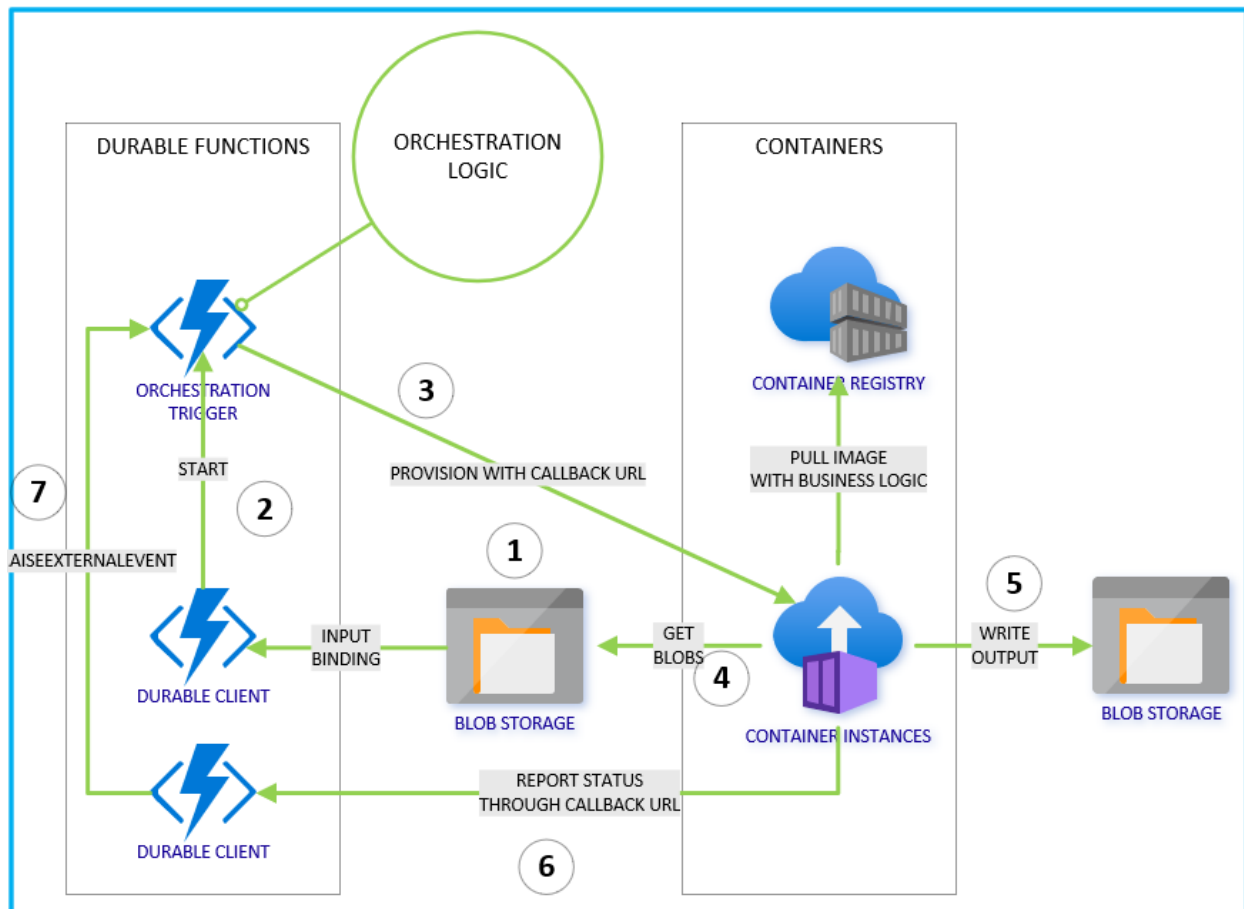


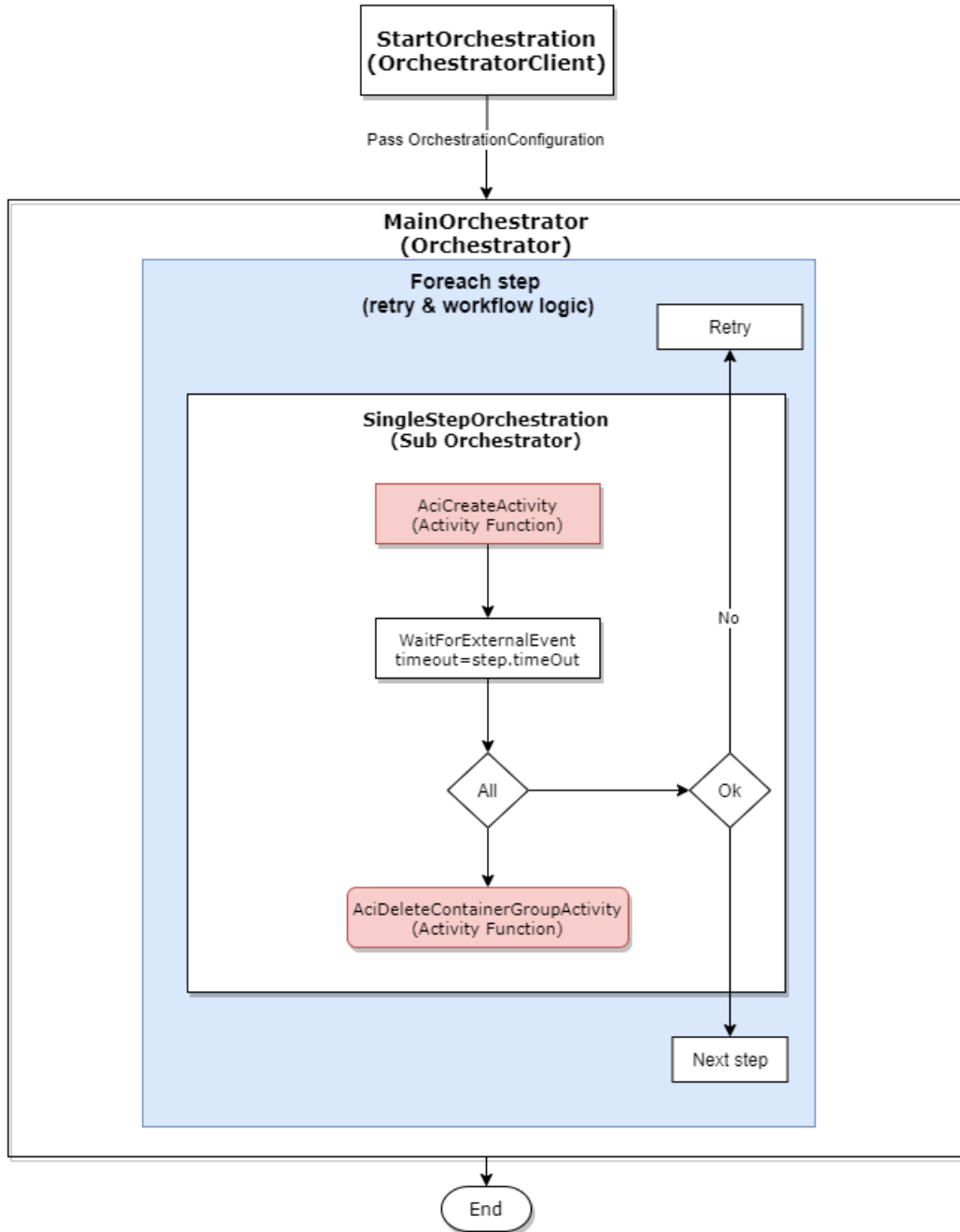




	Cost	Complexity	Operations
Azure Kubernetes Services	Medium to high	High	High
Azure Red Hat OpenShift	Medium to high	Medium	Medium
Service Fabric Mesh	Low	Medium	Low
Azure Container Instances	Low	Low	Low
Azure Batch	Medium to high	Medium	Low
Web App for Containers	Medium	Low	Low
Azure Functions on consumption	Low	Low	Low
Azure Functions on PaaS	Medium	Low	Low
Self-hosted containers versus a Microsoft-hosted counterpart	Higher	Higher	Higher







Azure Functions Core Tools (3.0.2912 Commit hash: bfcbb48ed6fdacdf9b309261ecc8093df3b83f2)
Function Runtime Version: 3.0.14287.0

Functions:

StartOrchestration: [POST] http://localhost:7071/api/StartOrchestration
stepResultCallback: [POST] http://localhost:7071/api/stepResultCallback
InitializeWithBreakingChange: activityTrigger
MainOrchestrator: orchestrationTrigger
SingleStepOrchestration: orchestrationTrigger

For detailed output, run func with --verbose flag.

Hosting environment: Production

Content root path: C:\Users\steph\Source\Repos\packt\PacktOrchestrationDemo\bin\Debug\netcoreapp3.1

Now listening on: http://0.0.0.0:7071

Application started. Press Ctrl+C to shut down.

Host lock lease acquired by instance ID '0000000000000000000000000921E07D7'.

Get Started  Statistics  Inspectors  AutoResponder  Composer

Use this page to compose a Request. You can clone a prior request by dragging and

Parsed Raw Scratchpad Options

POST

Host: localhost:7071
Content-Length: 324
|

<

Request Body

```
{
  "isRewindable": true,
  "defaultStepTimeout": 5,
  "steps": [
    {
      "stepName": "step1",
      "maxRetryCount": 1,
      "stopFlowOnFailure": true,
      "timeout": 5
    },
    {
      "stepName": "step2",
      "maxRetryCount": 0,
      "stopFlowOnFailure": false,
      "timeout": 3
    }
  ]
}
```

Azure Functions Core Tools (3.0.2912 Commit hash: bfcbb48ed6fdacdf9b309261ecc8093df3b83f2)
Function Runtime Version: 3.0.14287.0

Functions:

StartOrchestration: [POST] http://localhost:7071/api/StartOrchestration

stepResultCallback: [POST] http://localhost:7071/api/stepResultCallback

MainOrchestrator: orchestrationTrigger

SingleStepOrchestration: orchestrationTrigger

For detailed output, run func with --verbose flag.

Hosting environment: Production

Content root path: C:\Users\steph\Source\Repos\packt\PacktOrchestrationDemo\bin\Debug\netcoreapp3.1

Now listening on: http://0.0.0.0:7071

Application started. Press Ctrl+C to shut down.

Host lock lease acquired by instance ID '0000000000000000000000000921E07D7'.

-----CREATING ACI for step1 on instance e83c22e99c814e86b30ef83540548003:0 -----

-----CALLBACK URL http://localhost:7071/api/stepResultCallback?instanceId=e83c22e99c814e86b30ef83540548003:0&eventName=step1&status=

-----WAITING FOR EVENT step1 on instance e83c22e99c814e86b30ef83540548003:0-----

Use this page to compose a Request. You can clone a prior request by dragging and dropping a session from the Web Ses

Parsed Raw Scratchpad Options

POST

resultCallback?instanceId=e83c22e99c814e86b30ef83540548003:0&eventName=step1&status=nok

Host: localhost:7071

Content-Length: 0

RECEIVED FEEDBACK FROM step1 WITH STATUS nok

-----CREATING ACI for step1 on instance e83c22e99c814e86b30ef83540548003:0 -----

-----CALLBACK URL http://localhost:7071/api/stepResultCallback?instanceId=e83c22e99c814e86b30ef83540548003:0&eventName=step1&status=

-----WAITING FOR EVENT step1 on instance e83c22e99c814e86b30ef83540548003:0-----

-----STEP step1 INSTANCE IS e83c22e99c814e86b30ef83540548003:0 STATE IS nok-----

-----state NOK-----

-----deleting ACI-----

e83c22e99c814e86b30ef83540548003:0: Function 'SingleStepOrchestration (Orchestrator)' failed with an error. Reason: System.ApplicationException: Error in the application

at PacktOrchestrationDemo.Orchestration.SingleStepOrchestration(IDurableOrchestrationContext context, ILogger log) in C:\Users\steph\Source\Repos\packt\PacktOrchestra

tration.cs:line 94

at Microsoft.Azure.WebJobs.Host.Executors.FunctionInvoker`2.InvokeAsync(Object instance, Object[] arguments) in C:\projects\azure-webjobs-sdk-rqm4t\src\Microsoft.Azure

Executors\FunctionInvoker.cs:line 52

at Microsoft.Azure.WebJobs.Extensions.DurableTask.TaskOrchestrationShim.InvokeUserCodeAndHandleResults(RegisteredFunctionInfo orchestratorInfo, OrchestrationContext i

d:\a\r1\azure-functions-durable-extension\src\WebJobs.Extensions.DurableTask\Listener\TaskOrchestrationShim.cs:line 150. IsReplay: False. State: Failed. HubName: azur

ame: . SlotName: . ExtensionVersion: 2.2.2. SequenceNumber: 12.

Executed 'SingleStepOrchestration' (Failed, Id=c8297393-7e92-4169-a023-df5aab23d1fb, Duration=389ms)

System.Private.CoreLib: Exception while executing function: SingleStepOrchestration. System.Private.CoreLib: Orchestrator function 'SingleStepOrchestration' failed: Error

ation.

-----CREATING ACI for step1 on instance e83c22e99c814e86b30ef83540548003:2 -----

-----CALLBACK URL http://localhost:7071/api/stepResultCallback?instanceId=e83c22e99c814e86b30ef83540548003:2&eventName=step1&status=

-----WAITING FOR EVENT step1 on instance e83c22e99c814e86b30ef83540548003:2-----

Parsed Raw Scratchpad Options

POST

ResultCallback?instanceId=e83c22e99c814e86b30ef83540548003:2&eventName=step1&status=ok

HTTP/1.1

Host: localhost:7071

Content-Length: 0

-----WAITING FOR EVENT step1 on instance e83c22e99c814e86b30ef83540548003:2-----

RECEIVED FEEDBACK FROM step1 WITH STATUS ok

-----CREATING ACI for step1 on instance e83c22e99c814e86b30ef83540548003:2 -----

-----CALLBACK URL http://localhost:7071/api/stepResultCallback?instanceId=e83c22e99c814e86b30ef83540548003:2&eventName=step1&status=

-----WAITING FOR EVENT step1 on instance e83c22e99c814e86b30ef83540548003:2-----

-----STEP step1 INSTANCE IS e83c22e99c814e86b30ef83540548003:2 STATE IS ok-----

-----deleting ACI-----

-----CREATING ACI for step2 on instance e83c22e99c814e86b30ef83540548003:3 -----

-----CALLBACK URL http://localhost:7071/api/stepResultCallback?instanceId=e83c22e99c814e86b30ef83540548003:3&eventName=step2&status=

-----WAITING FOR EVENT step2 on instance e83c22e99c814e86b30ef83540548003:3-----

Parsed Raw Scratchpad Options

POST

ResultCallback?instanceId=e83c22e99c814e86b30ef83540548003:3&eventName=step2&status=ok

Host: localhost:7071

Content-Length: 0

```
RECEIVED FEEDBACK FROM step2 WITH STATUS ok
-----CREATING ACI for step2 on instance e83c22e99c814e86b30ef83540548003:3 -----
-----CALLBACK URL http://localhost:7071/api/stepResultCallback?instanceId=e83c22e99c814e86b30ef83540548003:3&eventName=step2&status=
-----WAITING FOR EVENT step2 on instance e83c22e99c814e86b30ef83540548003:3-----
-----STEP step2 INSTANCE IS e83c22e99c814e86b30ef83540548003:3 STATE IS ok-----
-----deleting ACI-----
-----ORCHESTRATION IS OVER-----
```

ParsedRawScratchpadOptions

POST⌵http://localhost:7071/api/StartOrchestration

<

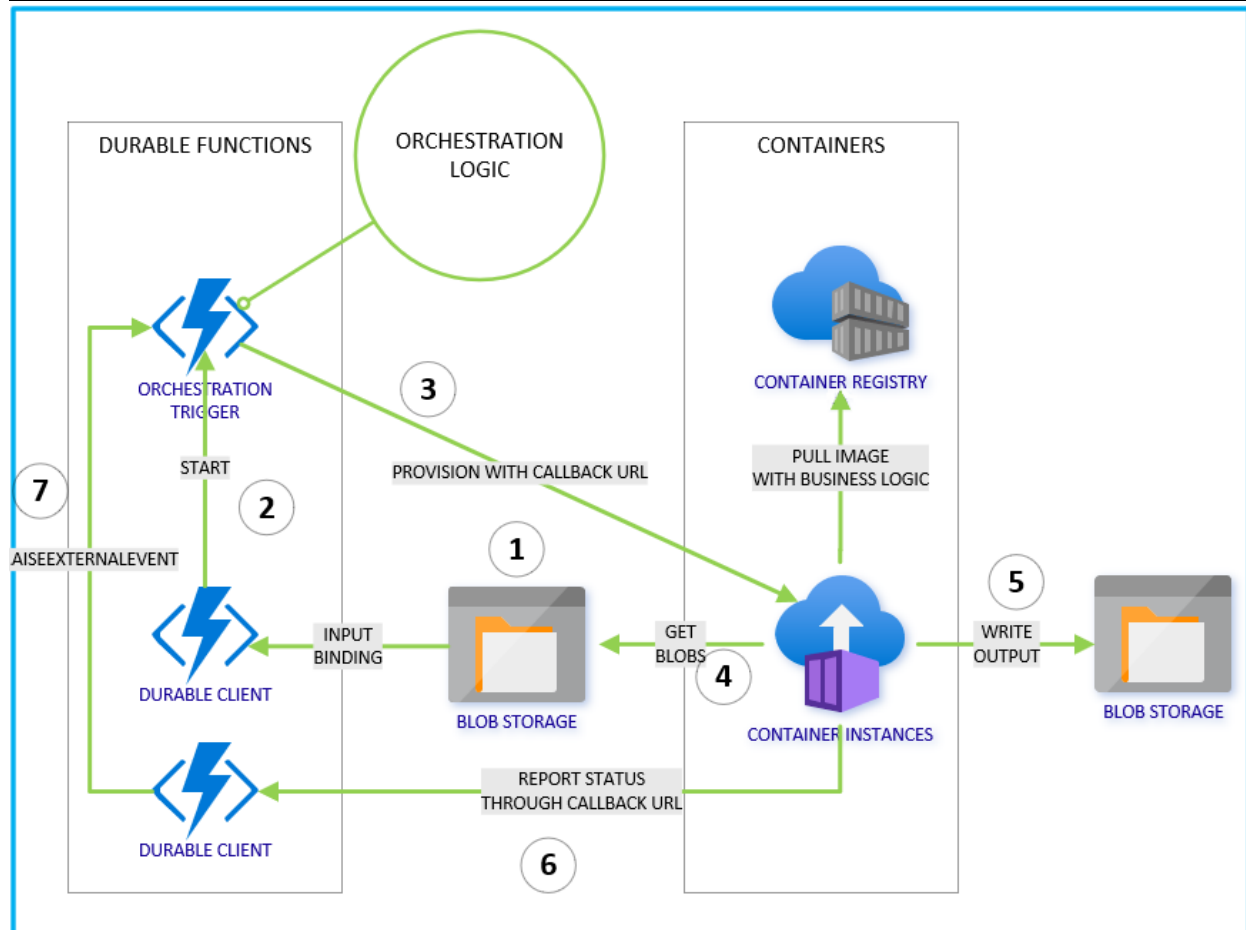
Request Body

```
{
  "isRewindable": true,
  "defaultStepTimeout": 5,
  "steps": [
    {
      "stepName": "step1",
      "maxRetryCount": 0,
      "stopFlowOnFailure": true,
      "timeOut": 1
    },
    {
      "stepName": "step2",
      "maxRetryCount": 0,
      "stopFlowOnFailure": false,
      "timeOut": 3
    }
  ]
}
```

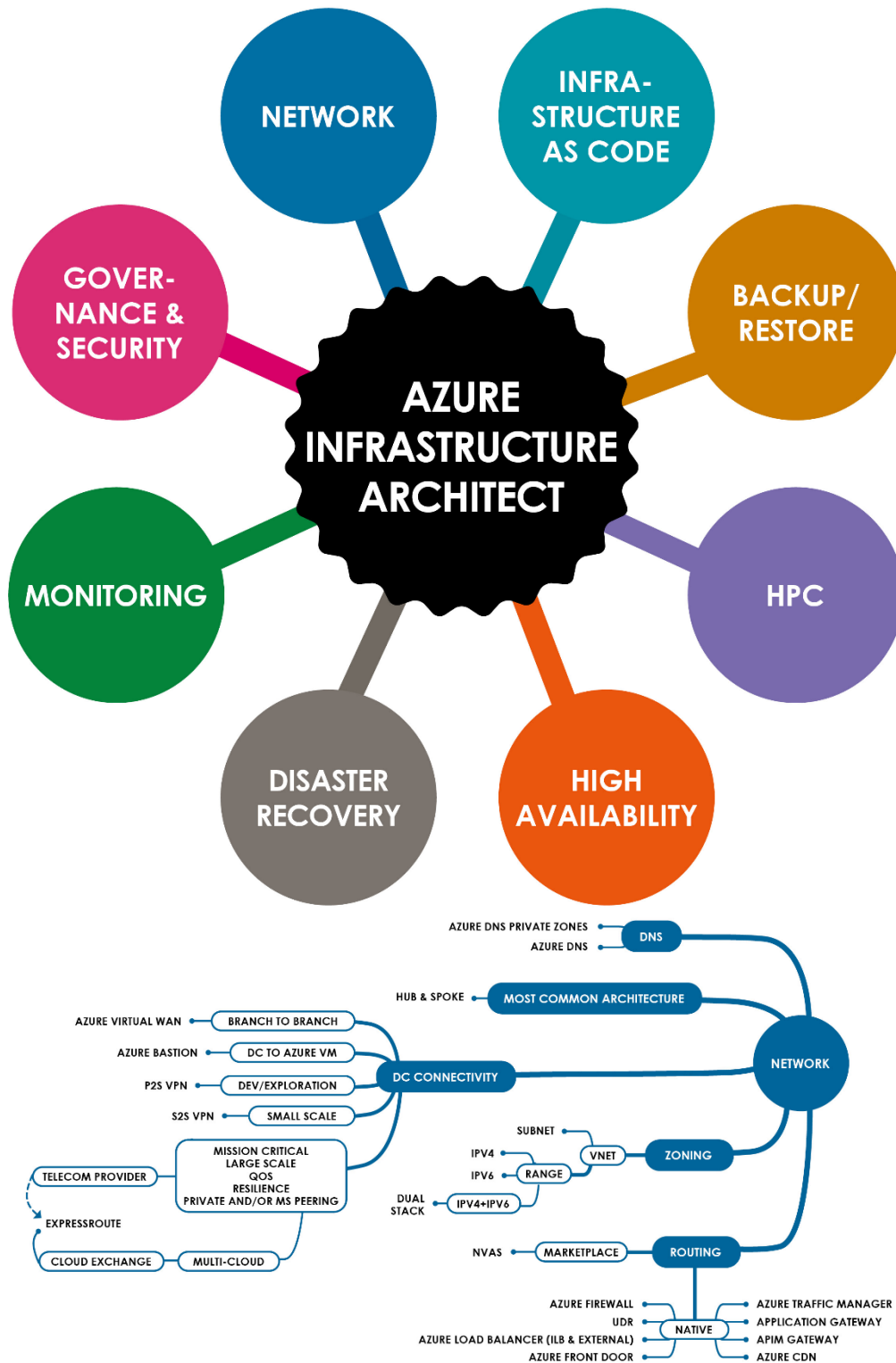
```

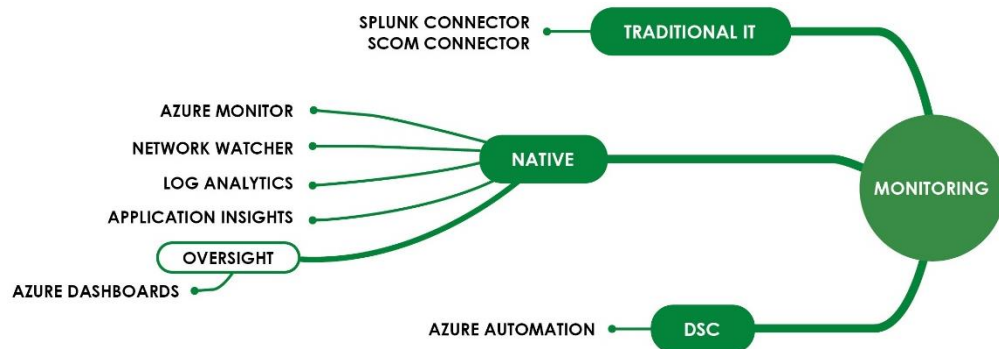
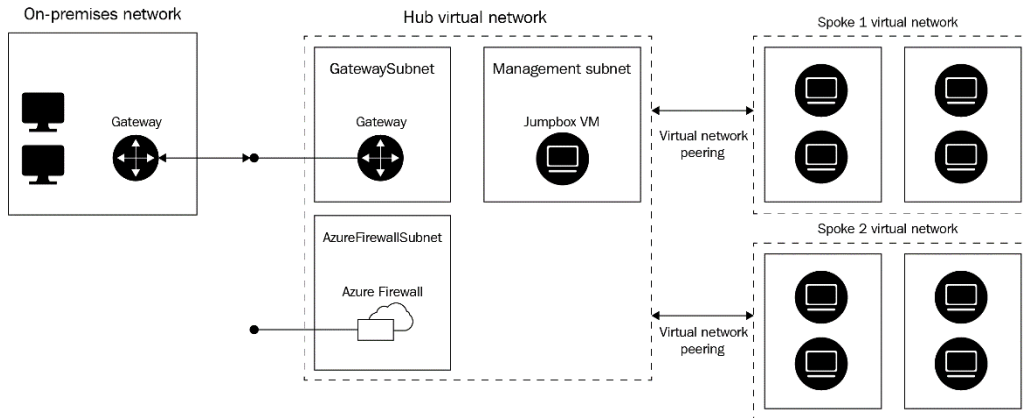
-----CREATING ACI for step1 on instance 6e2234d9a76a4c8faf07b6634fc62bd1:2 -----
-----CALLBACK URL http://localhost:7071/api/stepResultCallback?instanceId=6e2234d9a76a4c8faf07b6634fc62bd1:2&eventName=step1&status= -----
-----WAITING FOR EVENT step1 on instance 6e2234d9a76a4c8faf07b6634fc62bd1:2-----
-----TIMEOUT-----
-----deleting ACI-----
6e2234d9a76a4c8faf07b6634fc62bd1:2: Function 'SingleStepOrchestration (Orchestrator)' failed with an error. Reason: System.TimeoutException: Event step1 not received
at PacktOrchestrationDemo.Orchestration.SingleStepOrchestration(IDurableOrchestrationContext context, ILogger log) in C:\Users\steph\Source\Repos\packt\PacktOrches
tration.cs:line 81
at Microsoft.Azure.WebJobs.Host.Executors.FunctionInvoker`2.InvokeAsync(Object instance, Object[] arguments) in C:\projects\azure-webjobs-sdk-rqm4t\src\Microsoft.
Executors\FunctionInvoker.cs:line 52
at Microsoft.Azure.WebJobs.Extensions.DurableTask.TaskOrchestrationShim.InvokeUserCodeAndHandleResults(RegisteredFunctionInfo orchestratorInfo, OrchestrationConte
d:\a\r1\azure-functions-durable-extension\src\WebJobs.Extensions.DurableTask\Listener\TaskOrchestrationShim.cs:line 150. IsReplay: False. State: Failed. HubName:
ame: . SlotName: . ExtensionVersion: 2.2.2. SequenceNumber: 60.
Executed 'SingleStepOrchestration' (Failed, Id=2621191a-0a48-4527-9860-18d604a18d0d, Duration=716ms)
System.Private.CoreLib: Exception while executing function: SingleStepOrchestration. System.Private.CoreLib: Orchestrator function 'SingleStepOrchestration' failed:
eived in 00:01:00.
-----step step1 failed-----
-----step step1 failed-----
-----ORCHESTRATION IS OVER-----
-----ORCHESTRATION IS OVER-----

```



Chapter 3: Infrastructure Design





Category details

log

- ☐ Administrative
- ☐ Security
- ☐ ServiceHealth
- ☐ Alert
- ☐ Recommendation
- ☐ Policy
- ☐ Autoscale
- ☐ ResourceHealth

☐ SQLInsights

☐ AutomaticTuning

☐ QueryStoreRuntimeStatistics

☐ QueryStoreWaitStatistics

☐ Errors

☐ DatabaseWaitStatistics

☐ Timeouts

☐ Blocks

☐ Deadlocks

log

☐ FrontdoorAccessLog

☐ FrontdoorWebApplicationFirewallLog

Category details

log

☐ FrontdoorAccessLog

☐ FrontdoorWebApplicationFirewallLog

metric

☐ AllMetrics

Destination details

☐ Send to Log Analytics

☐ Archive to a storage account

☐ Stream to an event hub

▶ Run

Time range : Last 30 minutes

Save

Copy link

New alert rule

...

```
1 AzureDiagnostics
2 | where Category == "FrontdoorWebApplicationFirewallLog"
3 | project details_msg_s
```

Results

Chart

Columns

Display time (UTC+00:00)

Group columns

Completed. Showing results from the last 30 minutes. 00:00:00.383 106 rec

details_msg_s
SQL Injection Attack Detected via libinjection
SQL Injection Attack Detected via libinjection
SQL Injection Attack Detected via libinjection
SQL Injection Attack Detected via libinjection
SQL Injection Attack Detected via libinjection

Page 1 of 3 50 items per page

▶ Run

Time range : Last 7 days

Save

Copy link

New alert rule

Export

```
1 AzureDiagnostics | summarize count() by Category | render piechart
```

Results

Chart

Pie

Category

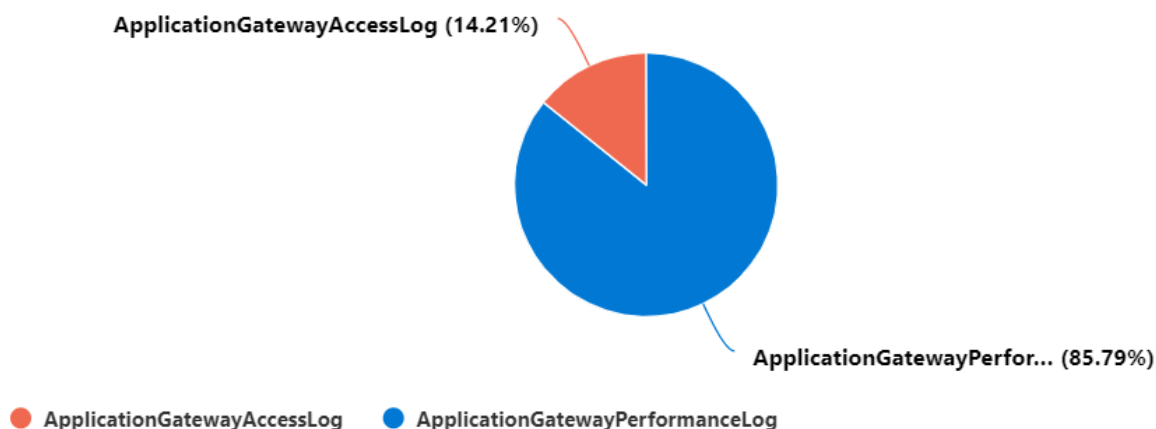
count_

Split by

Sum

Display time (UTC+0

Completed. Showing results from the last 7 days. 00:00.4 2 recor



Alert logic

 Monitoring 1 time series (\$0.10/time series)

Threshold 

Static

Dynamic

Operator 

Greater than

▼

Aggregation type * 

Average

▼

Threshold value * 

2000

✓

milliseconds

Condition preview

Whenever the average backend request latency is greater than 2000 milliseconds

Evaluated based on

Aggregation granularity (Period) * 

15 minutes

▼

Frequency of evaluation 

Every 5 Minutes

▼

Create action group

- Basics
- Notifications
- Actions
- Tags
- Review + create

Actions

Configure the method in which actions are performed when the action group tri associated details, and add a unique description. This step is optional.

Action type 

Name 

ITSM

Automation Runbook

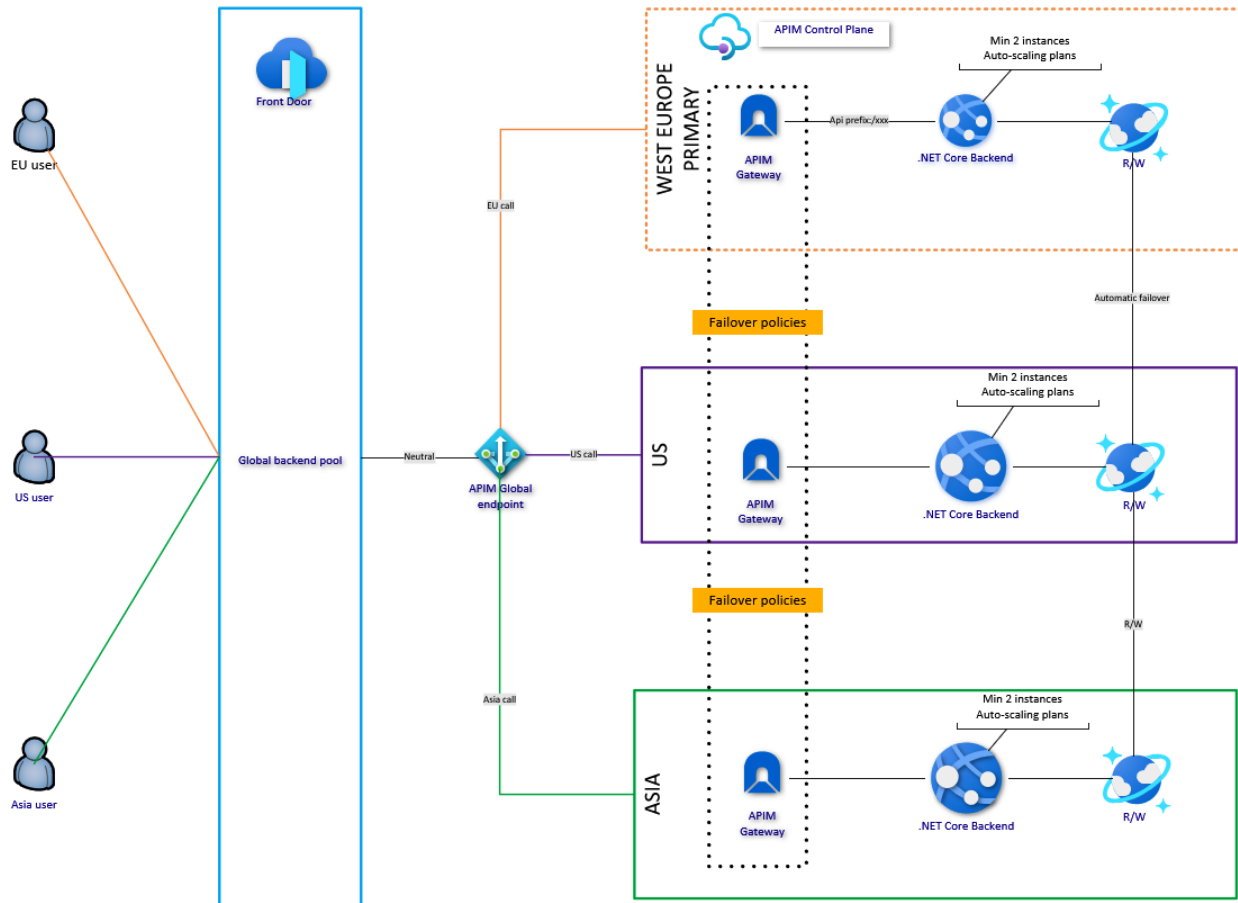
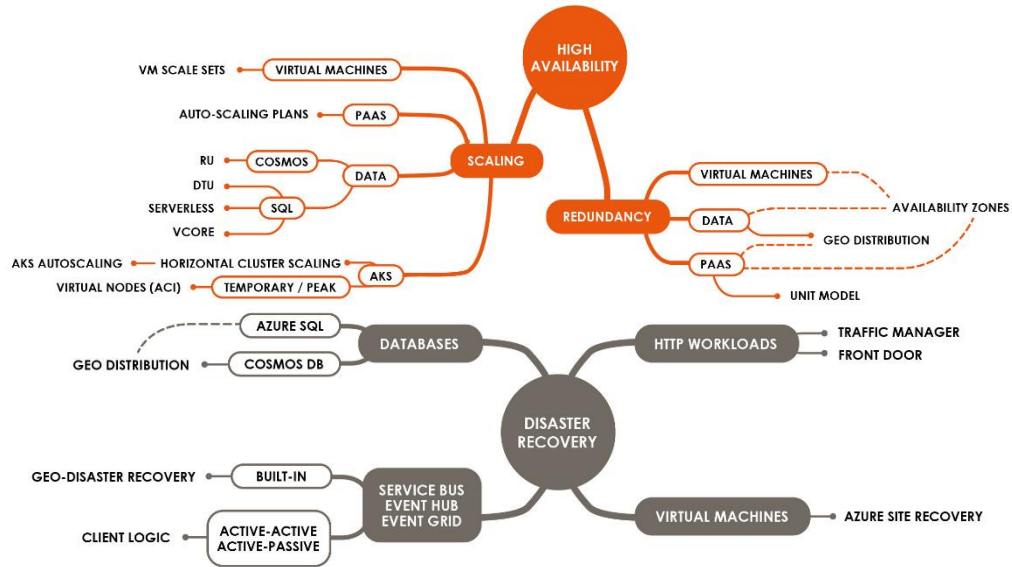
Azure Function

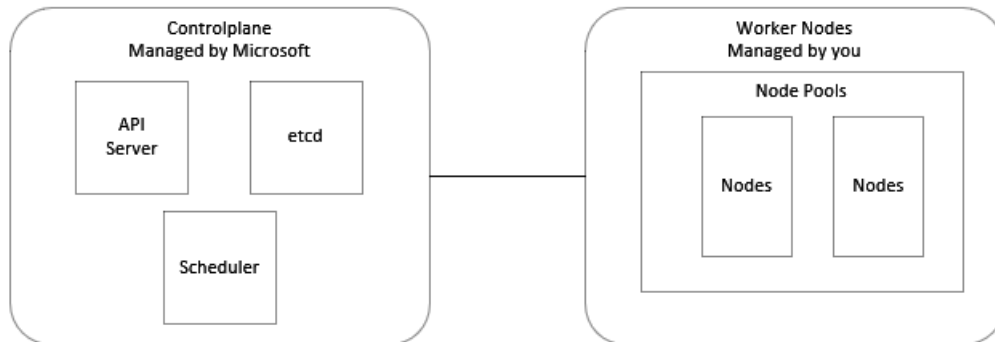
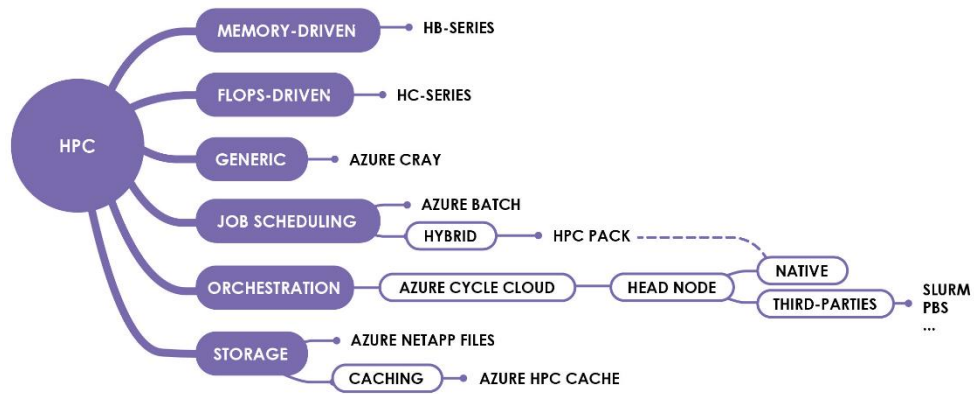
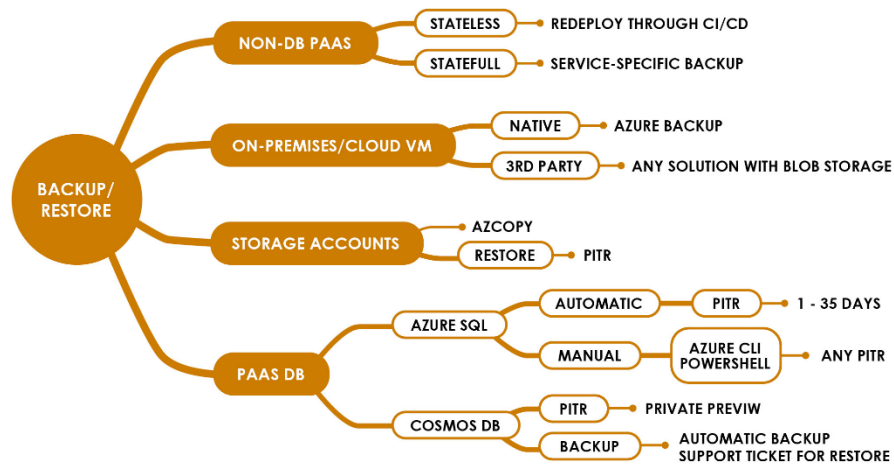
ITSM

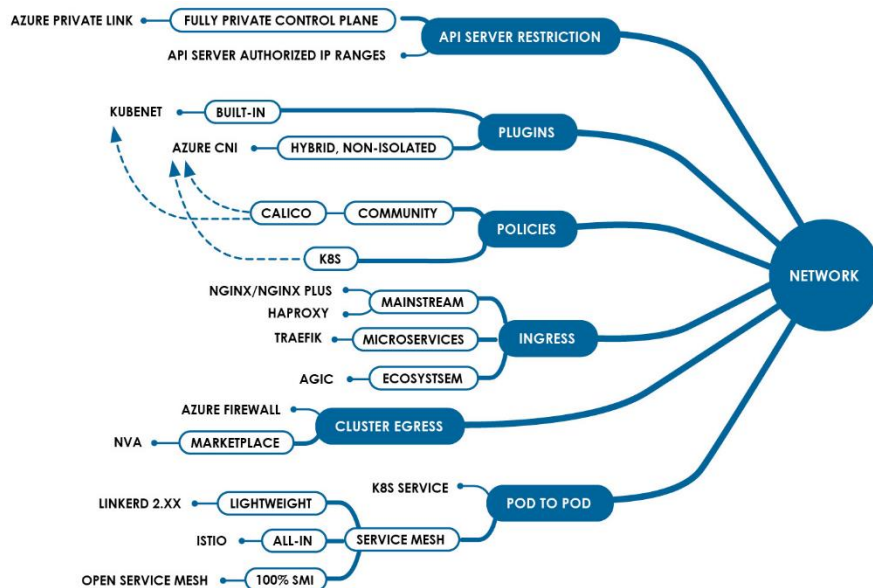
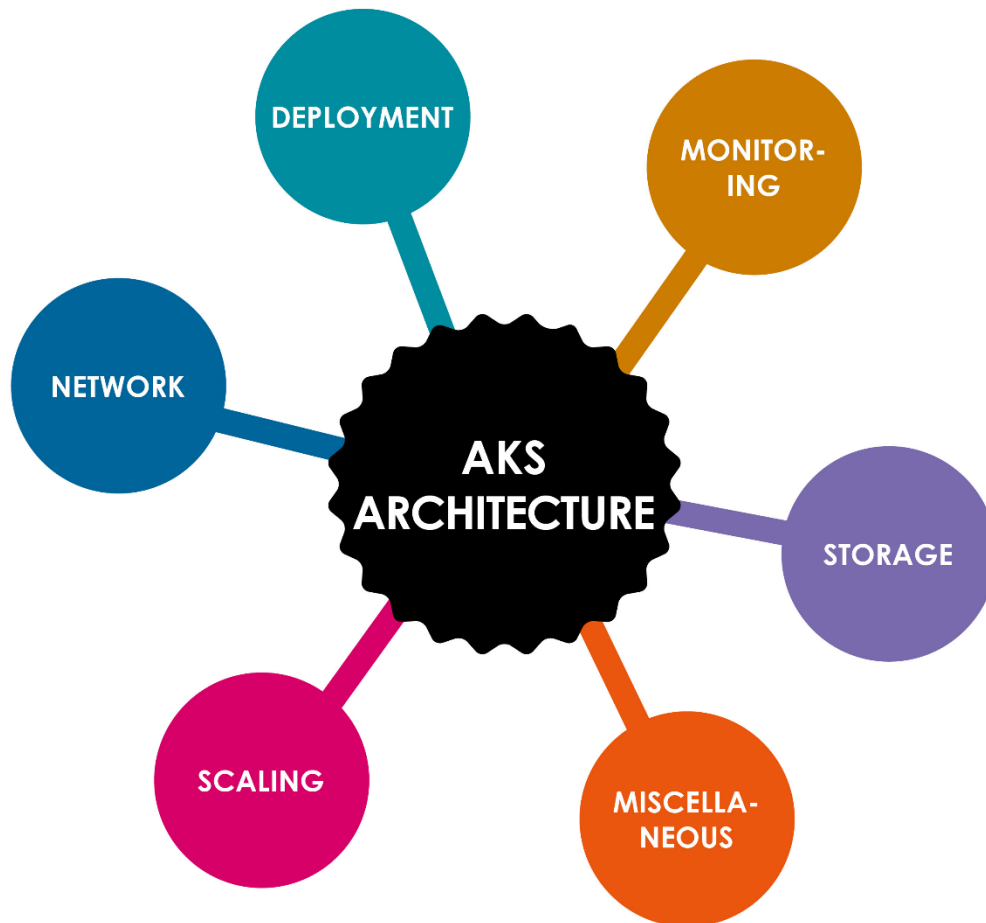
Logic App

Secure Webhook

Webhook

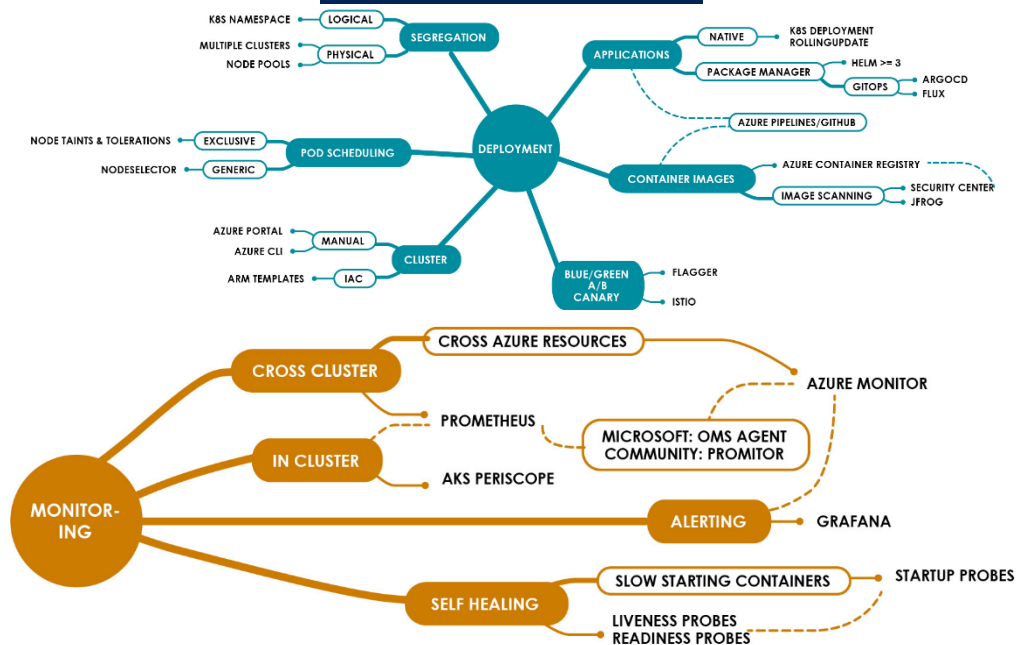


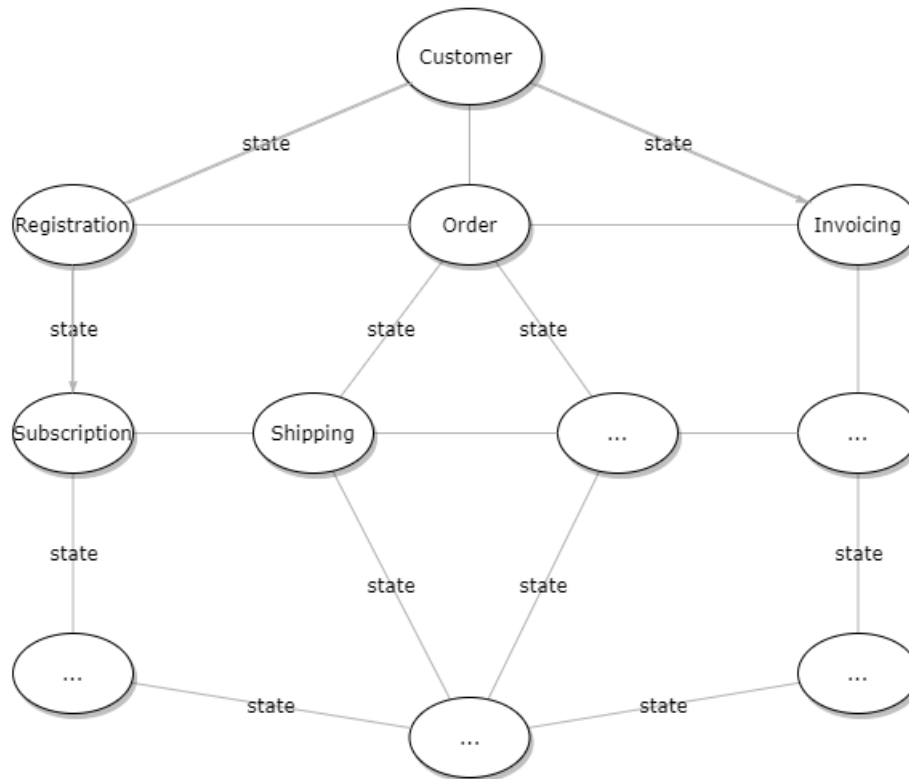
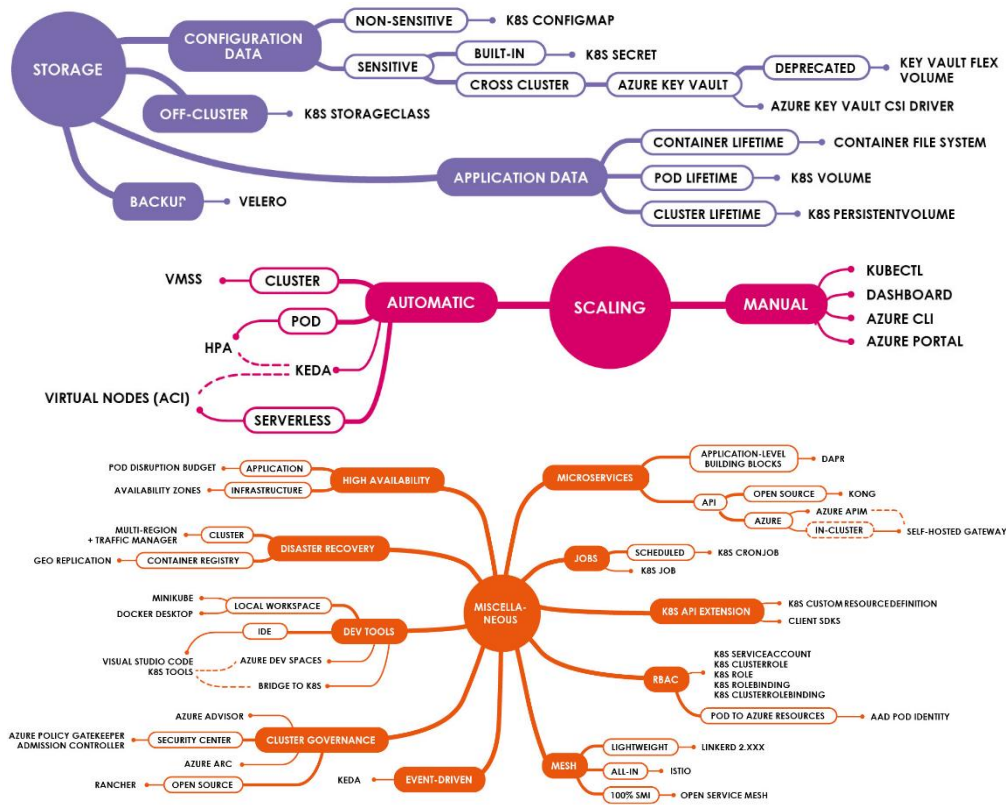


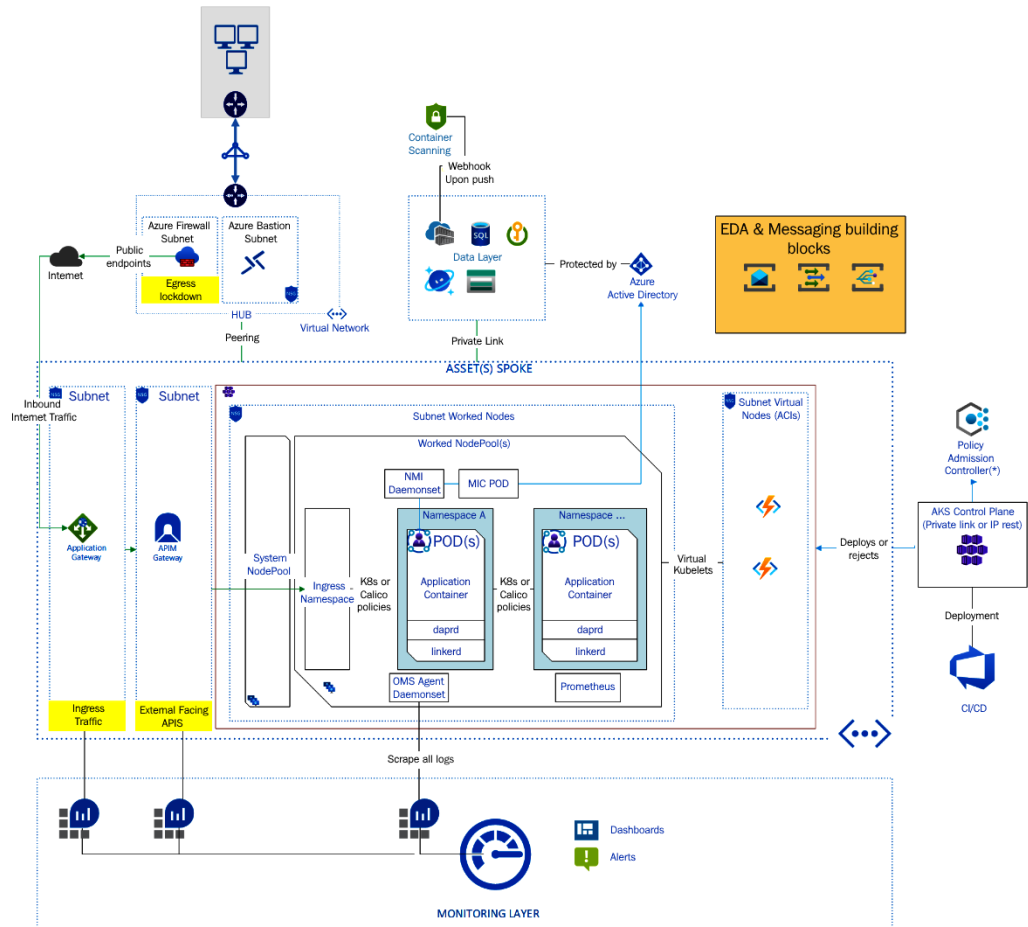


	Azure CNI	Kubenet
Subnet size	Considers both worker nodes and pods	Considers worker nodes only, meaning that a small subnet range might be enough
Autoscaling	Takes a margin when sizing the subnet, to allow both cluster growth and pod count growth	Considers the worker nodes growth only
Cluster upgrade (details follow)	Depends on the upgrade strategy, but in general, it requires more IPs	Depends on the upgrade strategy
Network policies (details follow)	K8s policies supported out of the box	Requires a plugin, such as Calico
Performance	Faster by default because no NAT (Network Address Translation) takes place	Slower by default because of NAT, but it can be mitigated with Calico
Virtual Kubelets (ACI in Azure)	Supported	Not supported (at the time of writing)

Capacity:
 cpu: 10k
 memory: 4Ti
 nvidia.com/gpu: 100
 pods: 5k







+ Initiative definition + Policy definition Refresh

Scope	Definition type	Type	Category	Search
Microsoft Azure Sp...	Initiative	All types	1 categories	Filter by name or ID..

Name	Definition location	Policies	Type	Definition type	Category
Kubernetes cluster pod security rest...		8	Built-in	Initiative	Kubernetes
Kubernetes cluster pod security bas...		5	Built-in	Initiative	Kubernetes

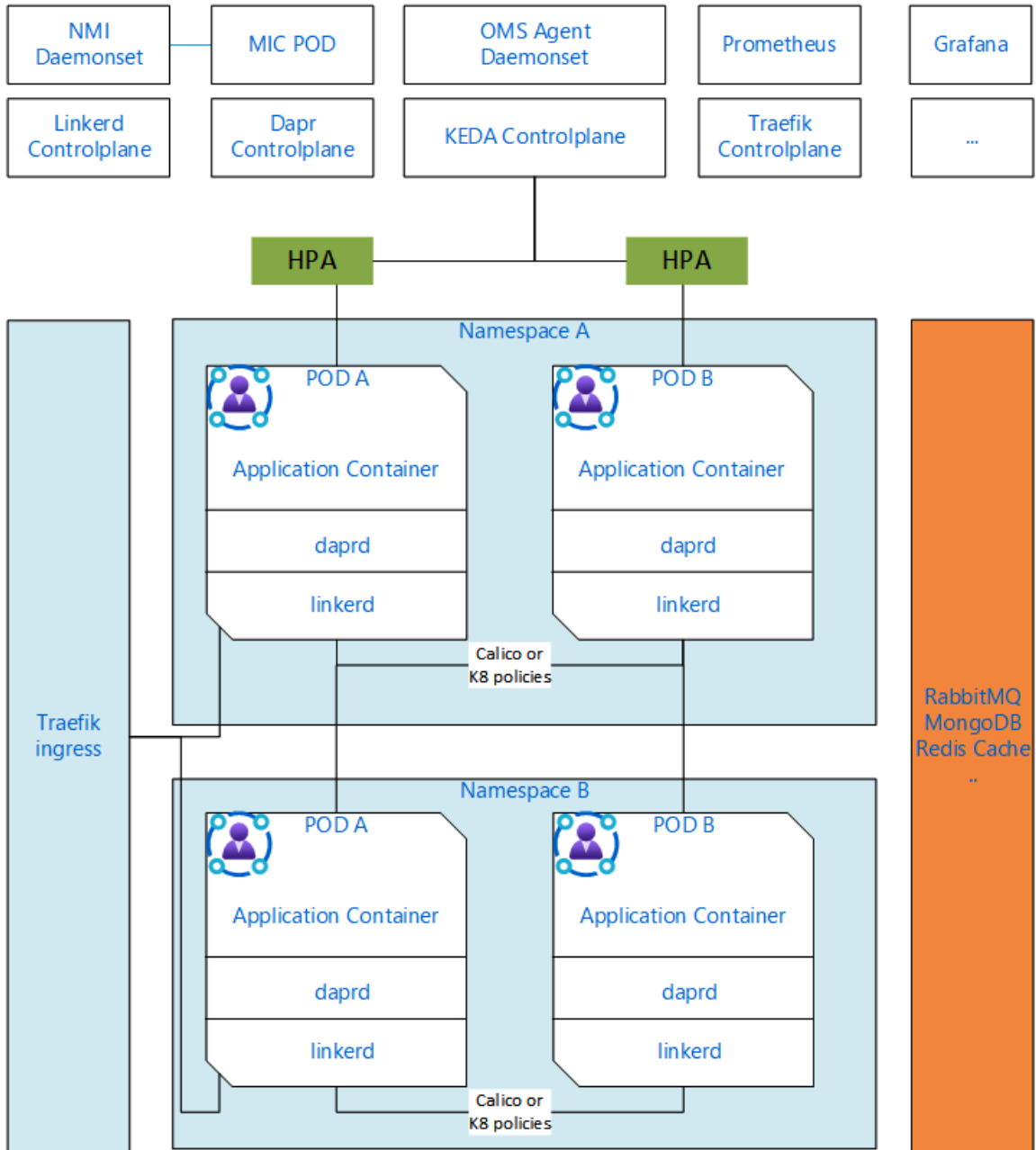
Policies Assignments (0) Parameters

Filter by reference ID, policy name or ID...

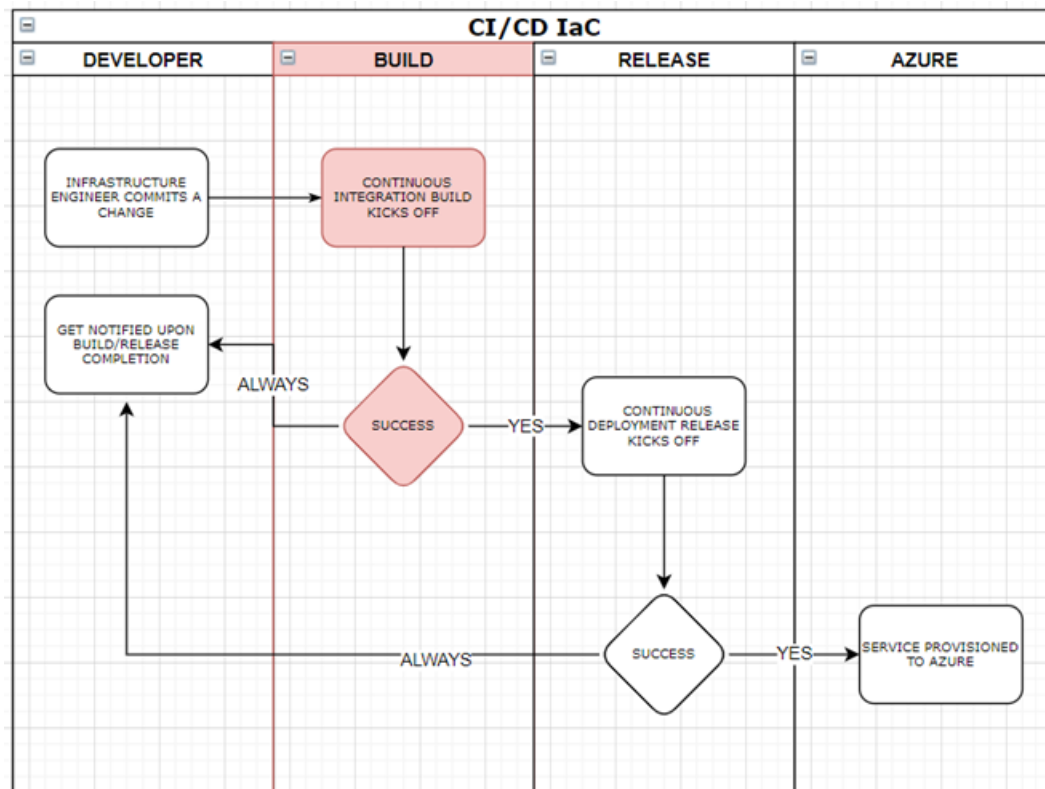
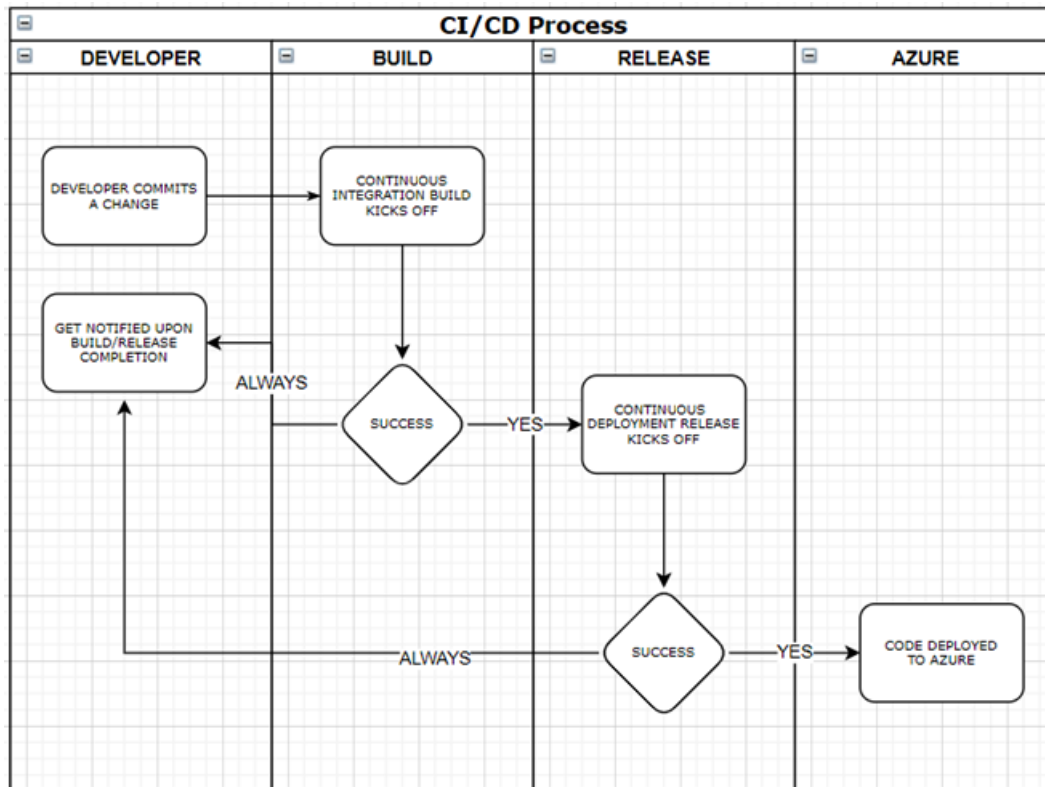
All effects

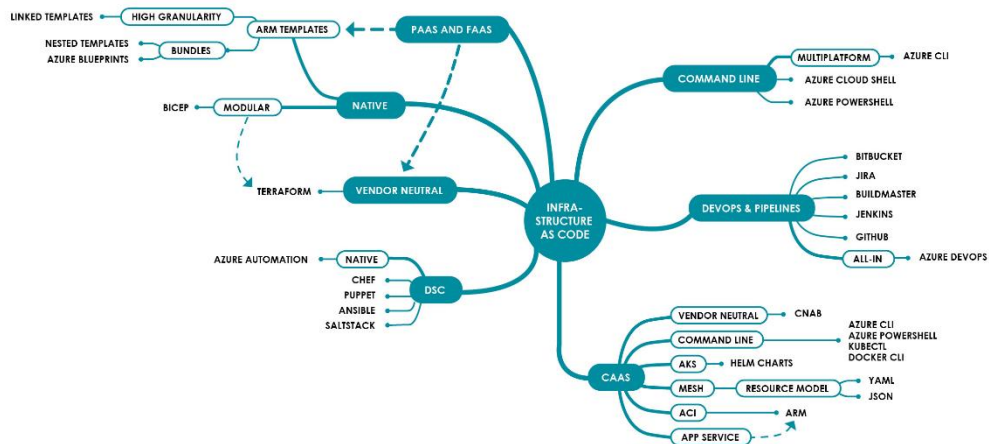
Policy

-  Do not allow privileged containers in Kubernetes cluster
-  Kubernetes cluster pods should only use approved host network and port range
-  Kubernetes cluster containers should not share host process ID or host IPC namespace
-  Kubernetes cluster containers should only use allowed capabilities
-  Kubernetes cluster pod hostPath volumes should only use allowed host paths



Chapter 4: Infrastructure Deployment





Microsoft Azure
Search resources, services, and docs (G+/)

PowerShell
?
?
?
?
?
?
?
?

```

PS /home/sponsor> az version
{
  "azure-cli": "2.14.0",
  "azure-cli-core": "2.14.0",
  "azure-cli-telemetry": "1.0.6",
  "extensions": {
    "ai-examples": "0.2.4",
    "mesh": "0.8.0rc2",
    "resource-graph": "0.1.8"
  }
}
PS /home/sponsor>

```

```
PowerShell | ? | ⚙ | 📄 | {} | 📄
PS /home/sponsor> az storage account list --query [].[name,sku.name]
[
  [
    "aciventsa",
    "Standard_RAGRS"
  ],
  [
    "365hookb27d",
    "Standard_LRS"
  ],
  [
    "aksqueues",
    "Standard_LRS"
  ],
  [
    "aksstoragesey",
    "Standard_LRS"
  ],
]

{
  "accessTier": "Hot",
  "allowBlobPublicAccess": null,
  "azureFilesIdentityBasedAuthentication": null,
  "blobRestoreStatus": null,
  "creationTime": "2020-01-17T17:44:50.300055+00:00",
  "customDomain": null,
  "enableHttpsTrafficOnly": true,
  "encryption": {
    "keySource": "Microsoft.Storage",
    "keyVaultProperties": null,
    "requireInfrastructureEncryption": null,
    "services": {
      "blob": {
        "enabled": true,
        "keyType": "Account",
        "lastEnabledTime": "2020-01-17T17:44:50.362540+00:00"
      },
      "file": {
        "enabled": true,
        "keyType": "Account",
        "lastEnabledTime": "2020-01-17T17:44:50.362540+00:00"
      },
      "queue": null,
      "table": null
    }
  },
}
```

```
PowerShell | ? | ? | ? | ? | ? | ? | ?
PS /home/sponsor> az storage account list --query [].[name,encryption.services.blob.enabled]
[
  [
    "aciventsa",
    true
  ],
  [
    "365hookb27d",
    true
  ],
  [
    "aksqueues",
    true
  ],
  [
    "aksstoragesey",
    true
  ],
]
```

PowerShell | ?

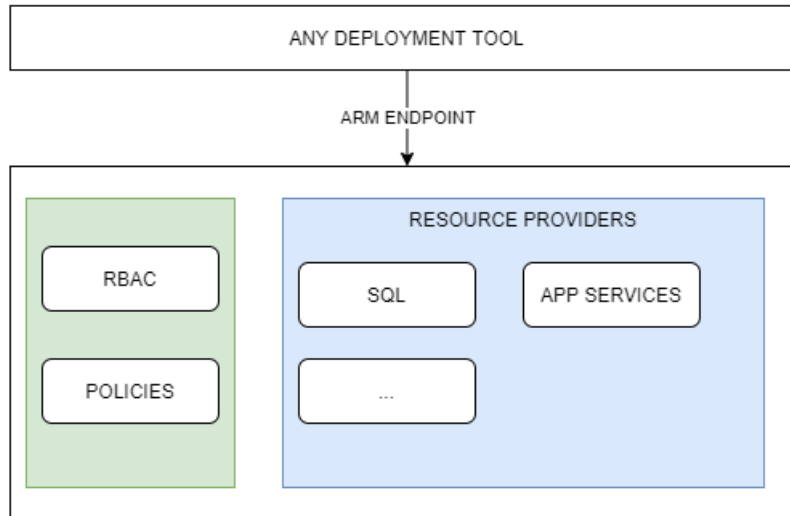
PS /home/sponsor>

```
PowerShell | ? | ⚙ | 📄 | {} | 📄  
PS /home/sponsor> Get-Host | Select-Object Version  
  
Version  
-----  
7.0.3  
  
PS /home/sponsor>
```

☰ Microsoft Azure 🔍 Search resources, services, and docs (G+ /)

```
PowerShell | ? | ⚙ | 📄 | {} | 📄  
PS /home/sponsor> Get-Module Az* | select Name  
  
Name  
----  
Az  
Az.Accounts  
Az.Advisor  
Az.Aks  
Az.AnalysisServices  
Az.ApiManagement  
Az.ApplicationInsights  
Az.Automation  
Az.Batch  
Az.Billing  
Az.Cdn  
Az.CognitiveServices  
Az.Compute
```

```
PowerShell | ? | ⚙ | 📄 | {} | 📄  
PS /home/sponsor> az storage account list | ConvertFrom-Json | where-object {$_.encryption.services.blob.enabled -eq $true} | select name  
  
name  
----  
aciventsa  
365hookb27d  
aksqueues  
aksstoragesey  
azsearchsey  
azsk20181119190851  
csbf63a908ac094x4d45xa1f  
dopoctest  
ecappsb7e
```



Progress Telerik Fiddler Web Debugger

File Edit Rules Tools View Help

WinConfig WinConfig Replay X Go Stream Decode

Keep: All sessions + Any Process Find Save Browse Clear Cache

Get Started Statistics Inspectors AutoResponder Composer Fiddler Orchest

Use Filters Note: Filters on this page are a simple subset of the filtering FiddlerScript offers (click Rules > Customize Rules).

Hosts

- No Zone Filter -
- No Host Filter -

Client Process

- ☐ Show only traffic from
- ☐ Show only Internet Explorer traffic
- ☐ Hide traffic from Service Host

Request Headers

- ☒ Show only if URL contains management.azure.com

WinConfig WinConfig Replay X Go Stream Decode

Keep: All sessions + Any Process Find Save Browse Clear

Get Started Statistics Inspectors AutoResponder Composer Fiddler

Headers Text View Syntax View WebForms Hex View Auth Cookies Raw

POST https://management.azure.com/batch?api-version=2015-11-01 HTTP/1.1

Host: management.azure.com

Connection: keep-alive

Content-Length: 4466

x-ms-client-session-id: efa1f3b1470c46a0822bf2f84f4c4db5

x-ms-command-name: f fx.Batch:0,Unknown:81

Accept-Language: en

Authorization: Bearer eyJ0eXA101JGV1Q1LC3hbGc101J3SUZlIN1IsIng1dCI6

Content-Type: application/json

Accept: */*

x-ms-client-request-id: 91a94582-a7c8-45ad-ab7e-3c4a4a6b000c

User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/53

Origin: https://portal.azure.com

Find... (press Ctrl+Enter to highlight all)

Transformer Headers Text View Syntax View Image View Hex View Web View

JSON

responses

0

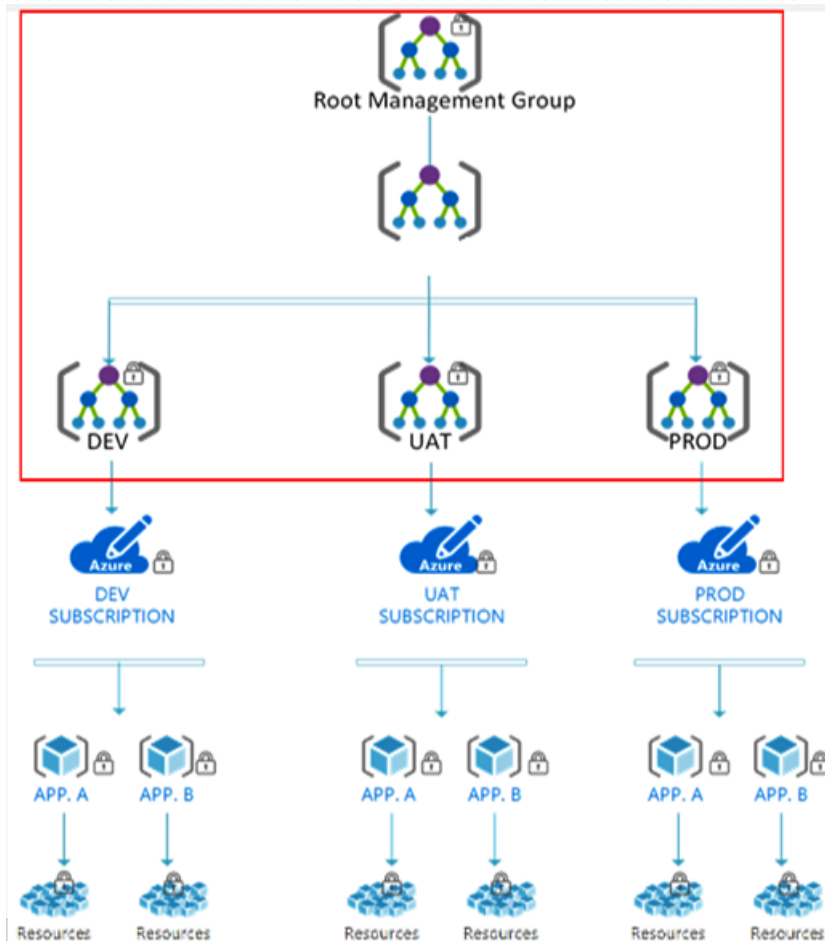
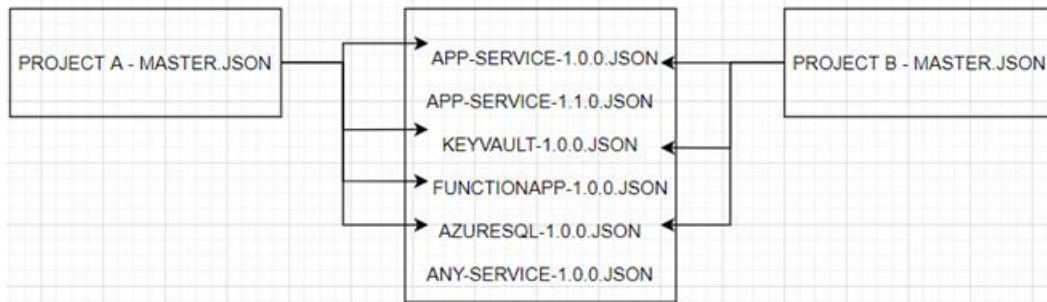
content

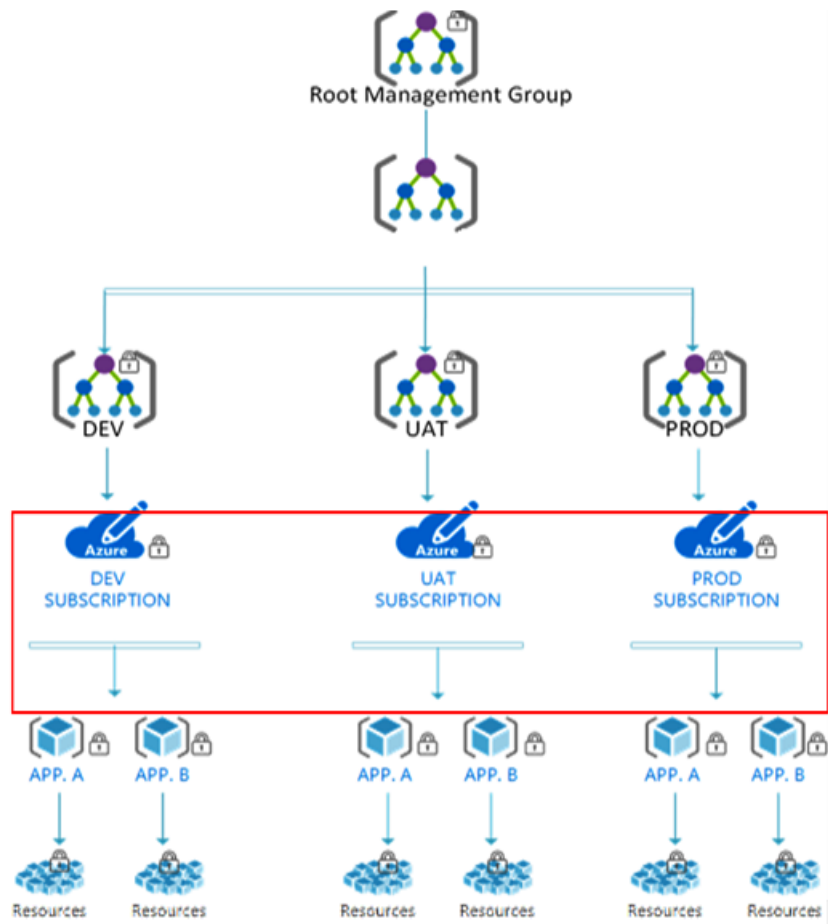
- cost=0
- interval=P1D
- namespace=microsoft.storage/storageaccounts
- resourcegroup=eastasia
- timespan=2020-11-11T20:00:00Z/2020-11-12T20:00:00Z

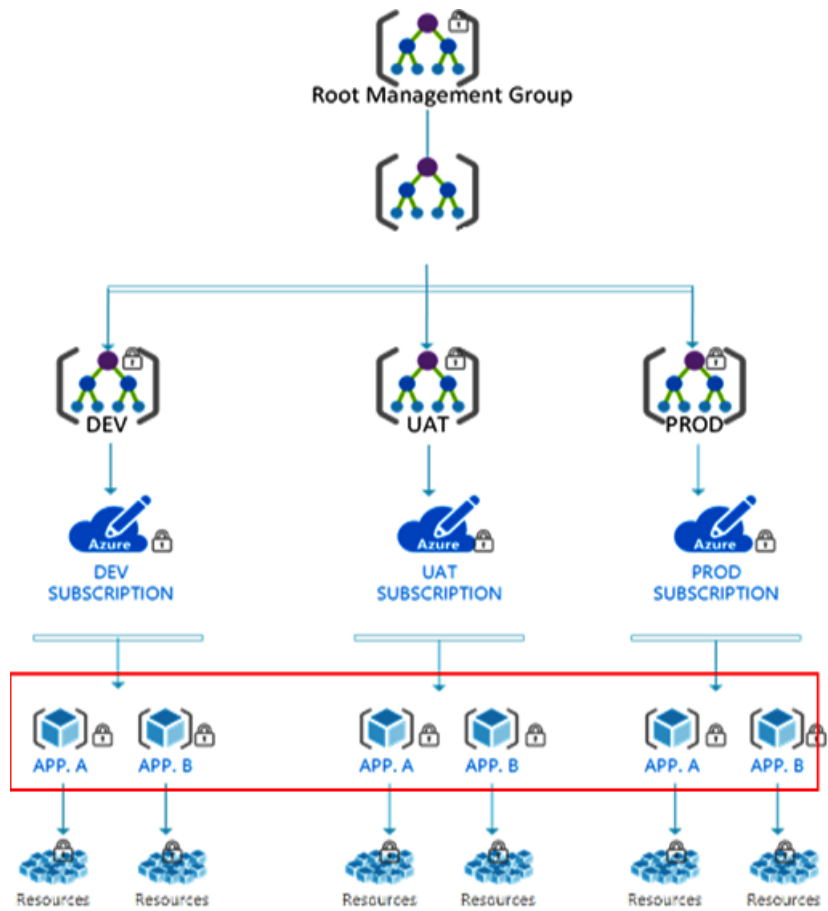
value

0

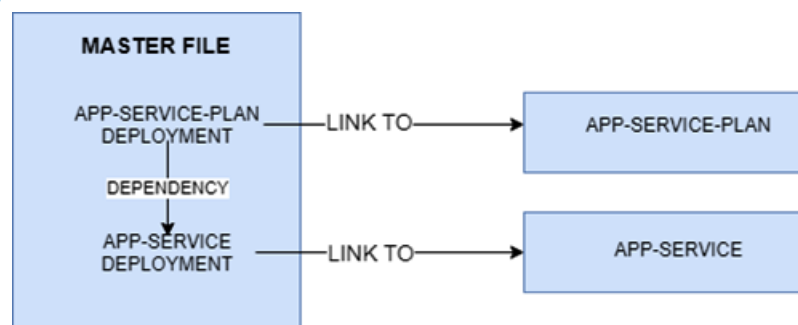
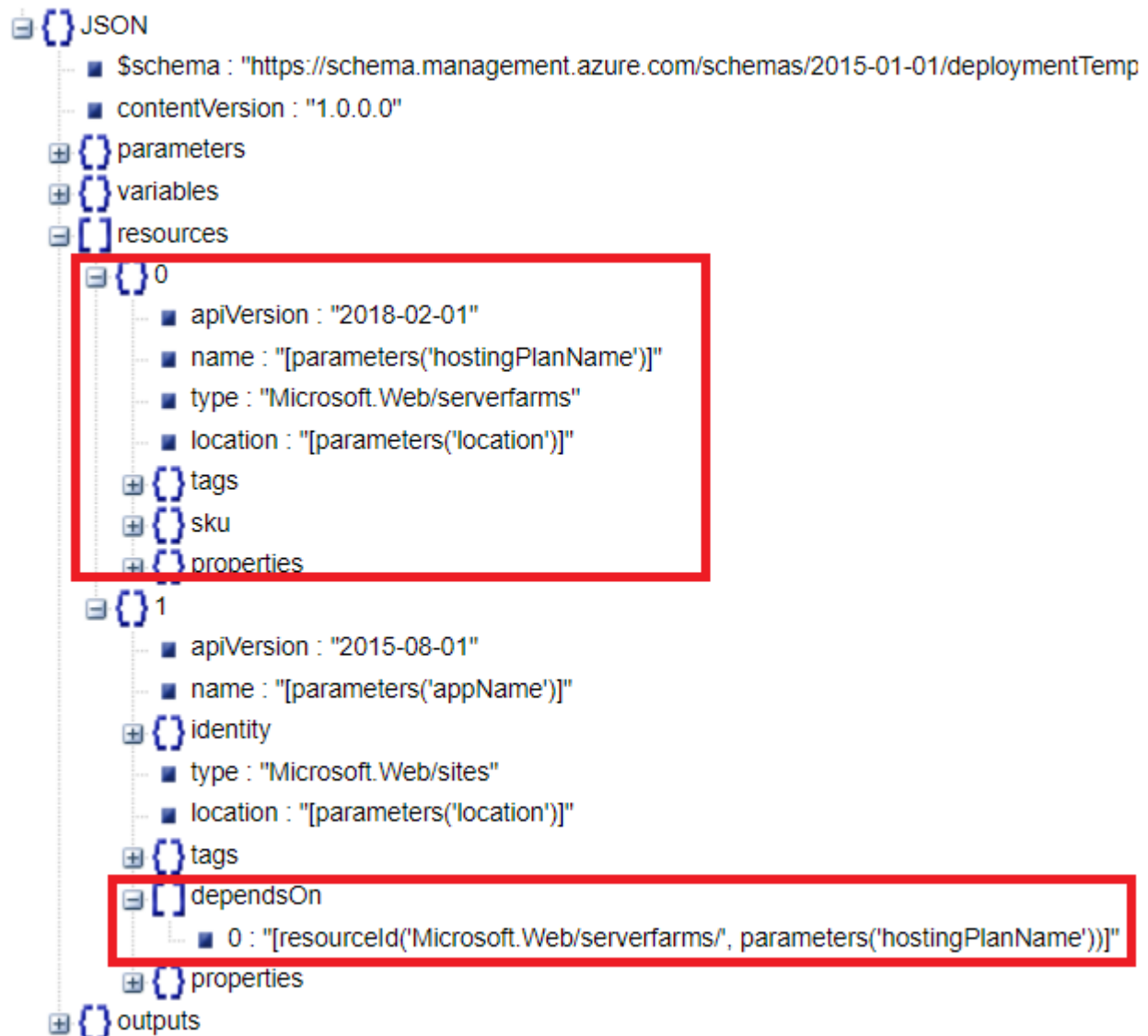
- displayDescription=The amount of egress data. This number includes
- errorCode=Success







```
JSON
├── $schema : "https://schema.management.azure.com/schemas/2015-01-01/deploymentTemp"
├── contentVersion : "1.0.0.0"
├── parameters
├── variables
├── resources
│   ├── 0
│   │   ├── apiVersion : "2018-02-01"
│   │   ├── name : "[parameters('hostingPlanName')]"
│   │   ├── type : "Microsoft.Web/serverfarms"
│   │   ├── location : "[parameters('location')]"
│   │   ├── tags
│   │   ├── sku
│   │   └── properties
│   ├── 1
│   │   ├── apiVersion : "2015-08-01"
│   │   ├── name : "[parameters('appName')]"
│   │   ├── identity
│   │   │   ├── type : "Microsoft.Web/sites"
│   │   │   └── location : "[parameters('location')]"
│   │   ├── tags
│   │   ├── dependsOn
│   │   │   └── 0 : "[resourceId('Microsoft.Web/serverfarms/', parameters('hostingPlanName'))]"
│   │   └── properties
│   └── outputs
```



[Home](#) >



Search (Ctrl+ /)

Overview

Activity log

Access control (IAM)

Tags

Events

Settings

Quickstart

Resource costs

Deployments



Search (Ctrl+ /)

Overview

Access Control (IAM)

Settings

Access policy

Properties

Metadata

+ Add Edit columns Delete resource group

Essentials

Subscription (change) : [Microsoft Azure Sponsorship](#)

Subscription ID : f63a908a-c054-4d45-a1fd-1eadae

Tags (change) : [Click here to add tags](#)

Filter by name...

Type == all

Location

Showing 1 to 1 of 1 records. ☐ Show hidden types

☐ Name ↑↓

☐ packtmapbook

Upload Change access level

Authentication method: Access key (Switch)

Location: templates

Search blobs by prefix (case-sensitive)

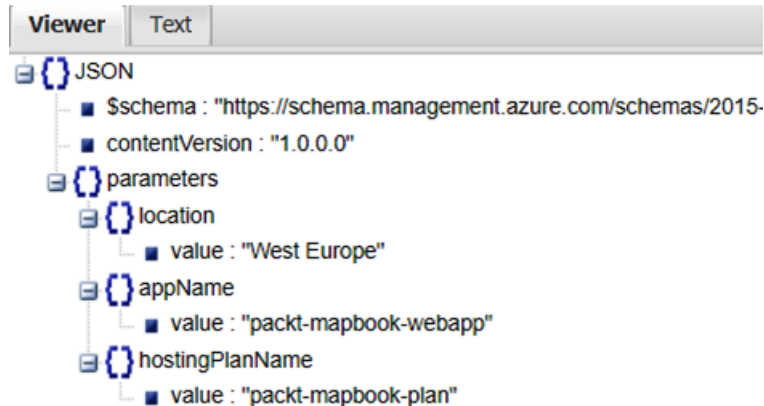
Name

☐ app-service-plan.json

☐ app-service.json



```
JSON
- $schema : "https://schema.management.azure.com/schemas/2015-01-01/deploymentTemplate.json#"
- contentVersion : "1.0.0.0"
- parameters
  - location
  - serverFarmId
  - appName
- resources
  - 0
    - apiVersion : "2015-08-01"
    - name : "[parameters('appName')]"
    - identity
      - type : "SystemAssigned"
      - type : "Microsoft.Web/sites"
      - location : "[parameters('location')]"
    - tags
    - properties
      - clientAffinityEnabled : false
      - httpsOnly : true
      - siteConfig
        - remoteDebuggingEnabled : false
        - name : "[parameters('appName')]"
        - serverFarmId : "[parameters('serverFarmId')]"
- outputs
  - principalId
    - type : "string"
    - value : "[reference(concat('Microsoft.Web/sites/', parameters('appName')), '2019-08-01', 'Full').identity.principalId]"
```



packt

Overview

Boards

Repos

Files

packt

master.json

master.parameters.json

MI README.md

packt

Overview

Boards

Repos

Pipelines

Pipelines

Environments

Connect

Select

Configure

New pipeline

Where is your code?

Azure Repos Git

Free private Git repositories, pull requests, and code search

Bitbucket Cloud

Hosted by Atlassian

GitHub

Home to the world's largest community of developers

New pipeline

Select a repository

Filter by keywords

packt



✓ Connect

✓ Select

Configure

Review

New pipeline

Configure your pipeline



Starter pipeline

Start with a minimal pipeline that you can customize to build and deploy your code.



Existing Azure Pipelines YAML file

Select an Azure Pipelines YAML file in any branch of the repository.

Show more

packt / azure-pipelines.yml *

```
1 # Starter pipeline
2 # Start with a minimal pipeline that you can customize to build and dep
3 # Add steps that build, run tests, deploy, and more:
4 # https://aka.ms/yaml
5
6 trigger:
7 - master
8
9 pool:
10 - vmImage: 'ubuntu-latest'
11
12 steps:
13 - script: echo Hello, world!
14   displayName: 'Run a one-line script'
15
16 - script: |
17   echo Add other tasks to build, test, and deploy your project.
18   echo See https://aka.ms/yaml
19   displayName: 'Run a multi-line script'
20
```

New pipeline

Review your pipeline YAML

Variables

Save and run



packt / azure-pipelines.yml *

```
1 # Starter pipeline
2 # Start with a minimal pipeline that you can customize to build and deploy your
3 # Add steps that build, run tests, deploy, and more:
4 # https://aka.ms/yaml
5
6 trigger:
7 - master
8
9 pool:
10 - vmImage: 'ubuntu-latest'
11
12 steps:
13
```

Tasks



arm



ARM template deployment

Deploy an Azure Resource Manager (ARM) templ...



AzSK ARM Template Checker

Scan ARM templates for security issues using AzSK.

← ARM template deployment ⓘ

Azure Details ^

Deployment scope * ⓘ

Resource Group



Azure Resource Manager connection * ⓘ

Microsoft Azure Sponsorship(f63a908a-...



Subscription * ⓘ

Microsoft Azure Sponsorship (f63a908a-...



Action * ⓘ

Create or update resource group



Resource group * ⓘ

packt



Location * ⓘ

West Europe



Template ^

Template location *

Linked artifact



Template * ⓘ

master.json

Template parameters ⓘ

master.parameters.json

Override template parameters ⓘ

About this task

Add

```

8
9 pool:
10   vmImage: 'ubuntu-latest'
11
12 steps:
13   - task: AzureResourceManagerTemplateDeployment@3
14     inputs:
15       deploymentScope: 'Resource Group'
16       azureResourceManagerConnection: 'Microsoft Azure Sponsorship(f63a908
17       subscriptionId: 'f63a908a-c054-4d45-a1fd-1eadaee67ffc'
18       action: 'Create Or Update Resource Group'
19       resourceGroupName: 'packt'
20       location: 'West Europe'
21       templateLocation: 'Linked artifact'
22       csmFile: 'master.json'
23       csmParametersFile: 'master.parameters.json'
24       deploymentMode: 'Incremental'
25

```

← Jobs in run #20201...
packt

Jobs

Job	1m 39s
Initialize job	2s
Checkout packt...	2s
AzureResou...	1m 33s
Post-job: Check...	<1s
Finalize Job	<1s
Report build st...	<1s

✓ AzureResourceManagerTemplateDeployment

View raw log

```

1 Starting: AzureResourceManagerTemplateDeployment
2 =====
3 Task : ARM template deployment
4 Description : Deploy an Azure Resource Manager (ARM) template to all the deployment scopes
5 Version : 3.176.0
6 Author : Microsoft Corporation
7 Help : https://docs.microsoft.com/azure/devops/pipelines/tasks/deploy/azure-resource-group-deplo
8 =====
9 ARM Service Connection deployment scope - Subscription
10 Checking if the following resource group exists: packt.
11 Resource group exists: true.
12 Creating deployment parameters.
13 The detected encoding for file '/home/vsts/work/1/s/master.json' is 'utf-8'
14 The detected encoding for file '/home/vsts/work/1/s/master.parameters.json' is 'utf-8'
15 Starting template validation.
16 Deployment name is master-20201113-090047-d181
17 Template deployment validation was completed successfully.
18 Starting Deployment.
19 Deployment name is master-20201113-090047-d181
20 Successfully deployed the template.
21 Finishing: AzureResourceManagerTemplateDeployment

```

 Search (Ctrl+/)

<<

 Add  Edit columns  Delete resource group

 Overview

 Activity log

 Access control (IAM)

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 Policies

 Properties

^ Essentials

Subscription ([change](#)) : [Microsoft Azure Sponsorship](#)

Subscription ID : f63a908a-c054-4d45-a1fd-1eadaee67

Tags ([change](#)) : [Click here to add tags](#)

Filter by name...

Type == **all** 

Location

Showing 1 to 3 of 3 records. ☐ Show hidden types 

☐ Name 

☐  [packt-mapbook-plan](#)

☐  [packt-mapbook-webapp](#)

☐  [packtmapbook](#)

```

packt.bicep > skuName
1  param hostingPlanName string
2  param appName string
3  param location string = resourceGroup().location
4  > param skuName string { ...
13 }
14 > param skuCapacity int { ...
18 }
19
20 resource plan 'Microsoft.Web/serverfarms@2020-06-01' = {
21   name: hostingPlanName
22   location: location
23   sku: {
24     name: skuName
25     capacity: skuCapacity
26   }
27 }
28
29 resource webapp 'Microsoft.Web/sites@2020-06-01' = {
30   name: appName
31   location: location
32   properties: {
33     serverFarmId: plan.id
34     siteConfig: {
35       remoteDebuggingEnabled: false
36     }
37   }
38 }

```

```

35     remoteDebuggingEnabled: false
36   }

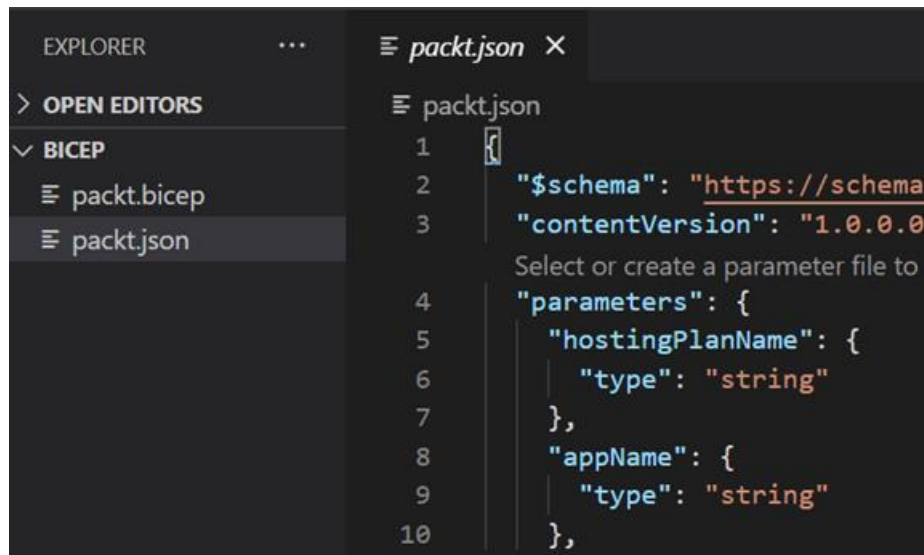
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```

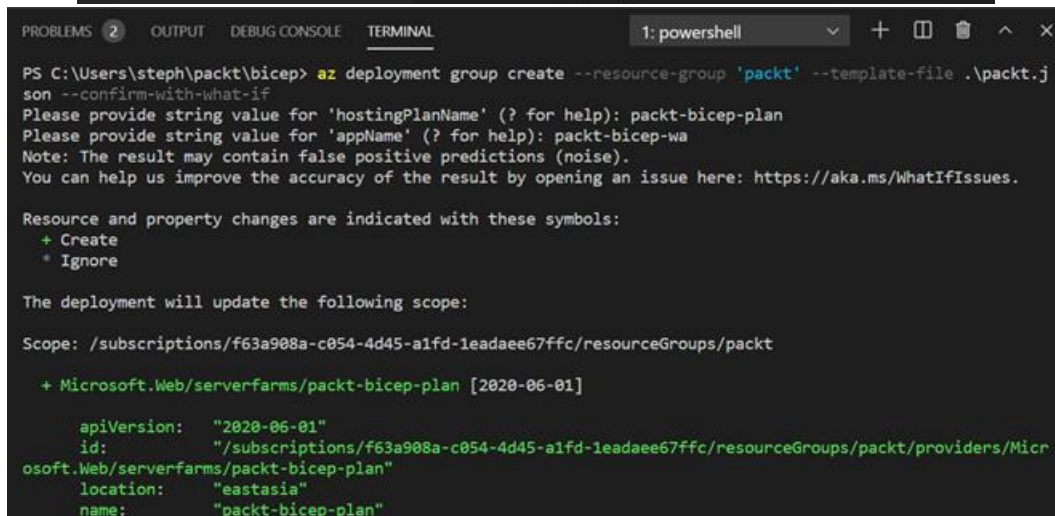
PS C:\Users\steph\packt\bicep> bicep build .\packt.bicep
PS C:\Users\steph\packt\bicep>

```



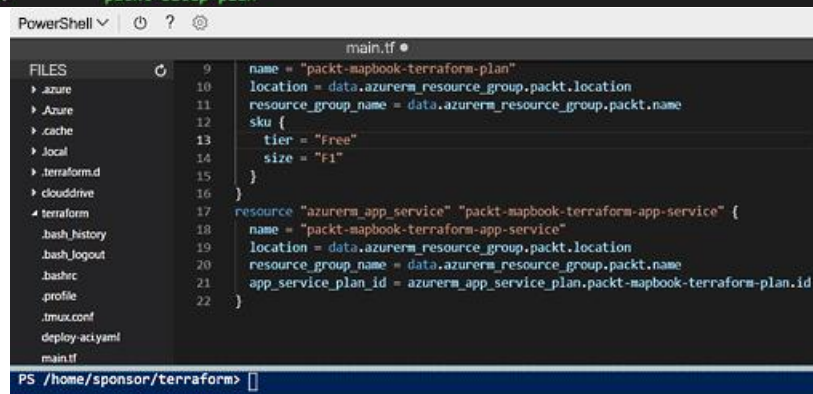
The screenshot shows the VS Code interface. The Explorer sidebar on the left has 'packt.json' selected under the 'BICEP' folder. The Editor pane shows the content of 'packt.json' with line numbers 1 through 10. The JSON content is as follows:

```
1 {  
2   "$schema": "https://schema  
3   "contentVersion": "1.0.0.0  
4   "parameters": {  
5     "hostingPlanName": {  
6       "type": "string"  
7     },  
8     "appName": {  
9       "type": "string"  
10    },
```



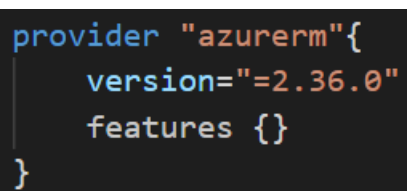
The screenshot shows a PowerShell terminal window with the following content:

```
PS C:\Users\steph\packt\bicep> az deployment group create --resource-group 'packt' --template-file .\packt.j  
son --confirm-with-what-if  
Please provide string value for 'hostingPlanName' (? for help): packt-bicep-plan  
Please provide string value for 'appName' (? for help): packt-bicep-wa  
Note: The result may contain false positive predictions (noise).  
You can help us improve the accuracy of the result by opening an issue here: https://aka.ms/WhatIfIssues.  
  
Resource and property changes are indicated with these symbols:  
+ Create  
* Ignore  
  
The deployment will update the following scope:  
  
Scope: /subscriptions/f63a908a-c054-4d45-a1fd-1eadaee67ffc/resourceGroups/packt  
  
+ Microsoft.Web/serverfarms/packt-bicep-plan [2020-06-01]  
  
  apiVersion: "2020-06-01"  
  id: "/subscriptions/f63a908a-c054-4d45-a1fd-1eadaee67ffc/resourceGroups/packt/providers/Micr  
osoft.Web/serverfarms/packt-bicep-plan"  
  location: "eastasia"  
  name: "packt-bicep-plan"
```



The screenshot shows the VS Code interface. The Explorer sidebar on the left has 'main.tf' selected. The Editor pane shows the content of 'main.tf' with line numbers 9 through 22. The Terraform HCL content is as follows:

```
9   name = "packt-mapbook-terraform-plan"  
10  location = data.azurem_resource_group.packt.location  
11  resource_group_name = data.azurem_resource_group.packt.name  
12  sku {  
13    tier = "Free"  
14    size = "F1"  
15  }  
16  }  
17  resource "azurerm_app_service" "packt-mapbook-terraform-app-service" {  
18    name = "packt-mapbook-terraform-app-service"  
19    location = data.azurem_resource_group.packt.location  
20    resource_group_name = data.azurem_resource_group.packt.name  
21    app_service_plan_id = azurerm_app_service_plan.packt-mapbook-terraform-plan.id  
22  }
```



```
provider "azurerm" {  
  version="=2.36.0"  
  features {}  
}
```

```
data "azurerm_resource_group" "packt"{
  name="packt"
}
```

```
resource "azurerm_app_service_plan" "packt-mapbook-terraform-plan" {
  name = "packt-mapbook-terraform-plan"
  location = azurerm_resource_group.packet.location
  resource_group_name = azurerm_resource_group.packet.name
  sku {
    tier = "Free"
    size = "F1"
  }
}

resource "azurerm_app_service" "packt-mapbook-terraform-app-service" {
  name = "packt-mapbook-terraform-app-service"
  location = azurerm_resource_group.appservice-rg.location
  resource_group_name = azurerm_resource_group.appservice-rg.name
  app_service_plan_id = azurerm_app_service_plan.packet-mapbook-terraform-plan.id
  tags = {
    environment = var.environment
  }
}
```

➤ clouddrive	8	resource "az
⊙ terraform	9	name = "pa
.bash_history	10	location =
.bash_logout	11	resource_gr

PS /home/sponsor/terraform> terraform init

Initializing the backend...

Initializing provider plugins...

- Finding hashicorp/azurerm versions matching "2.36.0"...
- Installing hashicorp/azurerm v2.36.0...
- Installed hashicorp/azurerm v2.36.0 (signed by HashiCorp)

Terraform has been successfully initialized!

```
PS /home/sponsor/terraform> terraform plan
Refreshing Terraform state in-memory prior to plan...
The refreshed state will be used to calculate this plan, but will not be
persisted to local or remote state storage.
```

```
data.azurerm_resource_group.packt: Refreshing state...
```

```
-----

An execution plan has been generated and is shown below.
Resource actions are indicated with the following symbols:
```

```
+ create
```

```
Terraform will perform the following actions:
```

```
# azurerm_app_service.packt-mapbook-terraform-app-service will be created
+ resource "azurerm_app_service" "packt-mapbook-terraform-app-service" {
  + app_service_plan_id      = (known after apply)
  + app_settings             = (known after apply)
  + client_affinity_enabled  = false
  + client_cert_enabled      = false
  + default_site_hostname    = (known after apply)
  + enabled                  = true
  + https_only               = false
  + id                      = (known after apply)
  + location                 = "eastasia"
  + name                     = "packt-mapbook-terraform-app-service"
```

```
PS /home/sponsor/terraform> terraform apply -auto-approve
data.azurerm_resource_group.packt: Refreshing state...
azurerm_app_service_plan.packt-mapbook-terraform-plan: Creating...
azurerm_app_service_plan.packt-mapbook-terraform-plan: Still creating... [10s elapsed]
azurerm_app_service_plan.packt-mapbook-terraform-plan: Creation complete after 13s [id=/subscriptions/packt/providers/Microsoft.Web/serverfarms/packt-mapbook-terraform-plan]
azurerm_app_service.packt-mapbook-terraform-app-service: Creating...
azurerm_app_service.packt-mapbook-terraform-app-service: Still creating... [10s elapsed]
azurerm_app_service.packt-mapbook-terraform-app-service: Still creating... [20s elapsed]
azurerm_app_service.packt-mapbook-terraform-app-service: Still creating... [30s elapsed]
azurerm_app_service.packt-mapbook-terraform-app-service: Still creating... [40s elapsed]
azurerm_app_service.packt-mapbook-terraform-app-service: Still creating... [50s elapsed]
azurerm_app_service.packt-mapbook-terraform-app-service: Creation complete after 58s [id=/subscriptions/packt/providers/Microsoft.Web/sites/packt-mapbook-terraform-app-service]

Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
PS /home/sponsor/terraform>
```

```
Apply complete! Resources: 2 added, 0 changed, 0 destroyed.
```

```
PS /home/sponsor/terraform> ls
```

```
main.tf  terraform.tfstate
```

```
PS /home/sponsor/terraform>
```

Bindings

Private Key Certificates (.pfx)

Public Key Certificates (.cer)



Protocol Settings

Protocol settings are global and apply to all bindings defined by your app.

HTTPS Only: ⓘ

Off On

Minimum TLS Version ⓘ

1.0 1.1 1.2



TLS/SSL bindings

Bindings let you specify which certificate to use when responding to requests to issued for the specific hostname. [Learn more](#)

```
client_cert_enabled      = false
default_site_hostname    = "packt-mapbook-
enabled                  = true
~ https_only             = true -> false
id                       = "/subscriptions
ook-terraform-app-service"
location                 = "easia"
name                     = "packt-mapbook-
outbound_ip_addresses    = "13.75.34.175,1
possible_outbound_ip_addresses = "13.75.34.175,1"
```

 v1.1

 / Type to find a file or folder...

 Filter branches

Branches

Tags

 master

Default

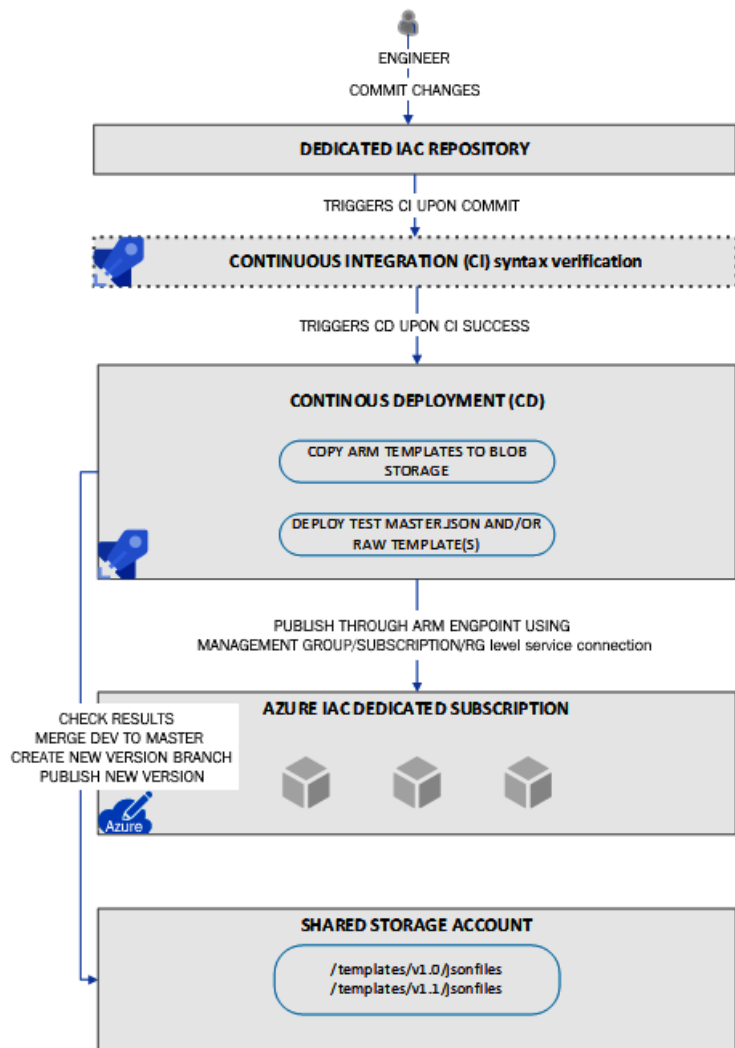
Mine

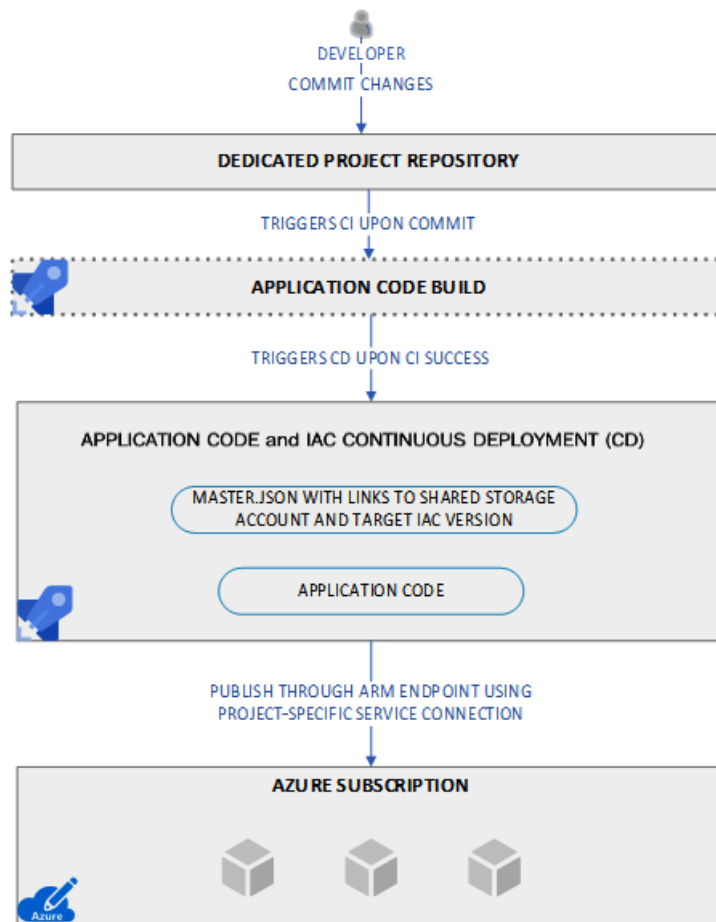
 dev

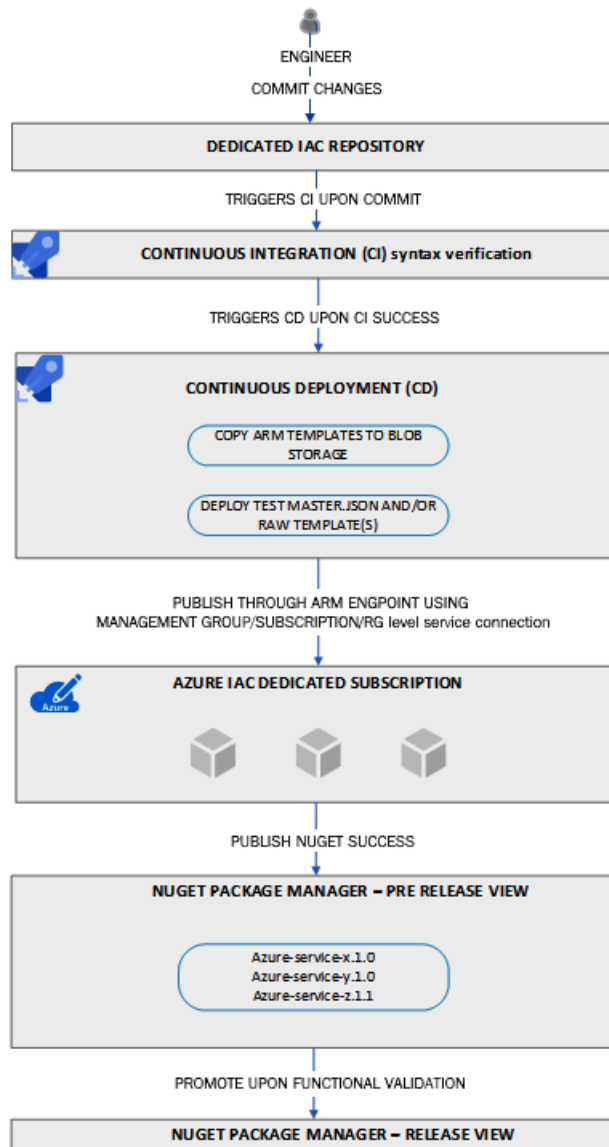
 v1.0

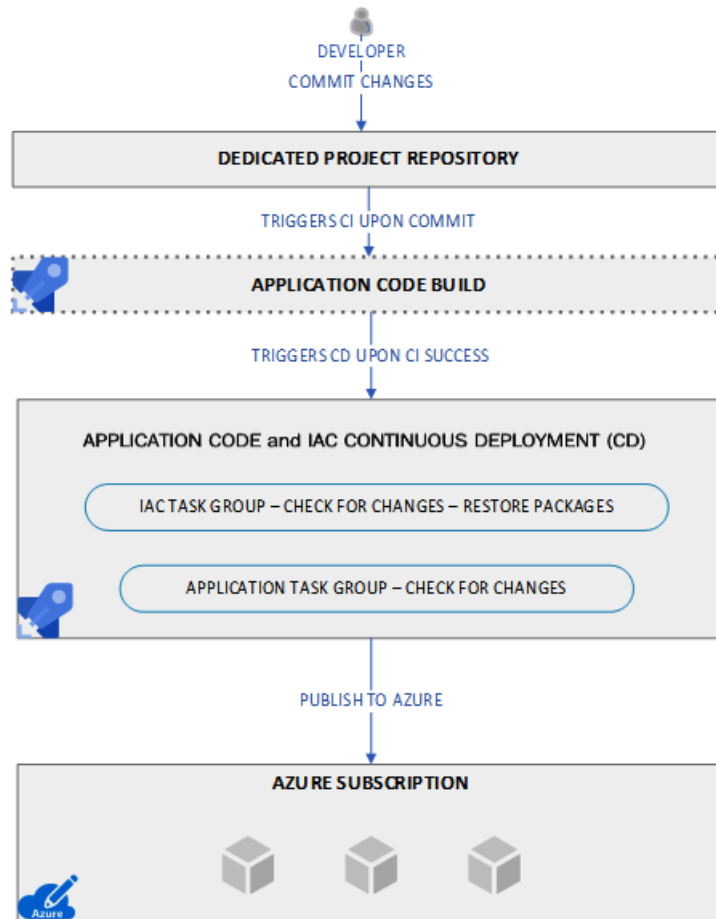
  v1.1

 New branch

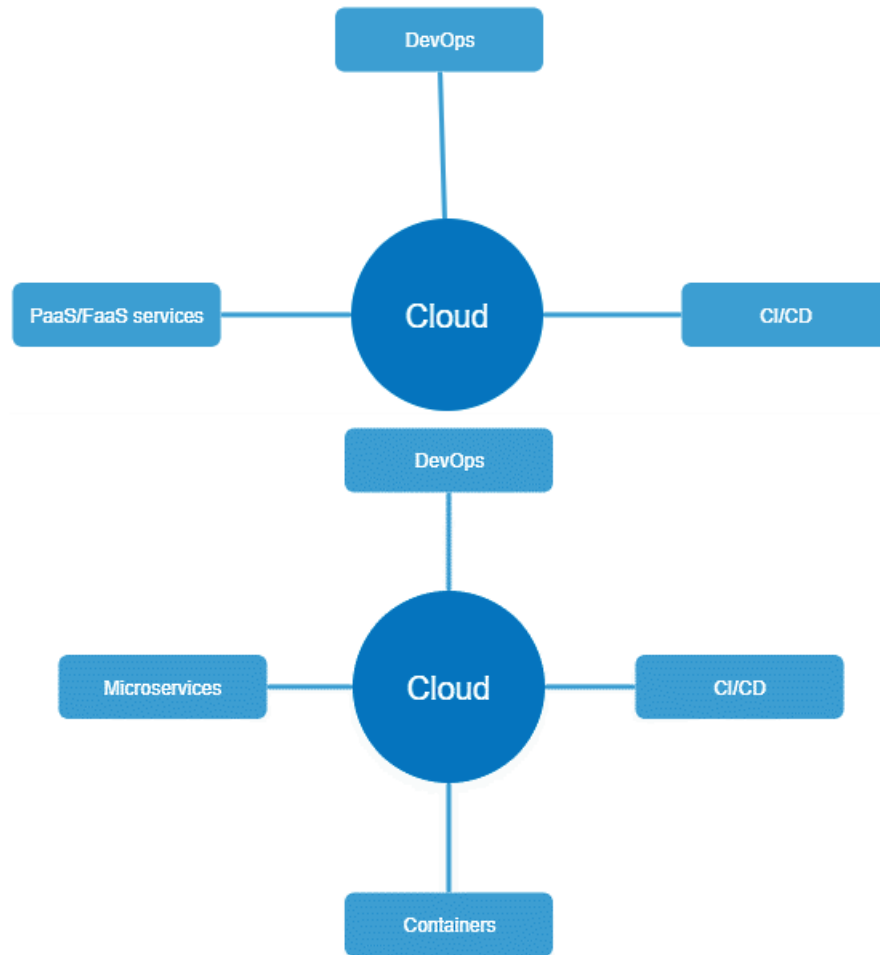


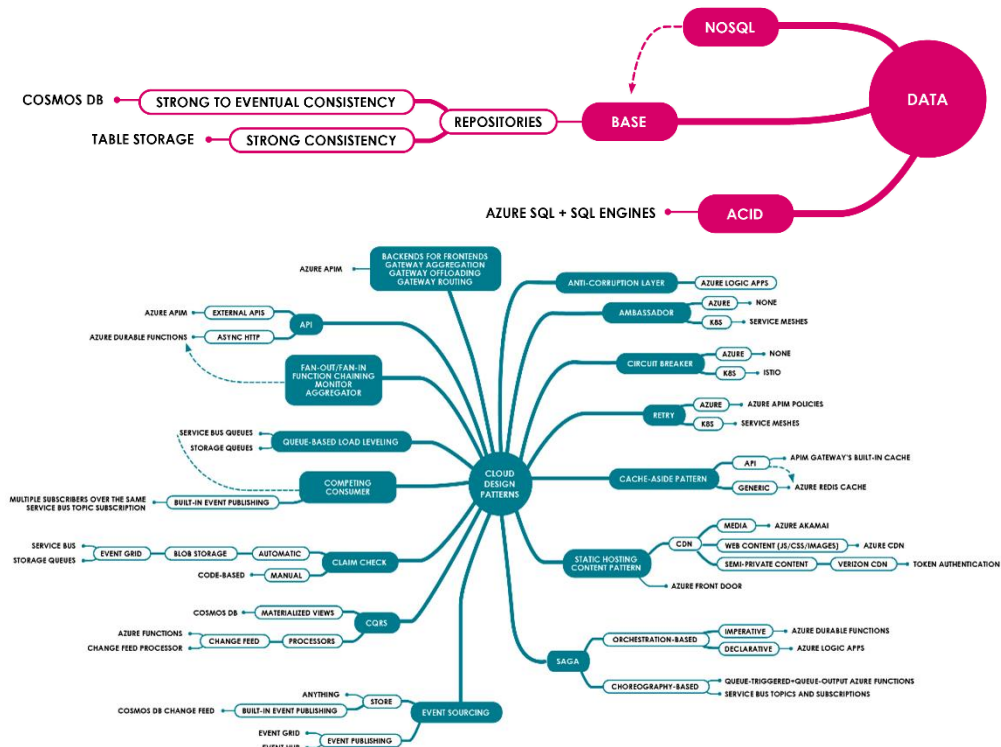
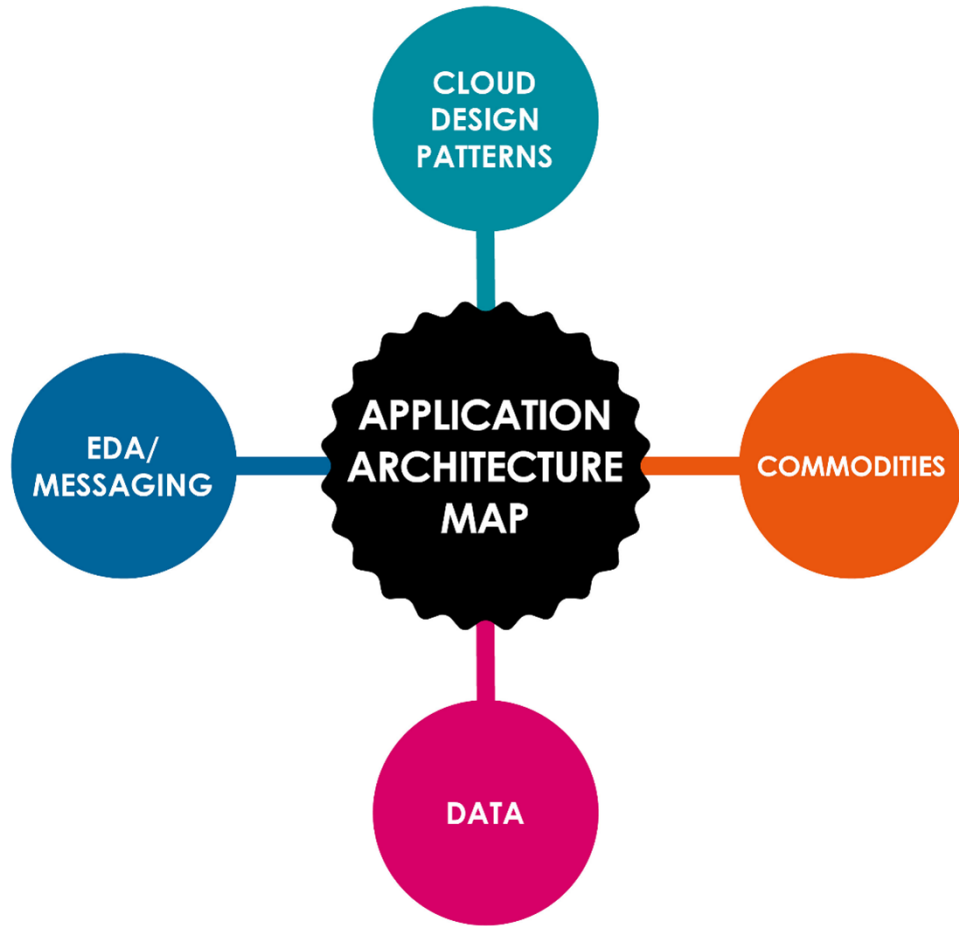


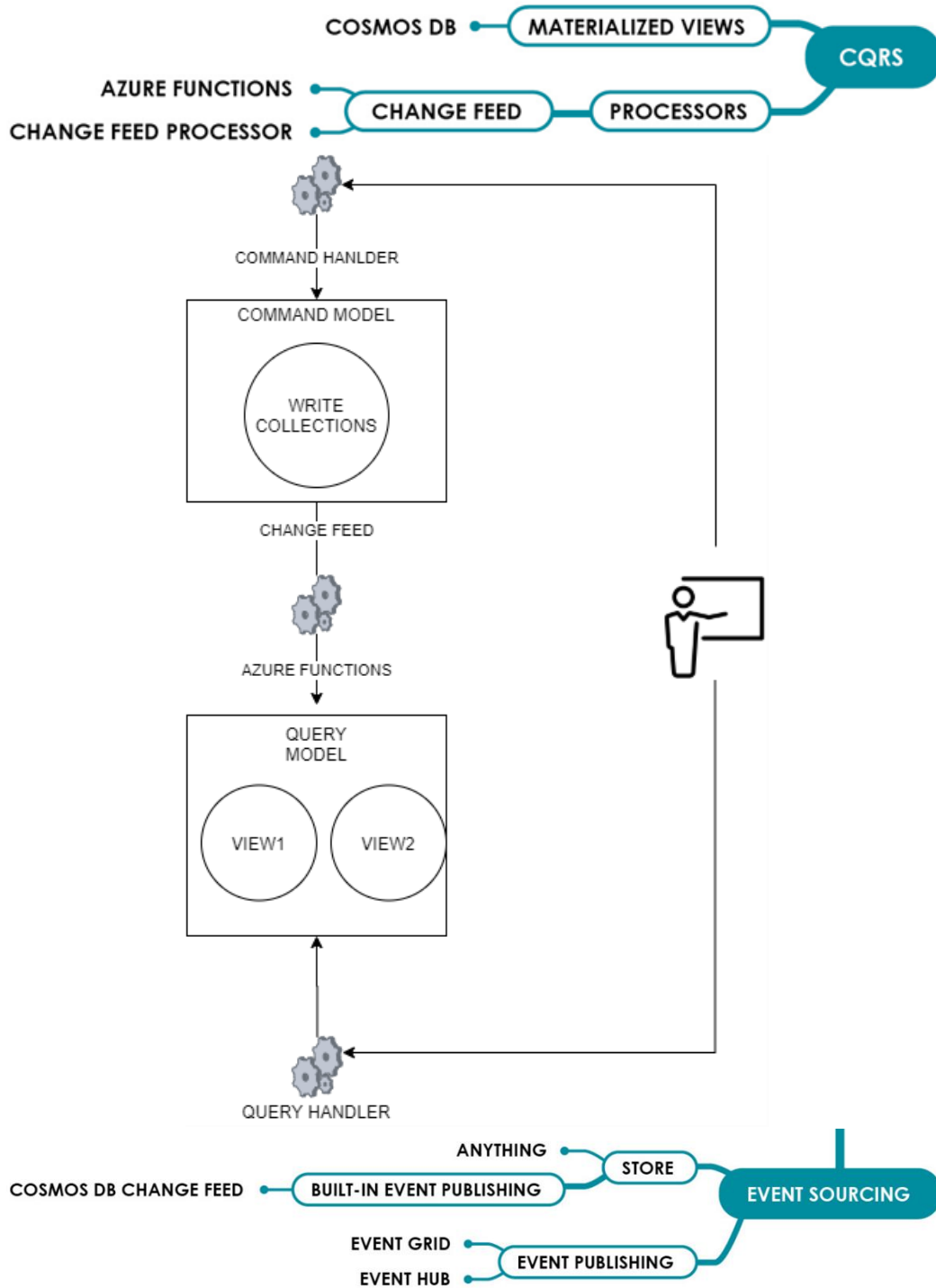


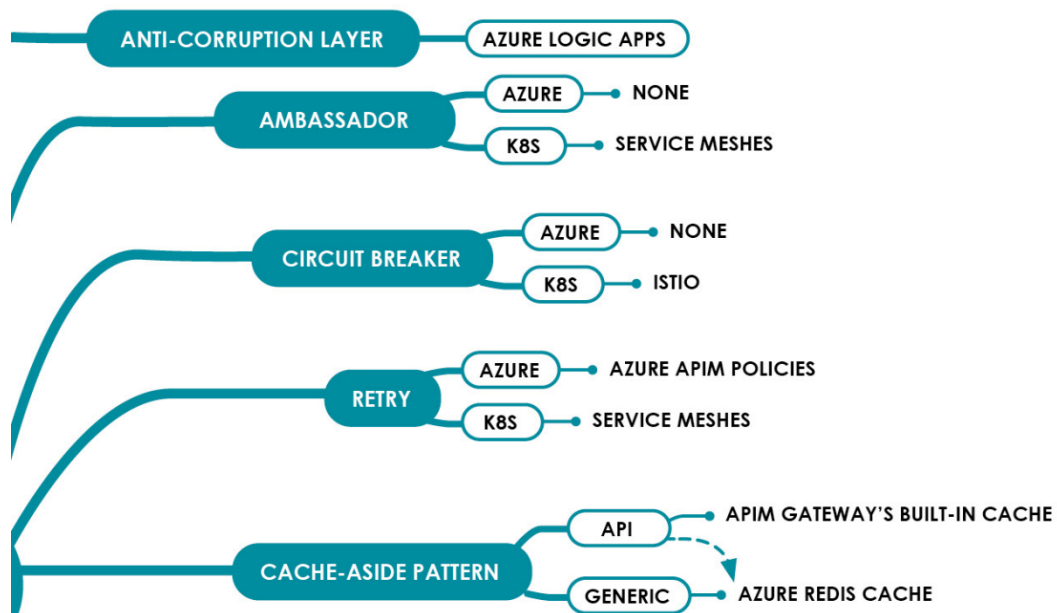
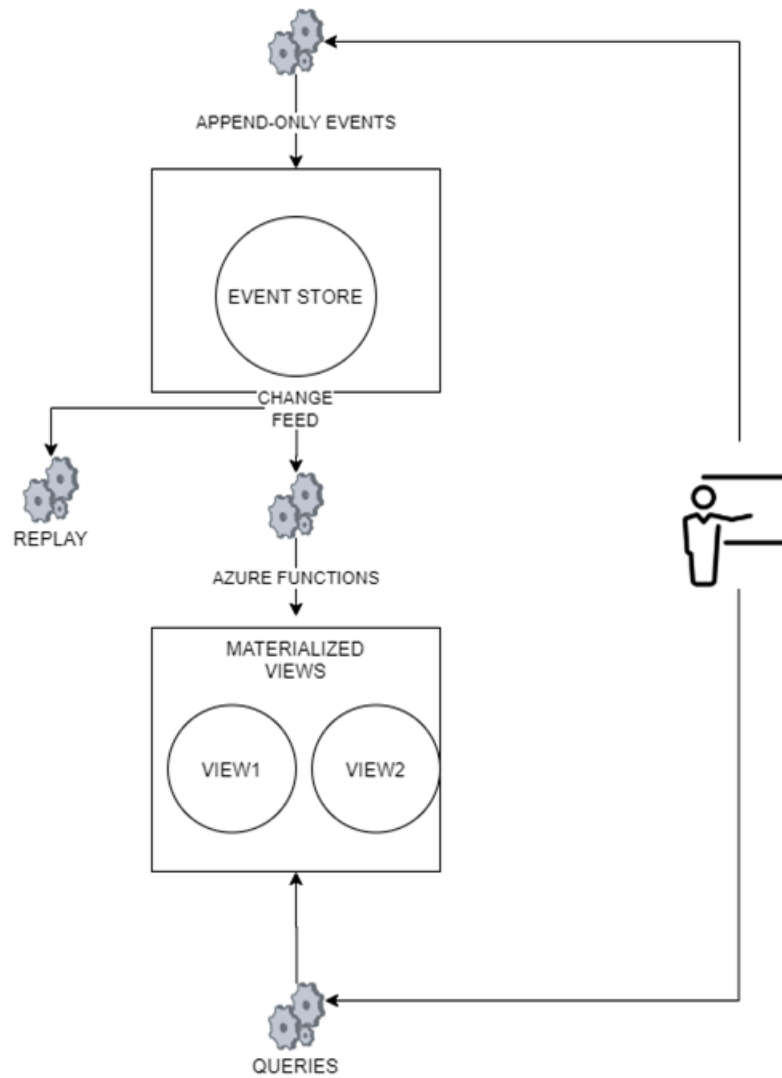


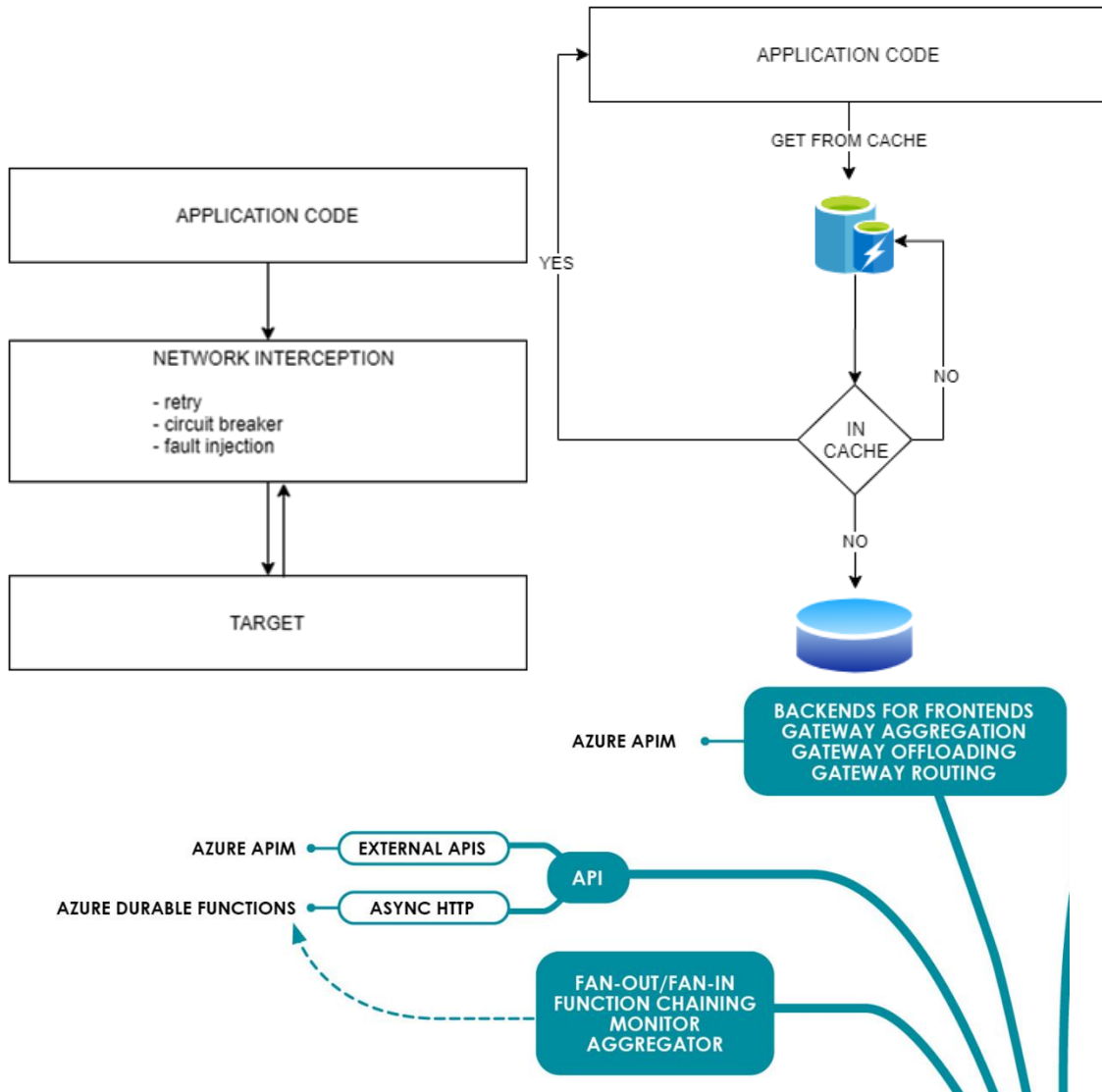
Chapter 5: Application Architecture

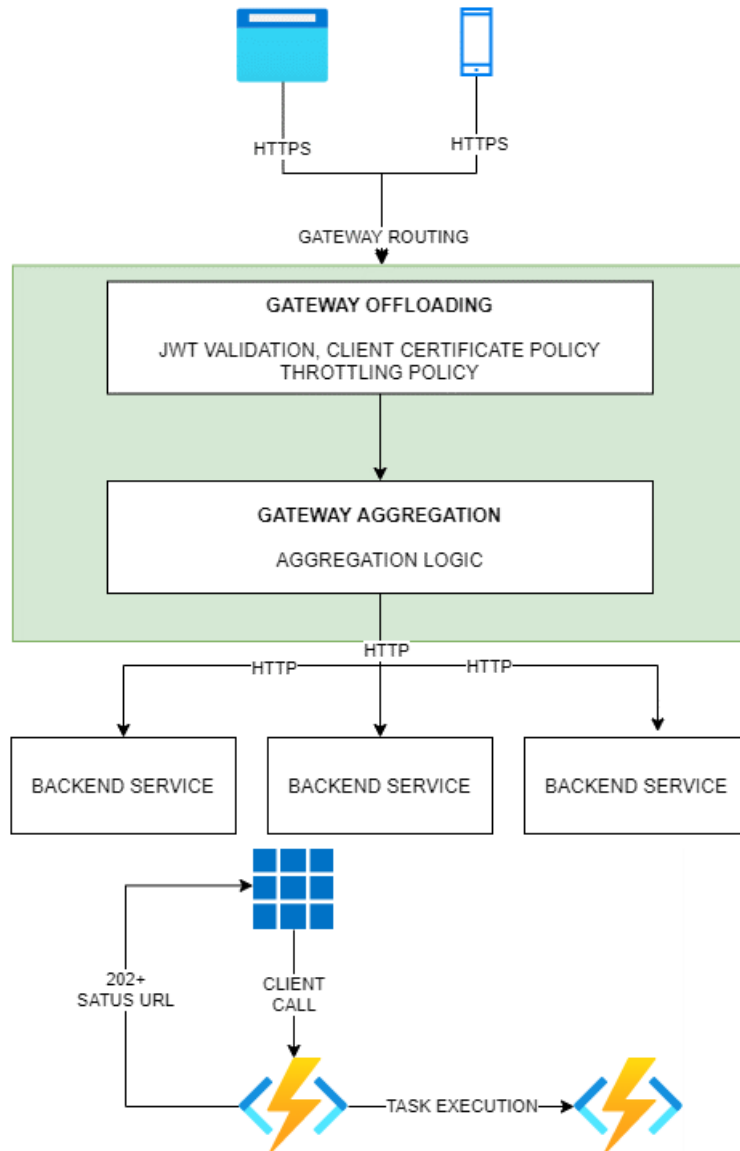


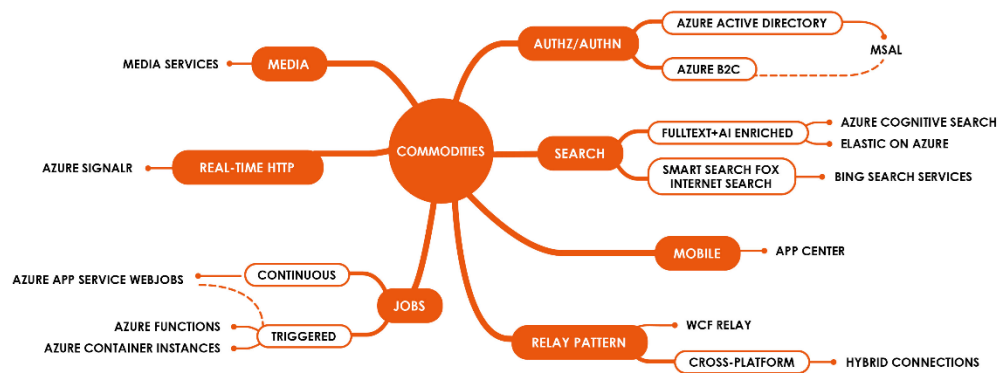
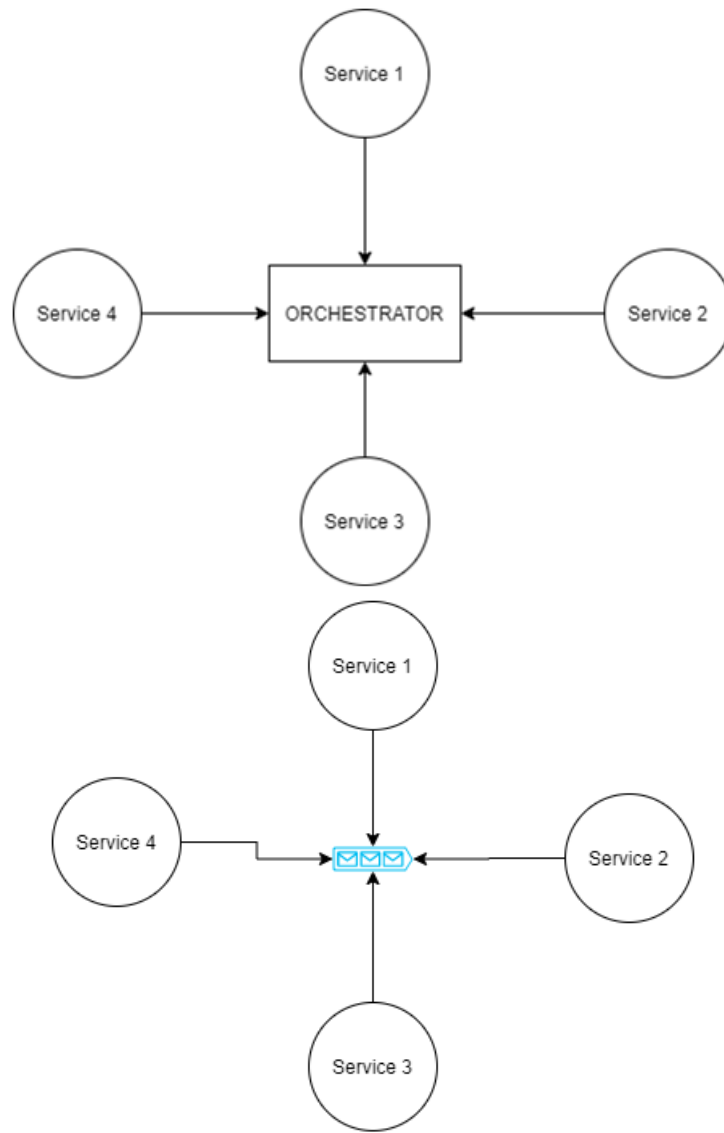


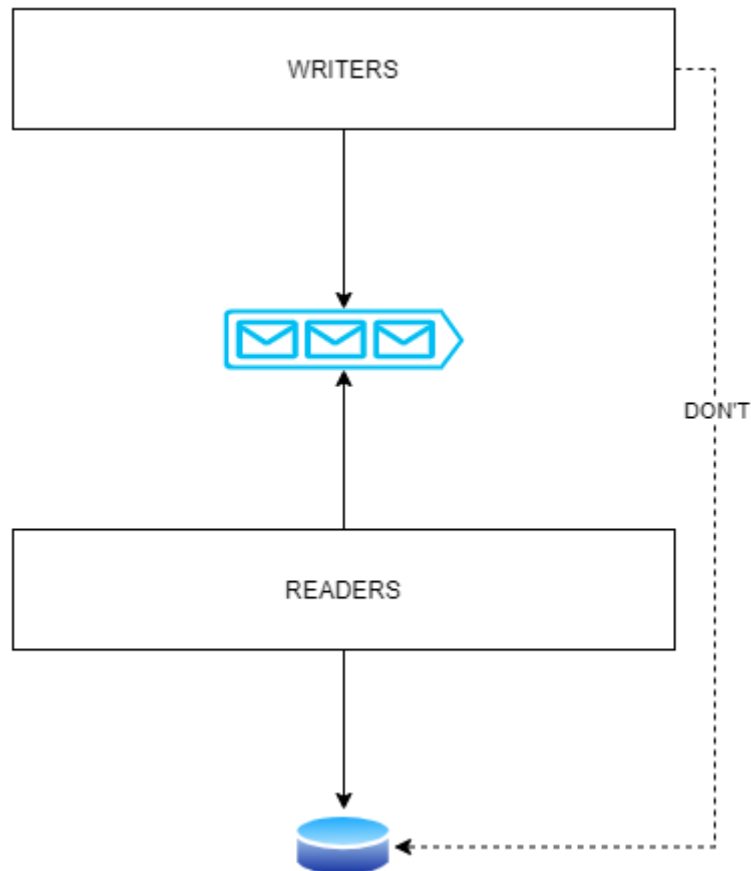
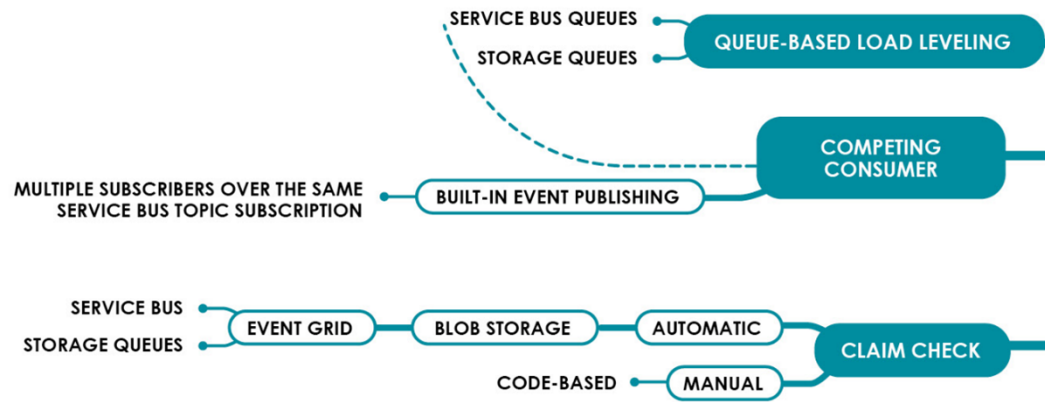


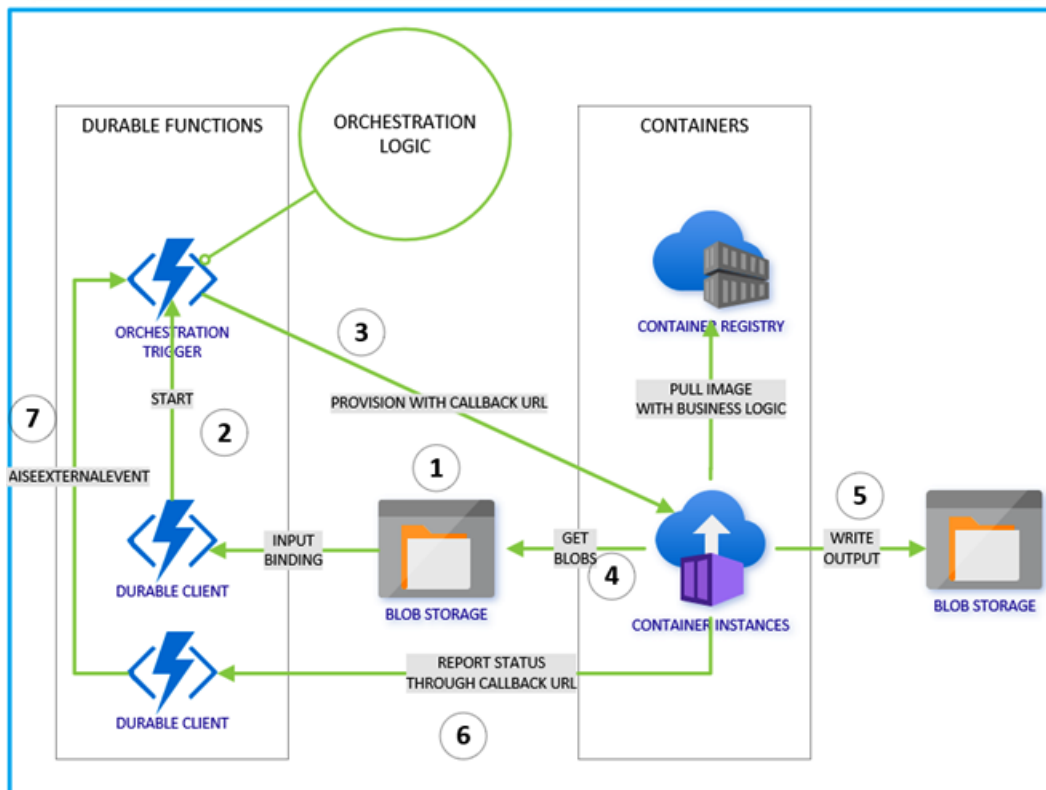
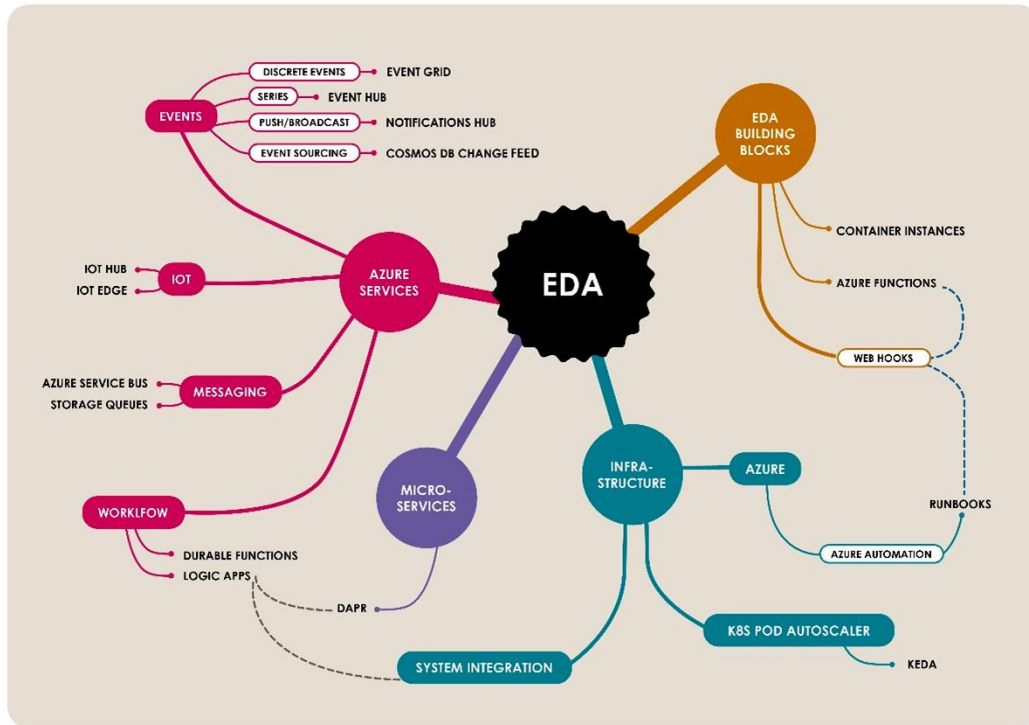


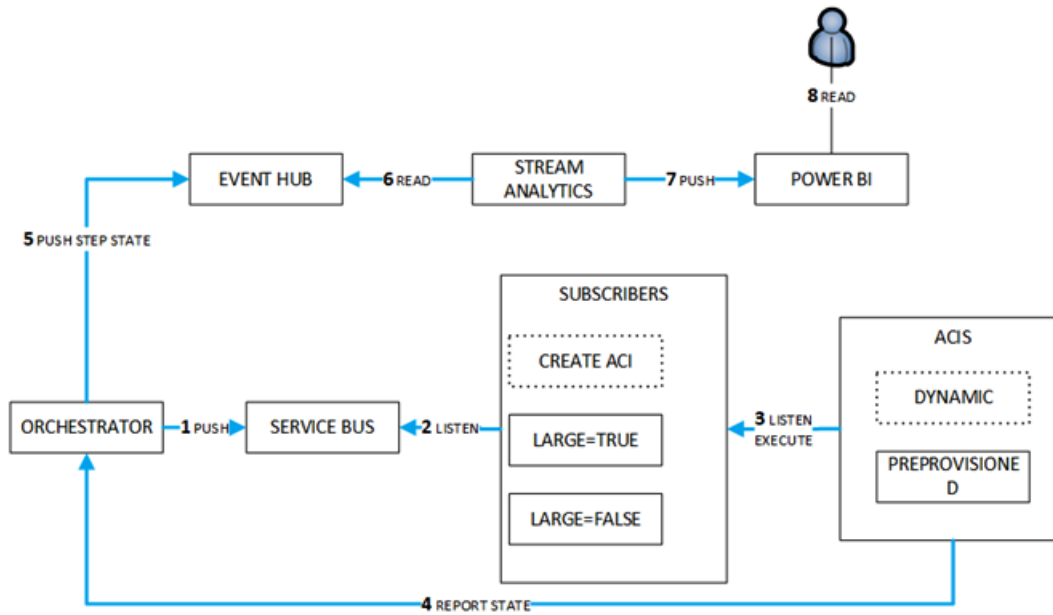












+ Topic Refresh

Search to filter items...

Name	Status	Max size	Subscription count
datasets	Active	1024 MB	3

+ Subscription Refresh

Search to filter items...

Name	Status	Message count	Max delivery count	Sessions Enabled
aci-large	Active	0	1	false
ds-large	Active	0	1	false
ds-smallmedium	Active	0	1	false

Max delivery count
1 (change)

Message time to live
10675199 DAYS (change)

Message lock duration
5 MINUTES (change)

Auto-delete
NEVER (change)

Active message count
0 MESSAGES

Dead-letter message count
0 MESSAGES

Transfer message count
0 MESSAGES

Scheduled message count
0 MESSAGES

Transfer message count
0 MESSAGES

FILTERS

+ Add filter

Name	Filter Type
aci-large	SqlFilter

aci-large

Filter Type

SQL Filter Correlation Filter

Filter messages by using a SQL-like expression that properties. [Learn more](#)

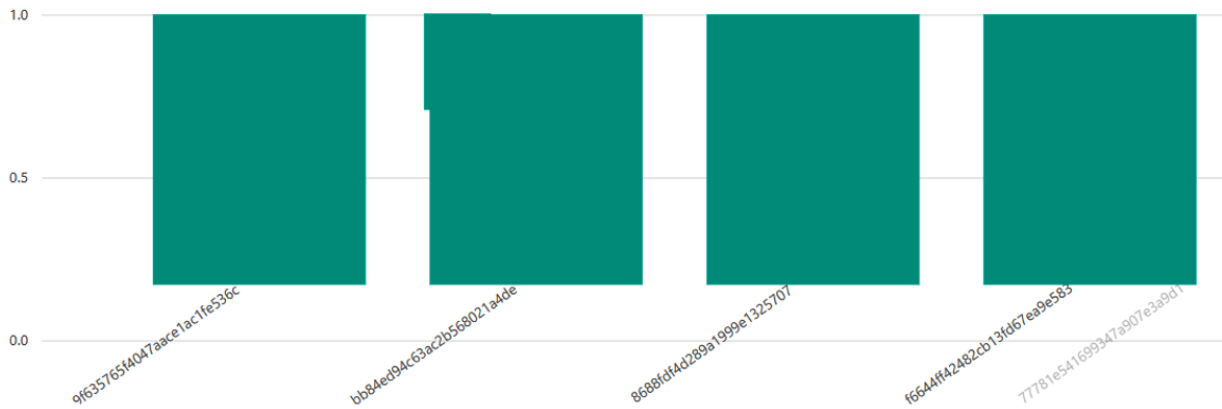
1 large = true

Test query Save query Discard changes

```
1 SELECT
2     *,CONCAT ( ParentOrchestrationId, '-', StepName) AS OrchestrationStep
3 INTO
4     [aci]
5 FROM
6     [orchestration]
7
8 SELECT
9     *,CONCAT ( ParentOrchestrationId, '-', StepName) AS OrchestrationStep
10 INTO
11     [failed]
12 FROM
13     [orchestration]
14     WHERE OrchestrationStatus = -1 or StepStatus = -1
```

Ask a question about your data

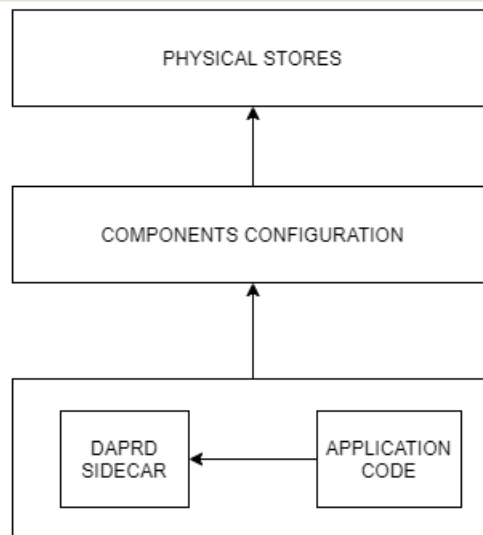
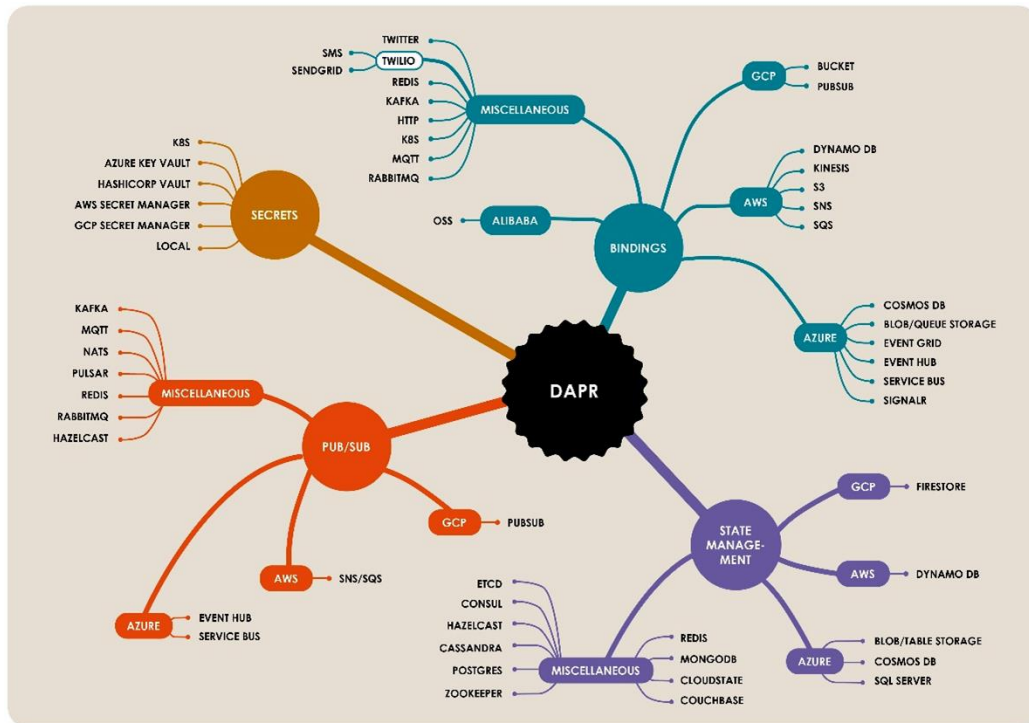
Ongoing orchestrations



Failed steps/orchestrations

Live steps





[Browse](#)

[Installed](#)

[Updates](#)

Dapr



Include prerelease



Dapr.Client by daprio, 28.6K downloads

v0.12.0-preview01

Prerelease

This package contains the reference assemblies for developing services using Dapr.



Dapr.AspNetCore by daprio, 24.3K downloads

v0.12.0-preview01

Prerelease

This package contains the reference assemblies for developing services using Dapr and ASP.NET Core.



Dapr.Actors by daprio, 12.7K downloads

v0.12.0-preview01

Prerelease

This package contains the reference assemblies for developing Actor services using Dapr.



Dapr.Actors.AspNetCore by daprio, 8.74K downloads

v0.12.0-preview01

Prerelease

This package contains the reference assemblies for developing Actor services using Dapr and ASP.NET Core.



Dapr.WebSockets by Reuben Bond, 14.8K downloads

v0.2.7

Portable Class Library for WebSockets, wrapping platform-specific APIs.

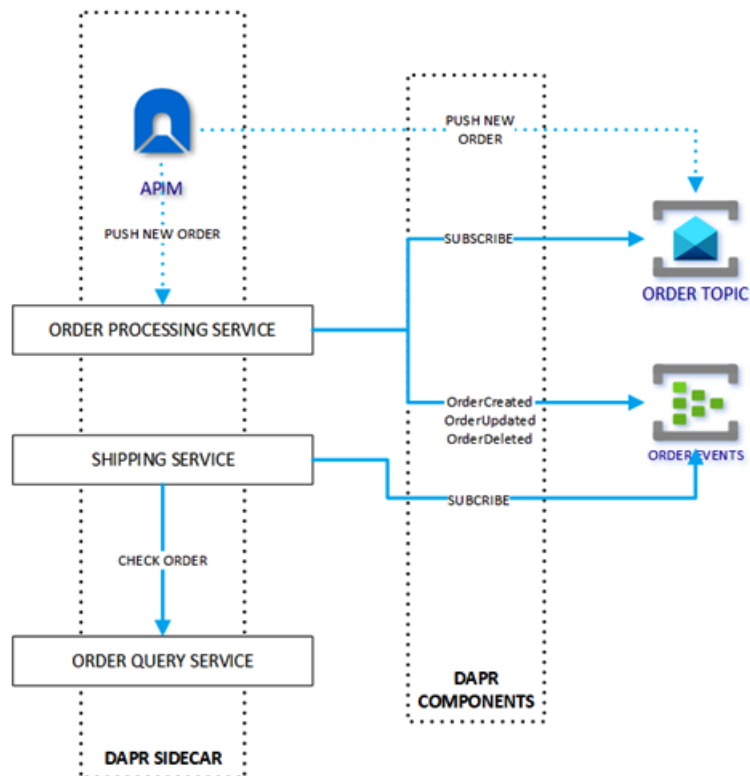


Dapr.AzureFunctions.Extension by daprio, 846 downloads

v0.10.0-preview01

Prerelease

This extension adds bindings for Dapr.



☐ Name ↑↓	Type ↑↓
☐  seyp-packtmapbook	Service Bus Namespace
☐  seyp-u4oceqdtieao6	Event Hubs Namespace
☐  seypu4oceqdtieao6	Storage account

```
Administrator: Windows PowerShell
[0m
[93;1m== DAPR == time="2020-12-10T18:15:46.1370072+01:00" level=info msg="dapr initialized. Status: Running. Init Elapsed 3325.4503999999997ms" app_id=shipping instance=DESKTOP-BICLHBI scope=dapr.runtime type=log ver=0.11.3
[0m
Updating metadata for app command: dotnet run
You're up and running! Both Dapr and your app logs will appear here.
```

```
Select Windows PowerShell
unning. Init Elapsed 3944.646ms" app_id=order instance=DESKTOP-BICLHBI scope=dapr.runtime type=log ver=0.11.3
[0m
Updating metadata for app command: dotnet run
You're up and running! Both Dapr and your app logs will appear here.
```

```
Select Administrator: Windows PowerShell
unlock" app_id=orderquery instance=DESKTOP-BICLHBI scope=dapr.runtime.actor type=log ver=0.11.3
[0m
Updating metadata for app command: dotnet run
You're up and running! Both Dapr and your app logs will appear here.
```

Parsed Raw Scratchpad Options

POST http://localhost:5002/order

content-type: application/json
Host: localhost:5002
Content-Length: 137
<

Request Body

```
{
  "Id": "4aad0f8-eeda-4ee7-9c26-a6d39cbfbc28",
  "Products": [{"Id": "5678f982-2ae4-408c-92ff-6af45118d159", "Name": "a product"}]
}
```

```
Windows PowerShell
+ [94;1m== APP == info: OrderService.Controllers.OrderController[0]
+ [0m
+ [94;1m== APP ==      Order with id 4aadcf8-eeda-4ee7-9c26-a6d39cbfbc28 created!
+ [0m
+ [94;1m== APP == info: OrderService.Controllers.OrderController[0]
+ [0m
+ [94;1m== APP ==      Order with id 4aadcf8-eeda-4ee7-9c26-a6d39cbfbc28 processed!

Administrator: Windows PowerShell
You're up and running! Both Dapr and your app logs will appear here.
+ [94;1m== APP == info: TrackingService.Controllers.ShippingController[0]
+ [0m
+ [94;1m== APP ==      Received new event
+ [0m
+ [94;1m== APP == info: TrackingService.Controllers.ShippingController[0]
+ [0m
+ [94;1m== APP ==      4aadcf8-eeda-4ee7-9c26-a6d39cbfbc28 OrderEvent Created
+ [0m
+ [94;1m== APP == info: TrackingService.Controllers.ShippingController[0]
+ [0m
+ [94;1m== APP ==      Starting shipping process for order 4aadcf8-eeda-4ee7-9c26-a6d39cbfbc28!
+ [0m

Administrator: Windows PowerShell
+ [94;1m== APP == info: OrderQueryService.Controllers.OrderQueryController[0]
+ [0m
+ [94;1m== APP ==      4aadcf8-eeda-4ee7-9c26-a6d39cbfbc28
+ [0m
+ [94;1m== APP == info: OrderQueryService.Controllers.OrderQueryController[0]
```

[+ Topic](#) [Refresh](#)

Name	Status	Max size
dapr	Active	1024 MB
datasets	Active	1024 MB

[+ Subscription](#) [Refresh](#)

Name	Status	Message count
all	Active	0

```
Select Administrator: Windows PowerShell
PS C:\Users\steph\Source\Repos\azug\dapr\.NET Core\deploy> kubectl apply -f .\packtsb.yaml
component.dapr.io/daprsb created
PS C:\Users\steph\Source\Repos\azug\dapr\.NET Core\deploy> 
0
\Users\steph\Source\Repos\azug\dapr\.NET Core\deploy> kubectl logs seyselhosted-5c488b9b4f-224mq dapr
"2020-12-06T17:50:55.1346578Z" level=info msg="found component daprsb (pubsub.azure.servicebus)" app_id=
```

Create a blank API

Basic | **Full**

* Display name	dapr
* Name	dapr
Web service URL	e.g. http://httpbin.org
API URL suffix	dapr-demo
Base URL	https://apimsp.azure-api.net/dapr-demo

Create

Cancel

REVISION 1 CREATED Dec 6, 2020, 6:59:28 PM

Design Settings Test Revisions Change log

Search operations

Filter by tags

☐ Group by tag

+ Add operation

All operations

No operations to display.

dapr > Add operation

Frontend

* Display name	order
* Name	order
* URL	POST order
Description	Post orders
Tags	e.g. Booking

```
<policies>
  <inbound>
    <base />
    <rate-limit calls="10" renewal-period="60" />
    <publish-to-dapr pubsub-name="daprsb" topic="dapr"
      response-variable-name="dapr-response">@(context.Request.Body.As<string>())</publish-to-dapr>
  </inbound>
  <backend />
  <outbound>
    <base />
  </outbound>
  <on-error>
    <choose>
      <when condition="@(context.Variables.Count > 0)">
        <return-response response-variable-name="dapr-response" />
      </when>
      <otherwise />
    </choose>
    <base />
  </on-error>
</policies>
```

ParsedRawScratchpadOptions

POSThttps://selfhostedgateway.silver-it.com/dapr-demo/order

Ocp-Apim-Subscription-Key: 37eddc1d7839c5f
Ocp-Apim-Trace: true

<

Request Body

{
 "ID": "acd9b7b4-ecd7-499a-8380-4f3b0817e850",
 "Products": [
 {"ProductId": "b6f66698-a635-4c35-891d-999500f761f9", "ProductName": "a product"}
]
}

Search to filter items...

Name	Status	Message count
all	Active	1

SendReceivePeek

Select Subscription on **dapr** *

all

Receive performs a destructive read (ReceiveAndDelete) from Subscription **all**. The mess
Subscription. Messages shown here are no longer stored.

Active0 MESSAGES

Dead-Lettered0 MESSAGES

Please Select Subscription or DeadLetter

☒ Subscription☐ DeadLetter

Receive

Can only receive 1 message.

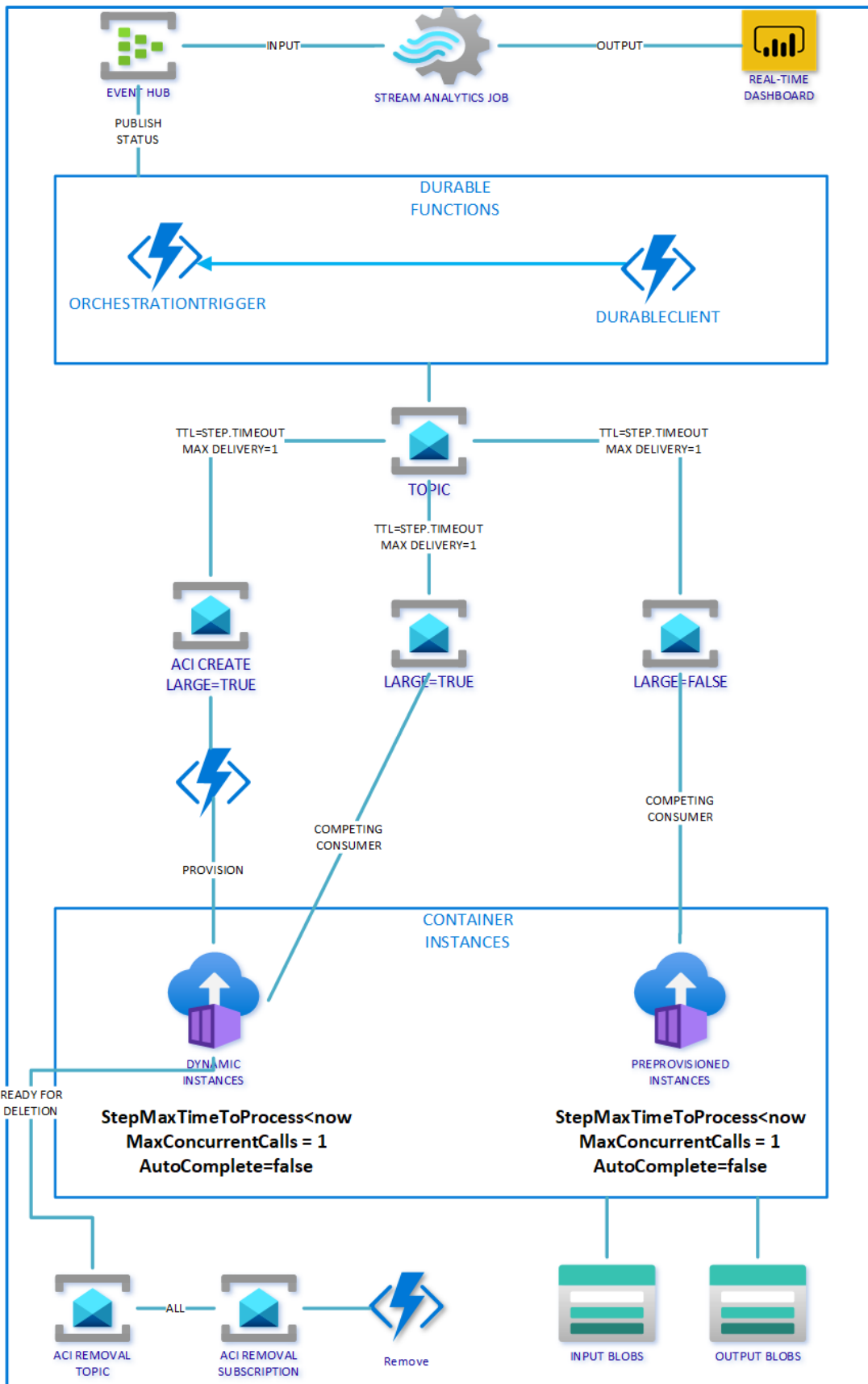
Sequence Number	Message ID
2	317e7c11-2f92-4c4e-acd0-07b527b99c8f

Progress Telerik Fiddler Web Debugger

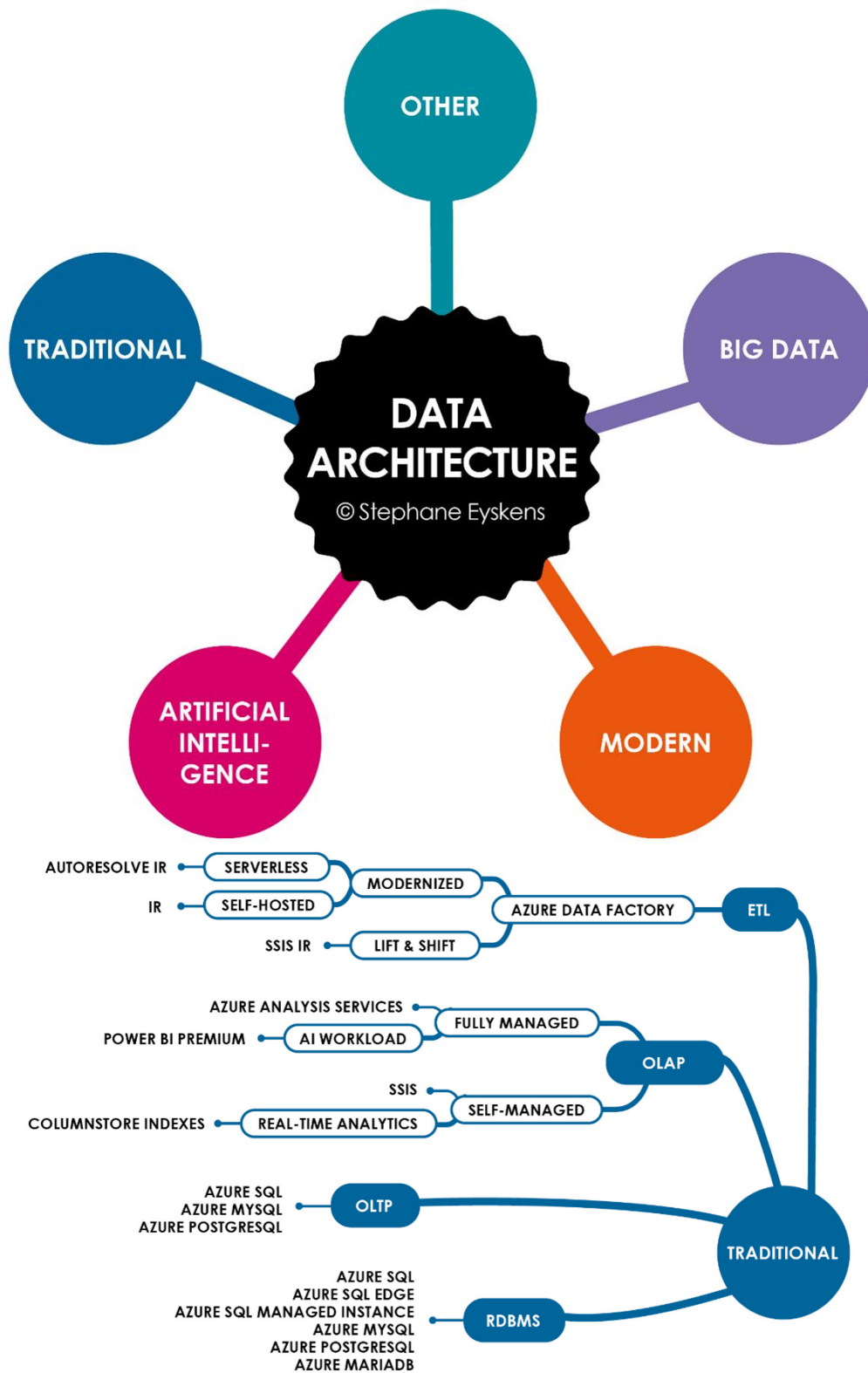
File Edit Rules Tools View Help

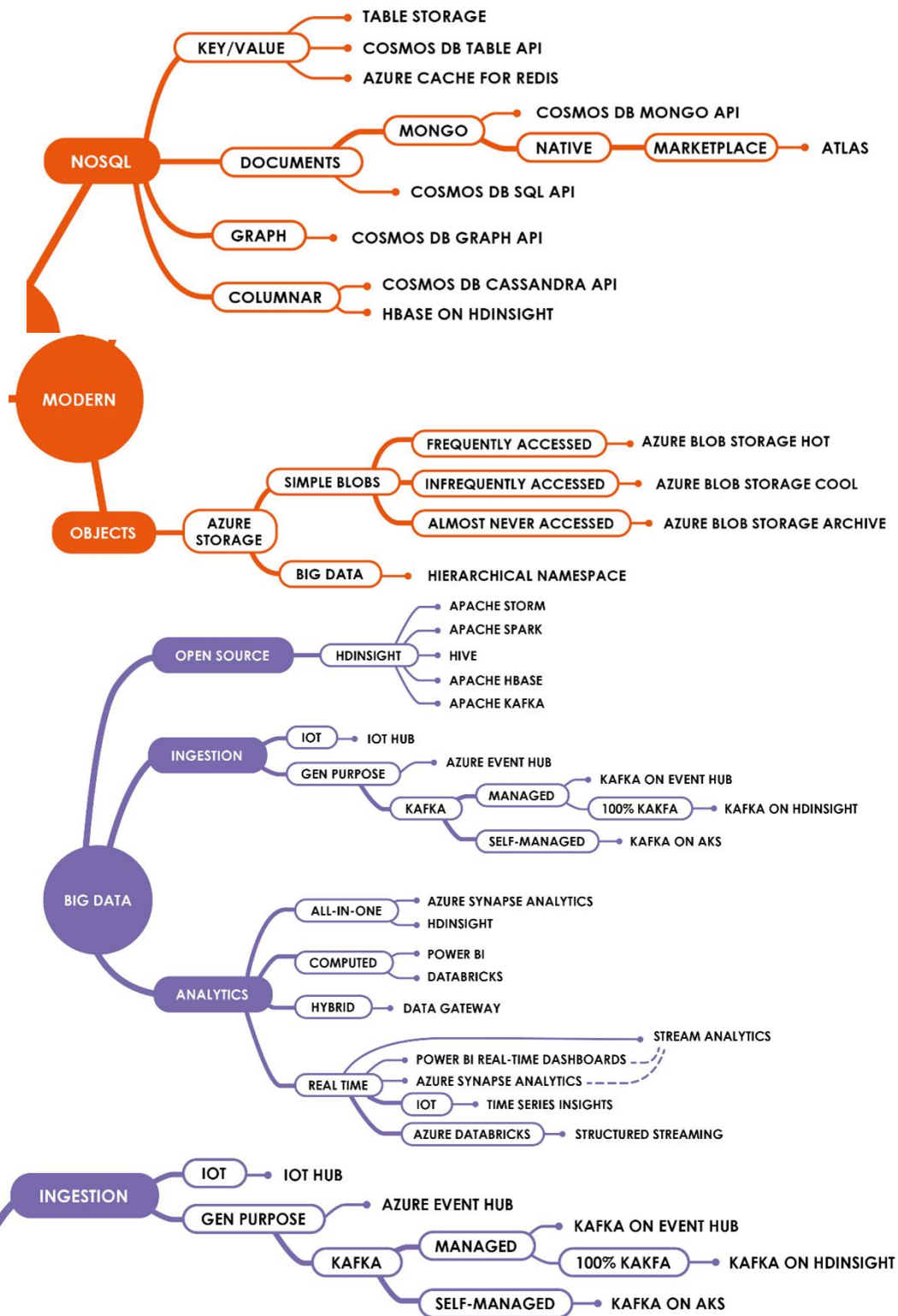
WinConfig WinConfig Replay X Go Stream Decode

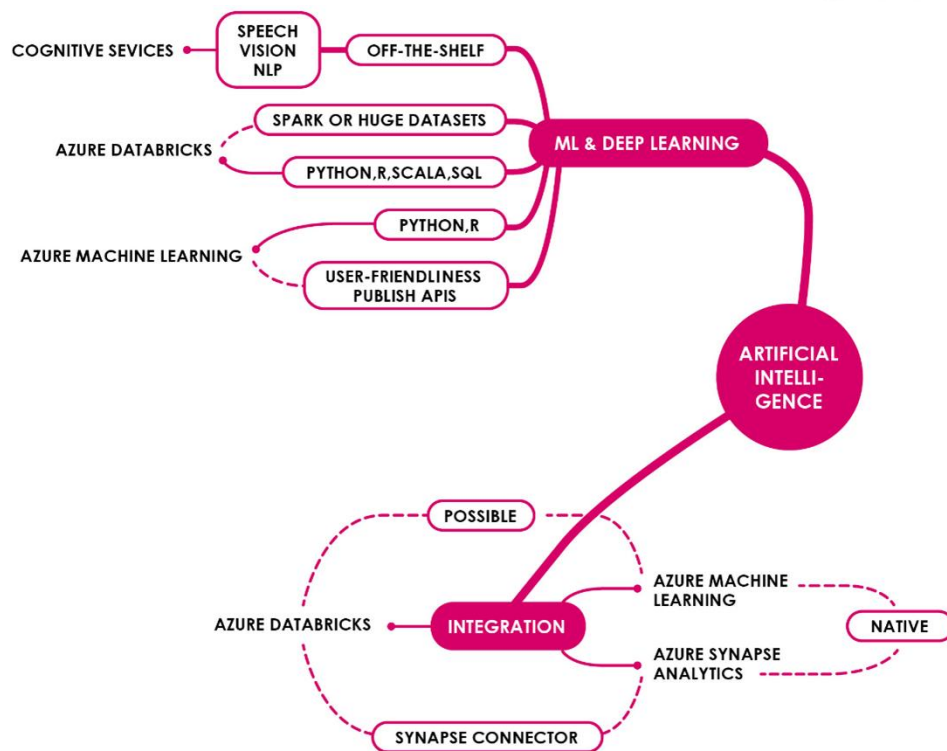
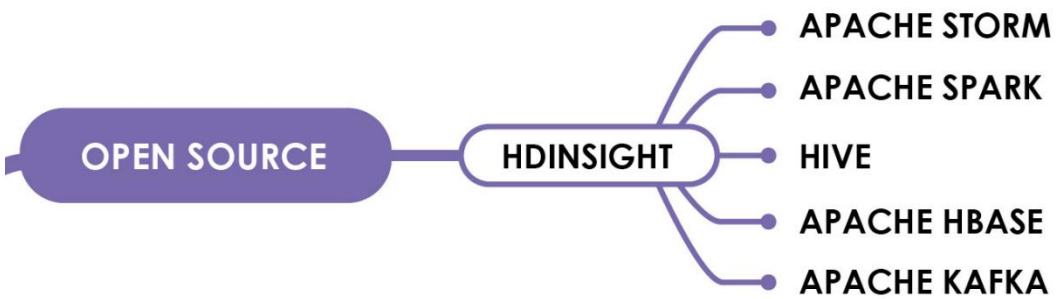
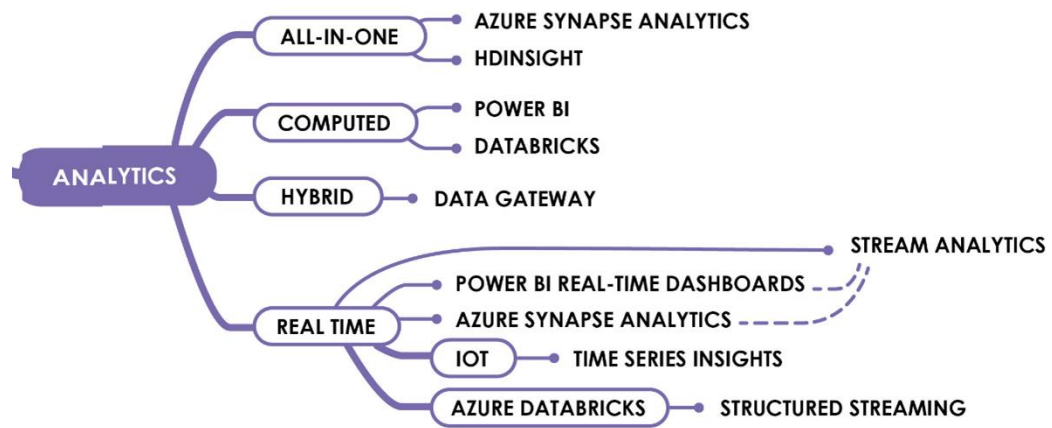
#	Result	Protocol	Host	URL
1	200	HTTPS	selfhostedgateway....	/dapr-demo/order
2	200	HTTPS	selfhostedgateway....	/dapr-demo/order
3	200	HTTPS	selfhostedgateway....	/dapr-demo/order
4	200	HTTPS	selfhostedgateway....	/dapr-demo/order
5	200	HTTPS	selfhostedgateway....	/dapr-demo/order
6	429	HTTPS	selfhostedgateway....	/dapr-demo/order
7	429	HTTPS	selfhostedgateway....	/dapr-demo/order
8	429	HTTPS	selfhostedgateway....	/dapr-demo/order
9	429	HTTPS	selfhostedgateway....	/dapr-demo/order
10	429	HTTPS	selfhostedgateway....	/dapr-demo/order
11	429	HTTPS	selfhostedgateway....	/dapr-demo/order
12	429	HTTPS	selfhostedgateway....	/dapr-demo/order
13	429	HTTPS	selfhostedgateway....	/dapr-demo/order
14	429	HTTPS	selfhostedgateway....	/dapr-demo/order
15	429	HTTPS	selfhostedgateway....	/dapr-demo/order

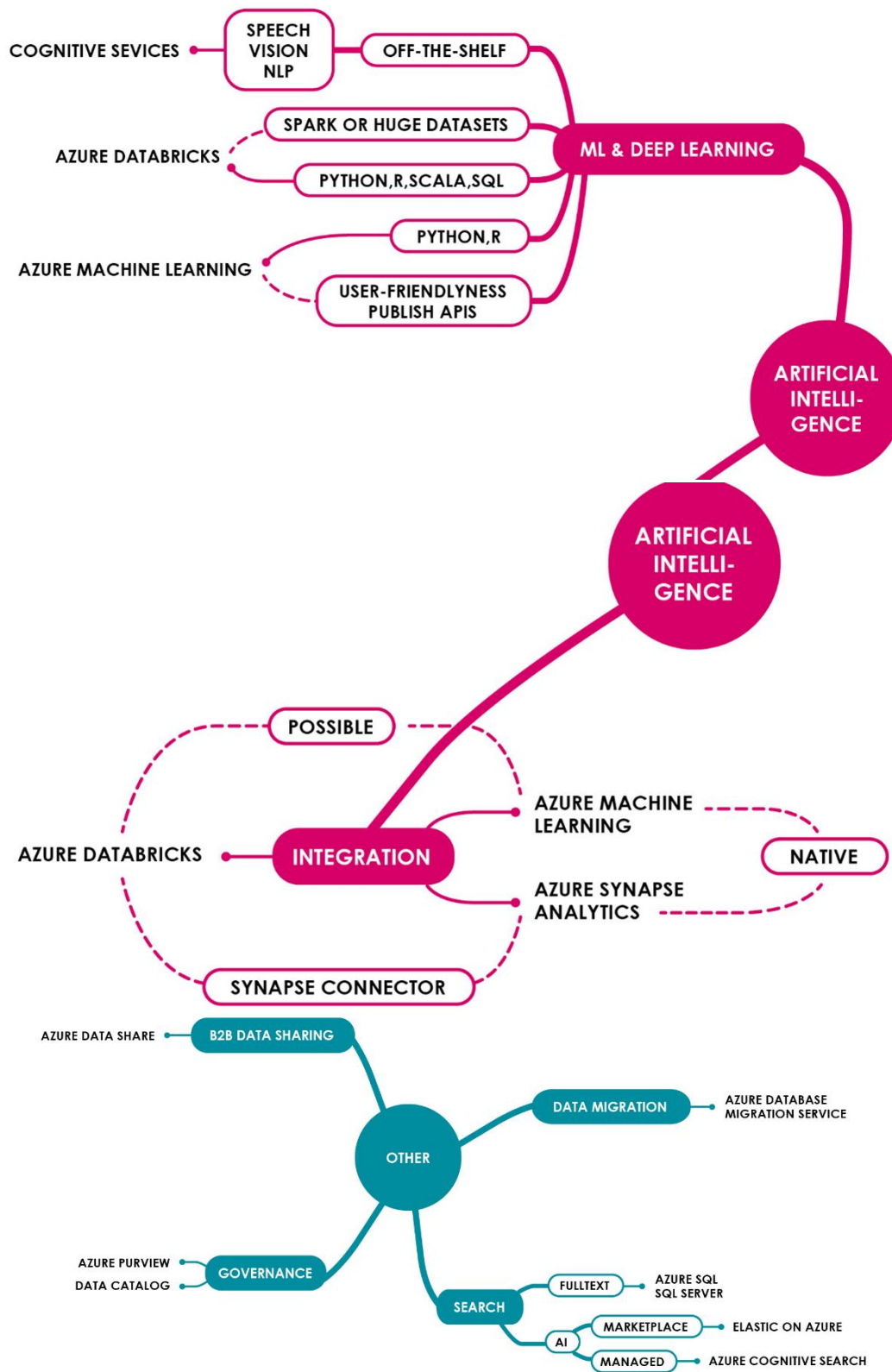


Chapter 6: Data Architecture









Add admins, members, or contributors. [Learn more](#)

pack

packt AppID: fdc4d913-ae62-4b73-ba94-1d50f33bd584

Member

Query language docs Open in Visual Studio UserVoice

Inputs (1)

hub

Outputs (2)

fine

powerbi

Test query Save query Discard changes

```
1 WITH
2 OK AS
3 (
4     SELECT sensorName, count(*) as TotalUnderMaxSpeed
5     FROM hub
6     WHERE speed <= 50
7     GROUP BY sensorName,
8     TumblingWindow(second,10)
9 ),
10 NOK AS
11 (
12     SELECT sensorName, count(*) as TotalOverMaxSpeed
13     FROM hub
14     WHERE speed > 50
15     GROUP BY sensorName,
16     TumblingWindow(second,10)
17 )
18 SELECT OK.sensorName, OK.TotalUnderMaxSpeed, NOK.TotalOverMaxSpeed INTO
19 powerbi from OK INNER JOIN NOK ON NOK.sensorName=OK.sensorName
20 AND DATEDIFF(second, OK, NOK) BETWEEN 0 AND 10
21 SELECT * INTO fine FROM hub WHERE speed > 50
```

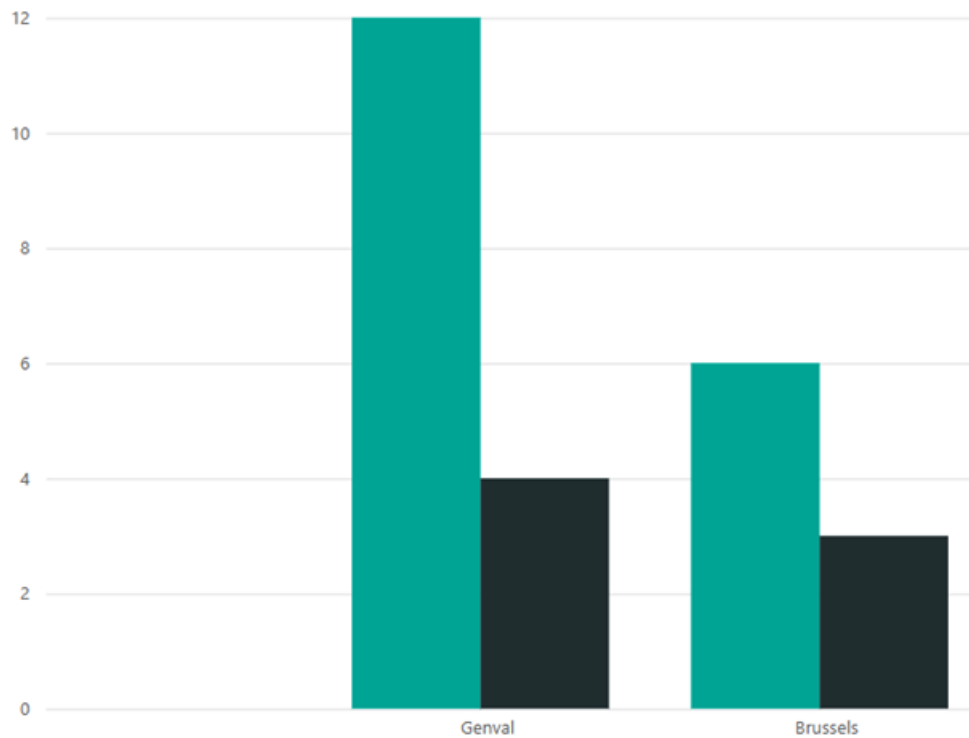
C:\WINDOWS\system32\cmd.exe

```
C:\Users\steph\Source\Repos\DeviceSimulator\DeviceSimulatorConsole\bin\Debug\netcoreapp3.1>DeviceSimulatorConsole.exe 5
Send 125 events in 0 seconds
```

```
C:\Users\steph\Source\Repos\DeviceSimulator\DeviceSimulatorConsole\bin\Debug\netcoreapp3.1>
```

⚡ TotalUnderMaxSpeed, TotalOverMaxSpeed
BY SENSORNAME

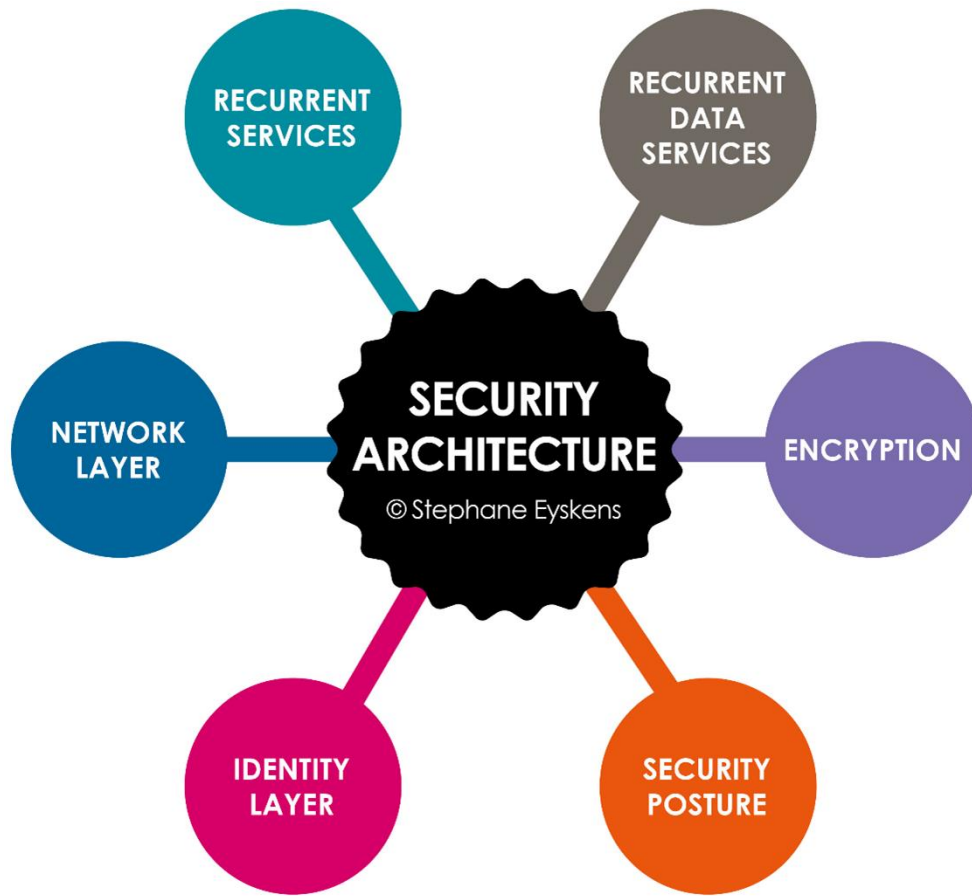
● TotalUnderMaxSpeed ● TotalOverMaxSpeed

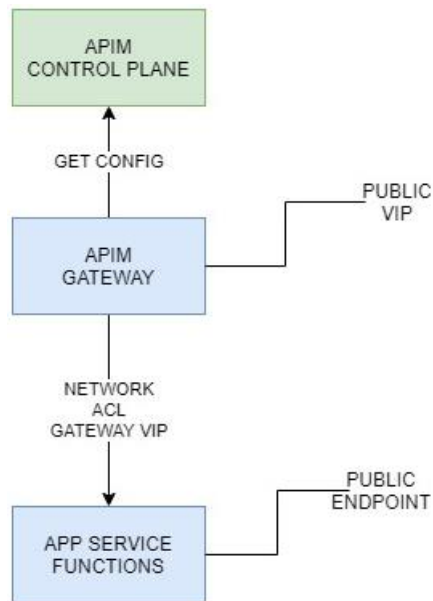
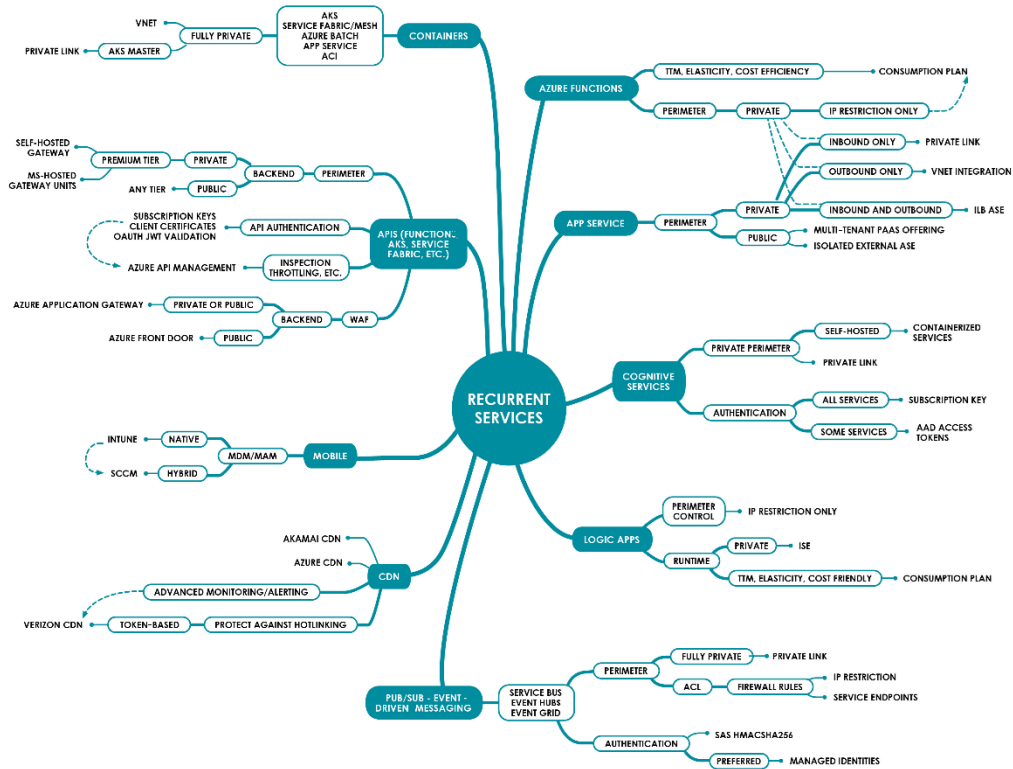


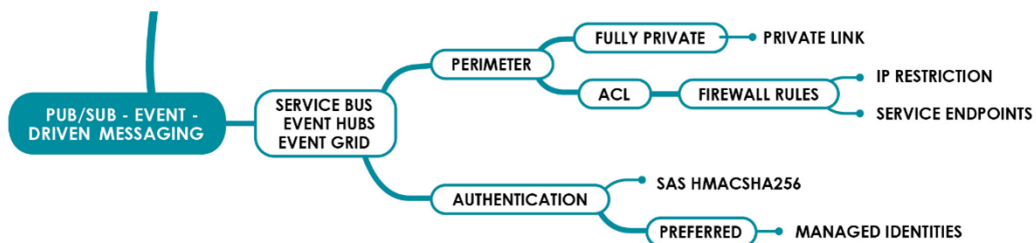
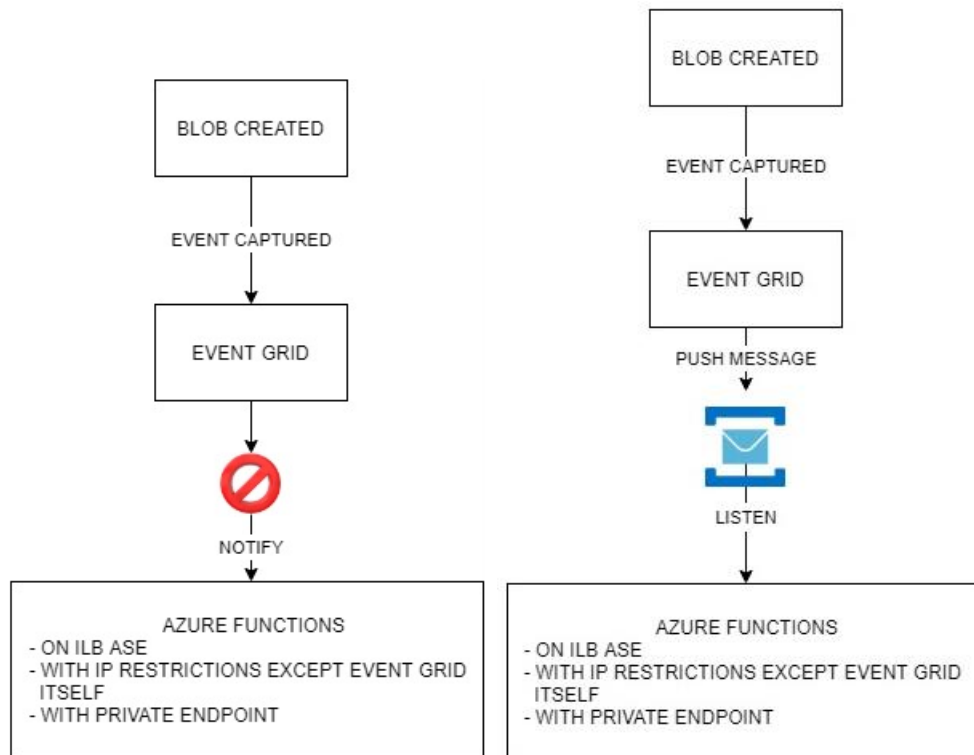
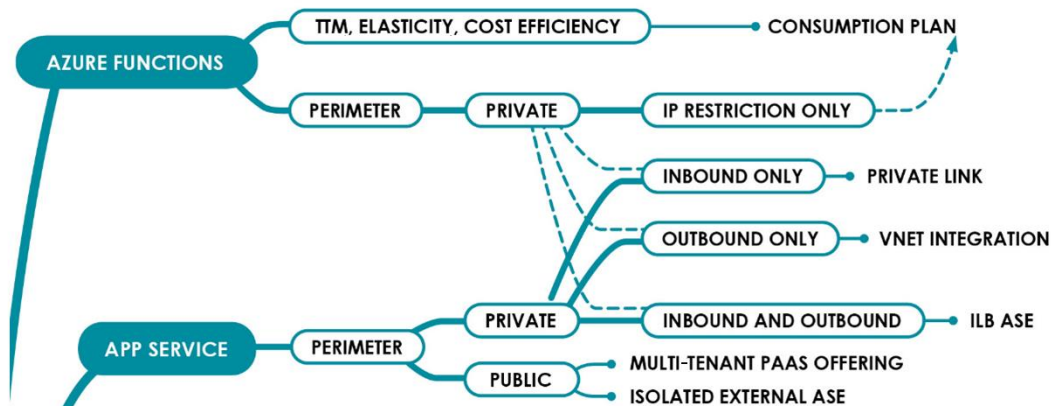
[Back to report](#)

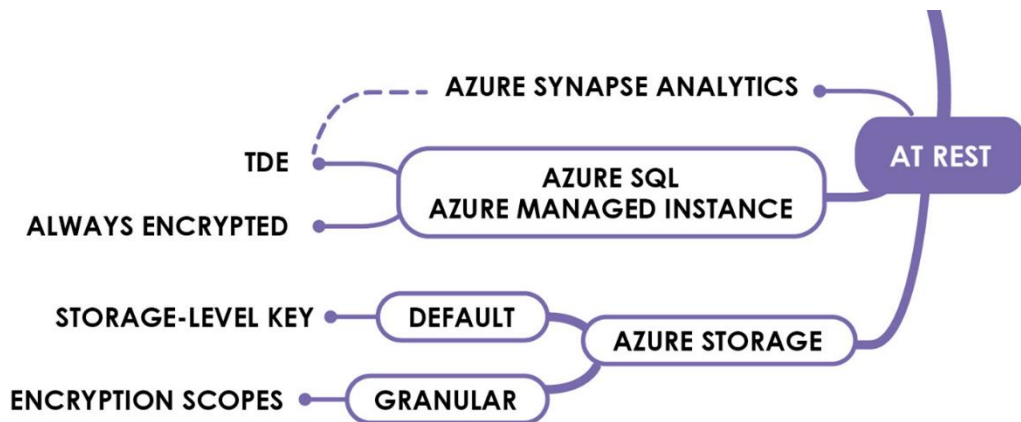
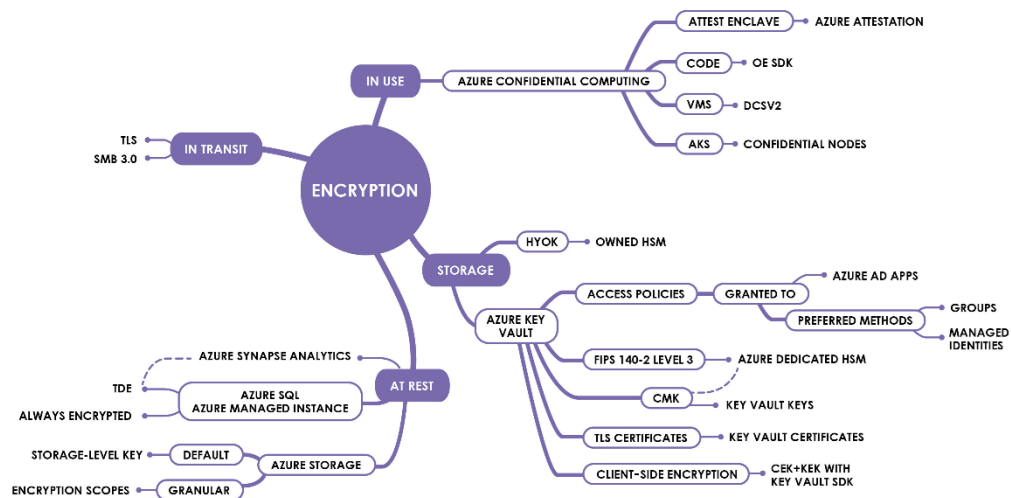
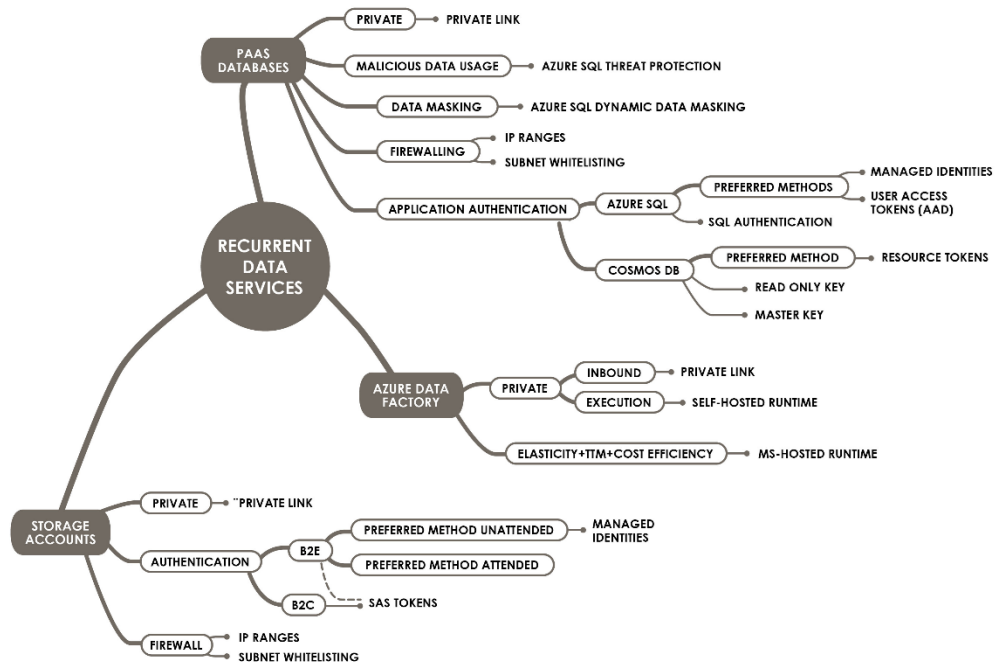
plateNumber	sensorName	speed
084647c6-c4ed-40b4-9953-775035319263	Brussels	80.33
49944c1b-b0f2-4996-a60a-207c900ee610	Brussels	87.47
9cee2149-81b3-46d5-bc85-990c0b05d735	Brussels	77.56
0cfc2f6f-6bc0-41a1-b330-76b2a7e75dd2	Genval	99.01
50e7172f-169d-4447-8f74-7947076fbd36	Genval	62.34
a1d6fcb0-3953-4c85-a687-cd5452d1e4f5	Genval	67.80
de07ade0-7ca1-409f-8c95-982970684fcb	Genval	74.21

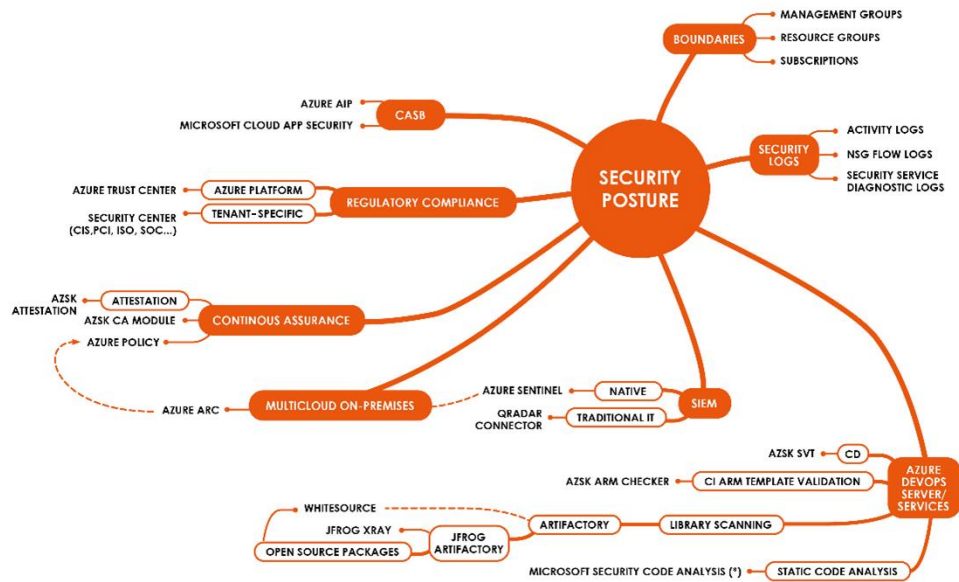
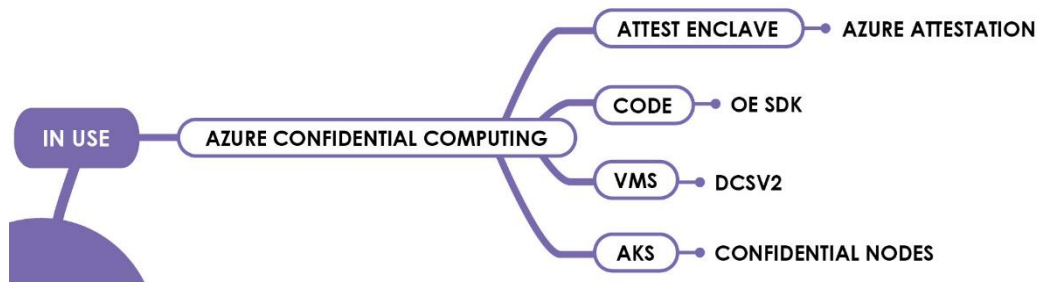
Chapter 7: Security Architecture







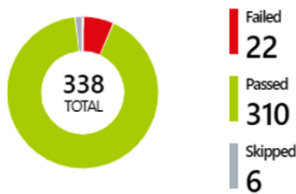




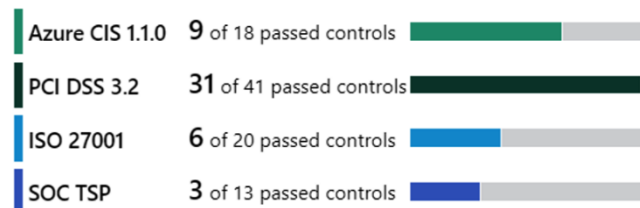
Total: 84 resources

	10 Servers	\$15	Server/Month
	1 App Service instances	\$15	Instance/Month
	1 Azure SQL Database	\$15	Server/Month
	52 Storage accounts	\$0.02	10k transactions
	10 Kubernetes cores	\$2	VM core/Month
	2 Container registries	\$0.29	Image
	8 Key Vaults	\$0.02	10k transactions
	0 SQL servers on machines ⓘ	\$15 \$0.015	Server/Month Core/Hour
	Resource Manager (Preview)	FREE during preview	ⓘ
	DNS (Preview)	FREE during preview	ⓘ

Regulatory compliance assessment



Regulatory standards compliance status



[Azure CIS 1.1.0](#) [PCI DSS 3.2](#) [ISO 27001](#) [SOC TSP](#) [All](#)

Under each applicable compliance control is the set of assessments run by Security Center that are associated with that control. If they are all green, it means those assessments are currently passing; this does not ensure you are fully compliant with that control. Furthermore, not all controls for any particular regulation are covered by Security Center assessments, and therefore this report is only a partial view of your overall compliance status.

Access to App Services should be restricted

Use Azure Security Center Standard for App Services to run advanced assessments and detect network threats on

^ Affected resources

Unhealthy resources (6) [Healthy resources \(0\)](#) [Unscanned resources \(0\)](#)

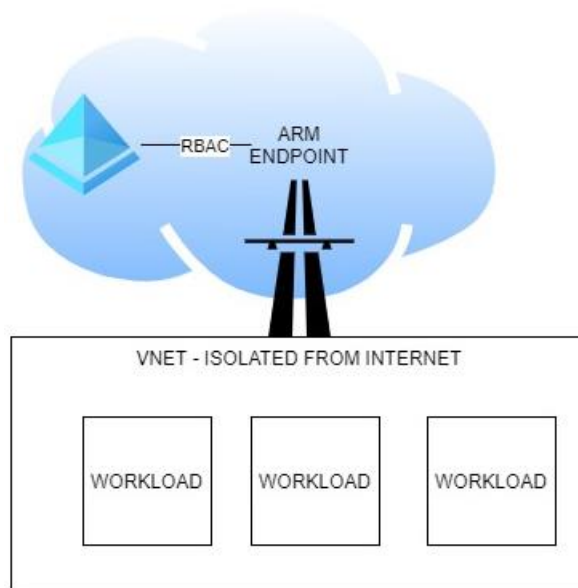
NAME

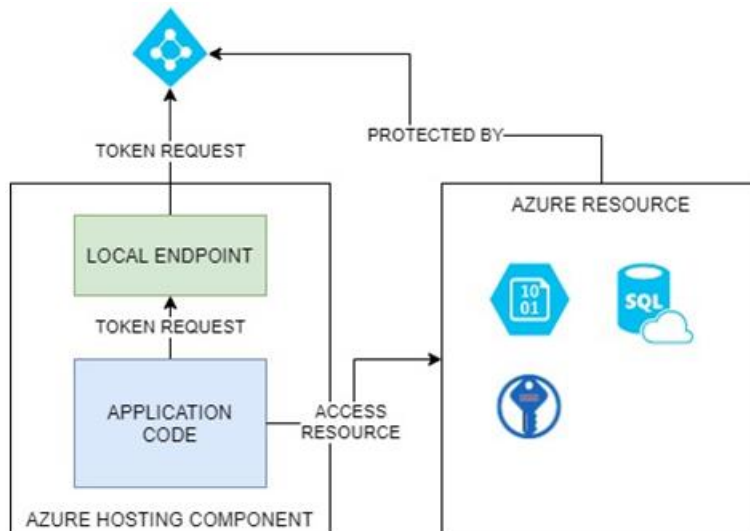
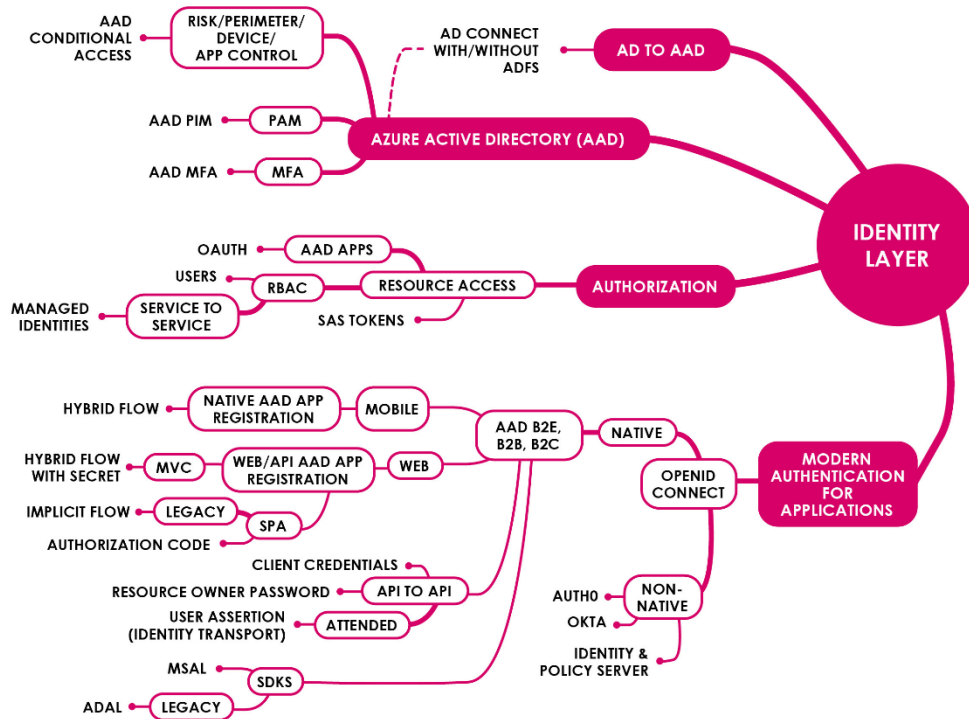
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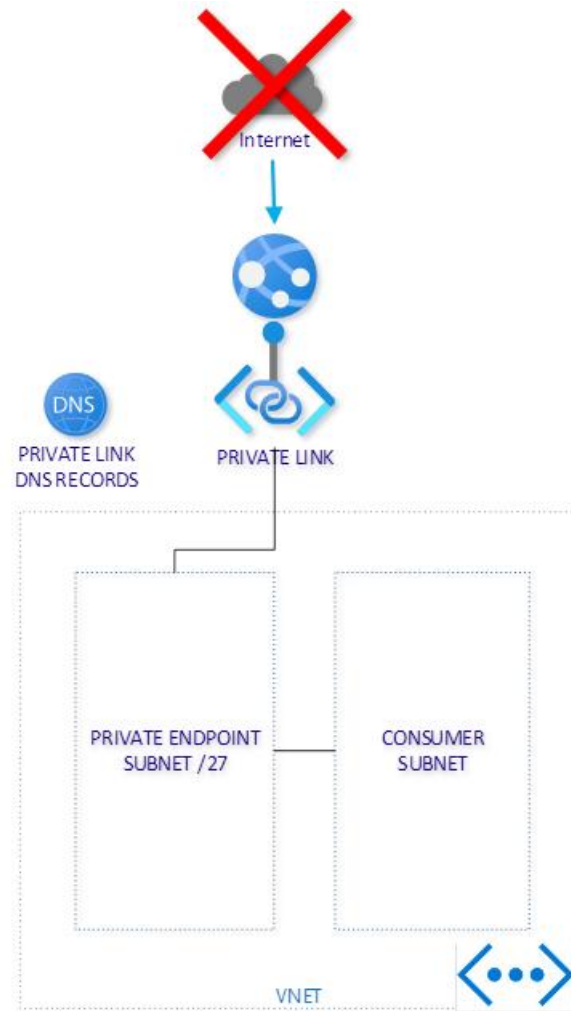
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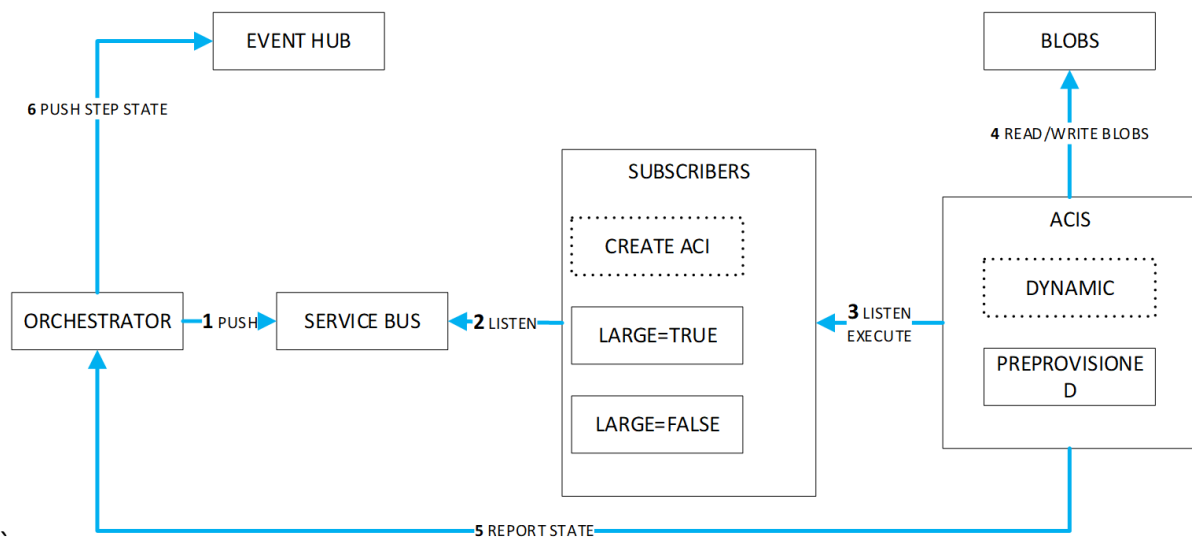
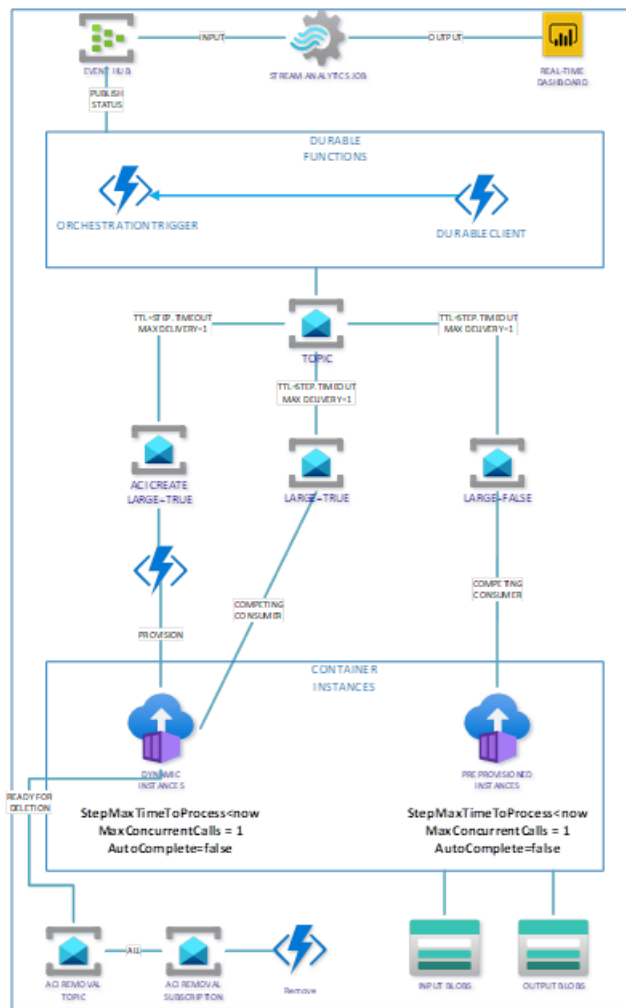
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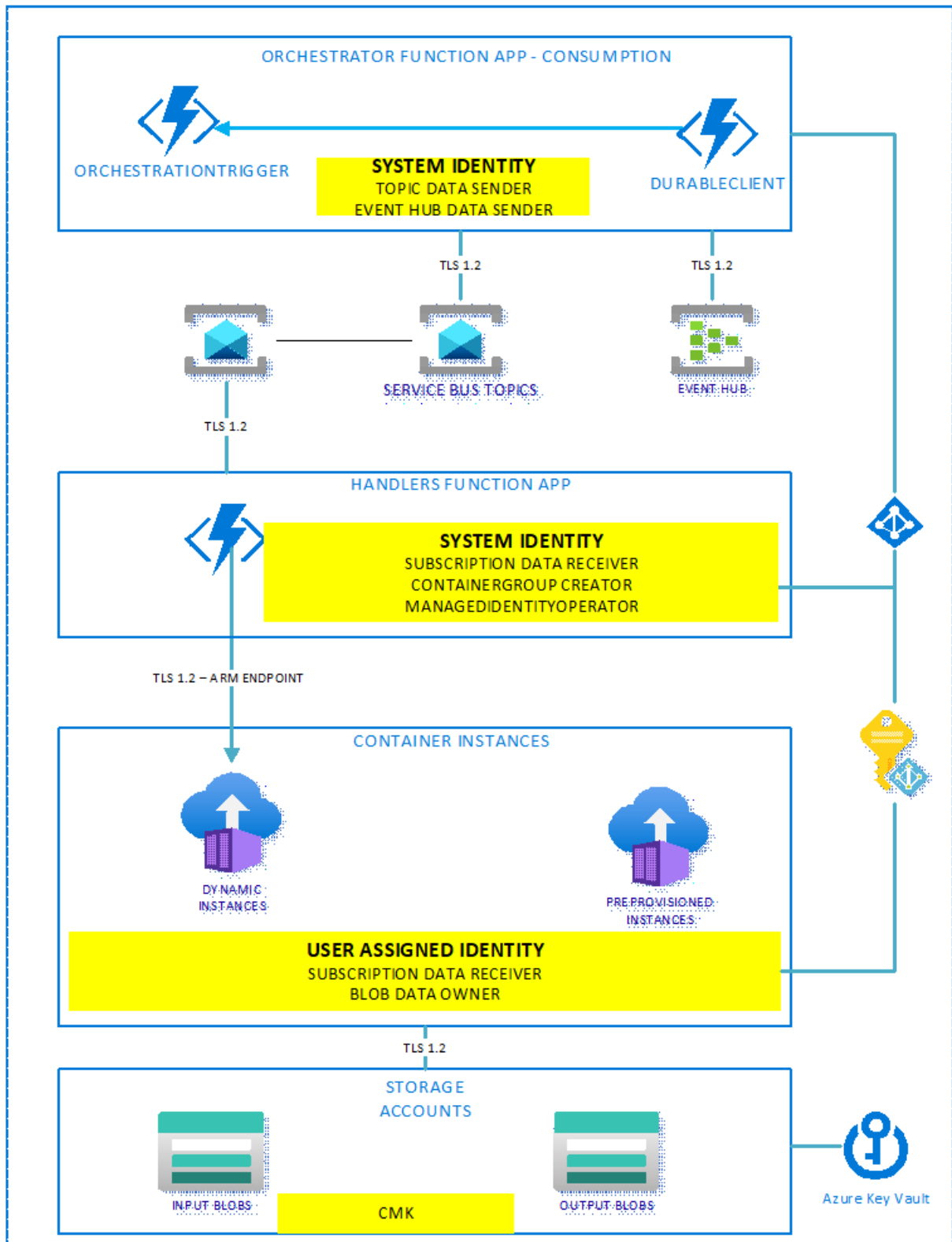
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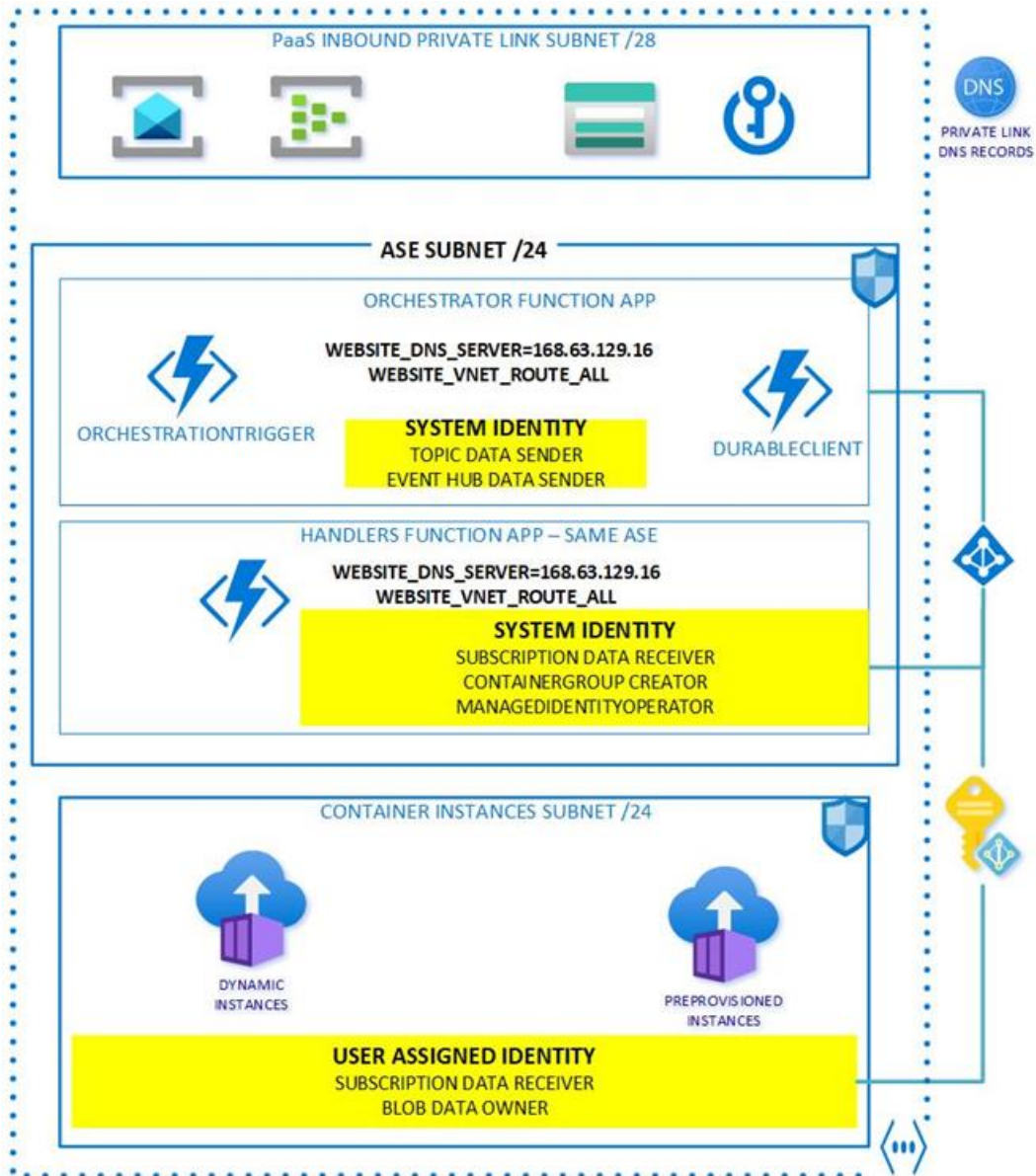












Chapter 8: Summary and Industry Scenarios

